

# NeuralCert: certified computational discovery of extremal mathematical constructions

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September 12, 2026

## Abstract

Neural networks are becoming popular in solving mathematical problems, but stochastic models do not provide mathematical exactness by themselves. This study introduces a discovery-to-certification framework in which high-dimensional variational trial functions are learned in a compact separable representation, spectrally diagnosed and pruned, and then certified exactly through multimodular evaluation. Exact certification makes the numerical proofs fully explicit and independently verifiable. This framework can be run on a standard personal computer.

Across three extremal problems, we show that neural optimization can contribute to rigorous mathematics in three distinct ways: by discovering improved constructions, by exposing empirical invariants that lead to proofs, and by revealing optimization barriers whose geometry motivates new analytic or numerical representations.

More broadly, these results suggest a path toward AI-assisted mathematics in which flexible computational discovery and exact certification become complementary components of a single rigorous workflow.

Machine learning can contribute to mathematical discovery by identifying latent relations between mathematical objects, guiding search over vast combinatorial or algebraic spaces, and generating candidate constructions or algorithms. Davies *et al.* used learned predictors and attribution methods to expose latent mathematical structure and guide conjecture formation [1]. AlphaTensor showed that reinforcement learning can search large spaces of tensor decompositions and recover exact matrix-multiplication algorithms, including previously unknown constructions [2]. FunSearch coupled neural generation to an executable evaluator, demonstrating that iterative proposal, evaluation and selection can produce new mathematical constructions [3]. Also, neural function approximators can transform a mixed discrete–continuous geometric problem into a differentiable optimization problem [4].

Despite these advances, a methodological gap persists for problems in which the unknown object is a function or extremizer in a large or infinite-dimensional space. Such problems are commonly reduced computationally by choosing a finite ansatz,

$$f(x) = \sum_{j=1}^m a_j \phi_j(x), \tag{1}$$

and optimizing over the coefficients  $a_j$ . While this makes the problem tractable, it also imposes an a priori structural restriction,

$$f^* \approx \text{span}\{\phi_1, \dots, \phi_m\}, \tag{2}$$

so that improvements in optimization accuracy cannot remove approximation error induced by an inadequate representation. However, neural parameterizations offer a complementary strategy: instead of fixing the shape of a candidate solution in advance, they provide flexible differentiable families in which large-scale functional search can be conducted. For example, in geometric colouring a mixed discrete–continuous search space was replaced by a probabilistic differentiable representation amenable to gradient optimization [4].

Here we introduce NeuralCert, a discovery-to-certification framework that deliberately separates the representation used for computational search from the representation used for mathematical proof. Flexible neural parameterizations are used to explore high-dimensional function spaces without prescribing a problem-specific finite ansatz. Discovered candidates are subsequently compressed into

explicit mathematical representations from which rigorous certificates can be constructed and independently verified. Across three extremal problems, we show that this separation enables neural optimization to contribute to rigorous mathematics through three distinct modes: discovering improved constructions, exposing structure that leads to analytic results, and identifying optimization barriers that motivate alternative representations.

Since a floating-point neural candidate does not constitute a proof, the discovery candidate is reduced to an independently checkable object [2, 3] such that verification establishes the mathematical claim. This idea builds on previous work where neural search produced tensor decompositions which could be checked algebraically [2], and candidate programs judged by an explicit executable evaluator [3]. These examples suggest that discovery and verification impose fundamentally different computational requirements. Discovery benefits from overparameterization, approximation and exploration, whereas verification benefits from low-dimensional structure, exactness and controlled numerical error.

## NeuralCert overview

NeuralCert is a Python framework for neural discovery of candidate functions followed by independent certification and verification. The architecture is deliberately separated into three components; discovery, certification and verification, which do not share numerical evaluation routines. Discovery uses flexible neural parameterizations to explore problem-specific function spaces and exports candidate functions and metadata, typically as NumPy `.npz` files. Certification is independent of the neural model: no network parameters, optimizer state or training data are required. Instead, the candidate is reconstructed in a problem-specific mathematical representation and converted into an explicit certificate. Verification is implemented as a standalone minimal code path that ingests only this certificate, typically as `.json`, and independently reconstructs the inequalities underlying the claimed result. Thus, neural optimization serves only as a proposal mechanism; final mathematical claims depend exclusively on independently reconstructed and verifiable proof objects.

## Discovery and certification of sieve constants improve prime gap bounds

The Maynard–Tao sieve reduces bounded gaps between primes to lower bounds on a variational constant  $M_k$ , which we approached through two independent computational routes. The first uses neural search over separable functional families, followed by reconstruction in a polynomial representation and rigorous Rayleigh–Ritz certification. At small  $k$ , this approach recovers known extremal behaviour ( $M_5 \geq 2.0071443298$  and  $M_{20} \geq 3.12755795$ ); at  $k = 25$  it improves the Polymath8b reference (certified 3.3221511 versus 3.3221426). Thus, high-quality extremizers can be discovered without prescribing the polynomial ansatz subsequently used for certification (Supplementary Tables 5 and 6).

At large  $k$ , an early neural run near  $k = 3600$  produced a numerically stable Rayleigh value above 8, remained below the analytic ceiling, and was stable under grid refinement. Exact reconstruction nevertheless failed: the optimizer had collapsed onto a sharply localized single channel whose discovery score was inflated by an inconsistent quadrature frame. The discovered profile approaches  $1/(\log k \cdot t)$ , a form that the polynomial channels cannot represent across the increasingly flat Rayleigh (Supplementary Methods 6.18). The artifact survived the available heuristic gates because its value did not cross the upper bound. Only independent reconstruction of the candidate (for certification) exposed the inconsistency, which motivated NeuralCert’s scalar cross-checks and support-aware guards.

Therefore, to access the much larger  $k$ , we developed a second route based on the analytically motivated rational family  $g(t) = 1/(c + nt)$ , whose evaluation can be certified in ball arithmetic at a computational cost nearly independent of  $k$ . This respectively yields  $M_{3655} \geq 8.000064869$ ,  $M_{208910} \geq 12.00000199$ ,  $M_{11655069} \geq 16.0000003924$ , and  $M_{644589002} \geq 20.0000009143$ , for 3, 4, 5 and 6 primes in a tuple, together with the uniform bound  $M_k \geq \log k - 0.307$  for  $100 \leq k \leq 1.88 \times 10^9$ . These results require only the Bombieri–Vinogradov theorem, assuming neither the Elliott–Halberstam conjecture nor Deligne-type distributional input, and through the sieve improve unconditional bounds for  $H_m$  as published on the Polymath8b Wiki [5] and [6] (see Figure 1 and Supplementary Data).

An asymptotic analysis of the single-channel rational family identifies its limiting convolution profile with a spectrally positive 1-stable law and proves  $M_k \geq \log k - C_* - o(1)$ ,  $C_* \approx 0.3343$  (Supplementary

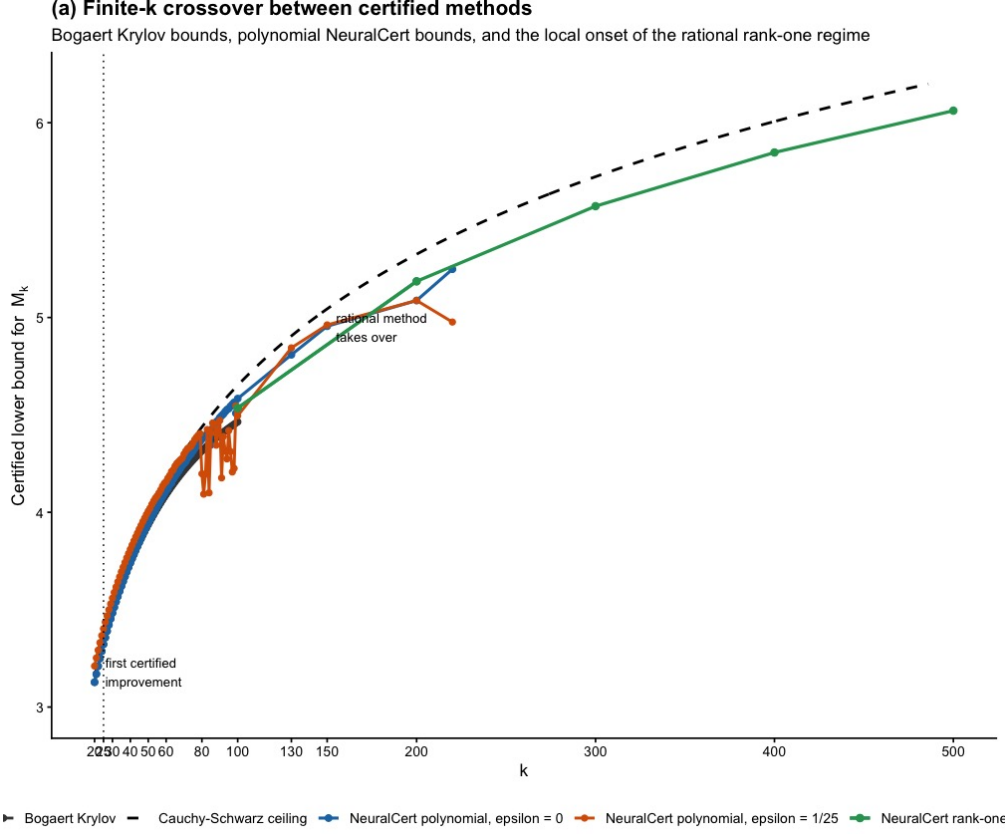


Figure 1: **Certified lower bounds for the Maynard variational constant across computational regimes.** Certified values obtained with Bogaert’s Krylov method are compared with NeuralCert polynomial constructions for the vanilla problem ( $\varepsilon = 0$ ) and enlarged support ( $\varepsilon = 1/25$ ), together with the certified rank-one rational family. The polynomial construction first exceeds the Krylov bounds at  $k = 25$ , but loses efficiency as  $k$  increases, whereas the rank-one rational representation becomes competitive and provides the continuation to larger  $k$ . The Cauchy–Schwarz bound is shown as a reference. Lines connecting discrete certified values are guides to the eye and do not represent interpolated certificates. Beyond  $k = 500$ , the rank-one rational family is the only representation that remained reliably optimizable and certifiable in the regime studied here.

Methods). This explains why the rank-one structure remains asymptotically competitive, attaining the optimal  $\log k$  growth rate with only a constant-order deficit, without implying global optimality of rank one. Consequently,  $0 \leq \liminf_{k \rightarrow \infty} (\log k - M_k) \leq C_*$ , where the lower endpoint follows from the classical bound  $M_k \leq \frac{k}{k-1} \log k$ . The finite-range constant 0.307 is smaller than  $C_*$  because, within this rational family, the finite-range deficit is numerically consistent with  $C_* - 1.047/\log k$ ; the directly evaluated deficit remains below 0.29 throughout the certified range.

Despite sharing no numerical evaluation, the two computational pipelines agree throughout their overlapping range. While the finite- $k$  results remain below Bogaert’s independently (Krylov subspace) computed values for  $k \neq 25$ ,  $k \leq 30$ , indicating the Krylov method is powerful for modest  $k$ , NeuralCert improves  $31 \leq k \leq 100$ . Every reported finite- $k$  inequality is accompanied by a machine-checkable certificate that can be independently verified.

## Enlarged sieve domains reveal a finite crossover

We also implemented the  $\varepsilon$ -enlarged variant proposed in Polymath8b. At  $k = 49$ , optimization within the certified trial family saturates at 3.98867, below  $M_{49} > 4$ , to obtain  $H_1 \leq 240$ . Providing further evidence that the enlarged sieve weight comes close to, but does not certify, this bound at Bombieri–

Vinogradov level. We report the first certified  $\varepsilon$ -values at  $k = 49, 52$  and above  $k = 54$ . Across the ladder  $50 \leq k \leq 220$  the advantage conferred by  $\varepsilon$  fluctuate and then decreases monotonically in  $[155, 192]$  and changes sign at a finite crossover  $k_*(\varepsilon)$ , which we locate at  $k_*(1/12) \approx 53.7$ ,  $k_*(1/16) \approx 80.7$  and  $k_*(1/25) \in [155, 192]$ . We conjecture that such a crossover exists for every  $\varepsilon > 0$  and that  $k_*(\varepsilon)$  grows superlinearly in  $1/\varepsilon$ , so that  $\varepsilon k_*(\varepsilon) \rightarrow \infty$  (Supplementary Methods 6.27 and Supplementary Table 7).

## Higher-order Delsarte discovery exposes a tensorization obstruction

Higher-order Delsarte hierarchies strengthen the classical LP for linear codes and show substantial finite-length gains, but asymptotic progress requires an explicit dual family rather than additional numerical optimization [7, 8]. We therefore treated the higher-order dual as a discovery problem, initially asking whether a low-complexity correction to the lifted level-one solution could improve the MRRW bound. Extending the symmetry-reduced hierarchy to  $r = 3$  confirmed that higher levels contain genuine information, including the level-3 bound  $V_3(12, 5) \leq 16$ , against the level-2 value  $V_2(12, 5) = 24.260255 \dots$  located numerically. However, comparison with the explicit lift showed that the source of the improvement changes with hierarchy level: at  $r = 2$  it can reside entirely in the partial-Fourier constraints, whereas at  $r = 3$  the full-transform rows already improve on the lift. A single learned correction was therefore not a stable asymptotic target.

We redirected discovery to the spectral construction of [8] where the blocklength dependence can be removed analytically. For a normalized configuration  $G$  on  $\mathbb{F}_2^\ell$ , the relevant objective is  $J_\ell(G) = H(G)/\ell$ . Although we initially intended to parameterize  $G$  neurally, symmetry and analytic reduction collapsed the problem faster than a generic network became useful: for  $\ell \leq 4$  the largest search space has only 15 free probability coordinates and no remaining dependence on  $n$ . We therefore optimized the reduced configuration directly. Across 28 instances and 13,600 randomized local starts, no configuration improved MRRW; instead, the recovered optima repeatedly matched the quasirandom tensor-product family to numerical precision.

This repeated null result exposed a structural obstruction. Defining the translation affinity

$$\Phi(G, v) = \sum_u \sqrt{G(u)G(u+v)},$$

the finite spectral walk condition implies  $\Phi(G, v) \geq \varepsilon$  on a spanning set of shifts (Methods, hypotheses (A1)–(A2)). For

$$h(\varepsilon) = H\left(\frac{1 - \sqrt{1 - \varepsilon^2}}{2}\right),$$

convexity of  $h$ , conditioning and Jensen’s inequality give

$$H(G) \geq \sum_{i=1}^{\ell} h(\Phi(G, e_i)) \geq \ell h(\varepsilon).$$

Hence, with  $\delta = (1 - \varepsilon)/2$ ,

$$J_\ell(G) \geq h(\varepsilon) = H\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right),$$

exactly the first MRRW expression. The tensor-product configuration  $\text{Bernoulli}(p_\varepsilon)^{\otimes \ell}$  saturates the entropy–affinity inequality, identifying the object repeatedly recovered computationally.

Hence, within the fixed single-orbit spectral ansatz, optimizing the configuration  $G$  cannot improve MRRW (Theorem 5 and Corollary 6).

## Convex discovery improves sign-uncertainty bounds

For the +1 Bourgain–Clozel–Kahane sign-uncertainty problem, radial polynomial–Gaussian Fourier eigenfunctions are the canonical ansatz. Cohn, Dong and Gonçalves [9] showed that sublinear-degree

members of this family encounter an asymptotic ceiling, motivating our initial search for more flexible Fourier-eigenfunction structures. We therefore treated the choice of envelope, scales and contact locations as a discovery problem, with learned proposals followed by independent high-precision evaluation.

The flexible search improved rapidly but stalled in dimension  $d = 1$ . Gaussian-mixture constructions reached  $\rho = 0.578532$ , above the published  $A_+(1) \leq 0.572990$  bound. Neural optimization did not remove the plateau: apparently improving trajectories systematically entered coefficient systems with condition numbers  $10^{17}$ – $10^{18}$ , where no reliable digits could be guaranteed from the floating-point solve. Multiprecision reevaluation rejected these candidates. Multistart runs also converged to distinct plateaus rather than a common value, consistent with changes in root multiplicity and contact structure in the last-sign-change functional. We therefore treated the failure as a diagnostic of the search representation rather than evidence that the function class itself had been exhausted.

The decisive simplification was to return to the Fourier-eigenfunction constraint. With  $u = \pi|x|^2$ , every radial  $+1$  polynomial–Gaussian eigenfunction in the truncated Laguerre space has the form

$$f(x) = P(u)e^{-u}, \quad P \in \text{span} \left\{ L_0^{(d/2-1)}(2u), L_2^{(d/2-1)}(2u), \dots \right\}.$$

The Fourier constraint is linear in the Laguerre coefficients and  $e^{-u} > 0$ , so for fixed  $u_0$  the remaining condition is simply

$$P(0) = 0, \quad P(u) \geq 0 \quad (u \geq u_0).$$

Thus the search can be reformulated as a convex semi-infinite feasibility problem in the coefficients, with bisection in  $u_0$ , rather than a non-convex optimization over forced contacts. An adaptive exchange LP locates the active tangencies; these are then rationalized, used to reconstruct an exact Laguerre polynomial over  $\mathbb{Q}$ , and verified on the complete ray by exact factorization and a Sturm chain over  $\mathbb{Z}$ . The numerical search therefore determines sharpness, whereas admissibility of each quoted bound follows independently from the rational certificate.

In  $d = 1$  the degree ladder first reproduces the published configuration and then improves it:

$$0.572989678 \quad (\text{deg } 22), \quad 0.572706699 \quad (\text{deg } 30), \quad 0.572588699 \quad (\text{deg } 38).$$

The degree-22 search numerically resolves the same five-contact configuration as the published construction to its stated precision; the new improvement occurs at higher degree. The degree-38 candidate is certified by an explicit rational polynomial, giving  $A_+(1) \leq 0.572588699$ . The same pipeline gives  $A_+(2) \leq 0.756206236$ , which is only  $7.6 \times 10^{-7}$  below the published 0.756207 value and is therefore treated as a certified reproduction with a marginal gain rather than a substantive improvement (see Figure 2).

Higher-dimensional runs approach, rather than cross, the known polynomial–Gaussian asymptotic ceiling; their interpretation is limited by the numerical degree ceiling of the floating-point discovery stage (Methods). Thus, the new small-dimensional bounds do not evade the Cohn–Dong–Gonçalves obstruction. Instead, they show that the finite polynomial–Gaussian family had not been numerically exhausted.

## Discussion

Can modern neural optimization rediscover and improve extremal constructions in hard analytic and combinatorial mathematics? Across the three problems studied here, the answer is a set of distinct discovery modes. In the Maynard–Tao sieve, neural optimization identifies stronger extremizers, which can be compressed into explicit structure, extended to a new asymptotic kernel and certified to yield improved bounds. In the higher-order Delsarte hierarchy, repeated computational failure is itself informative: the search collapses onto a tensor-product family, and the resulting empirical invariant becomes a proof that the chosen spectral degree of freedom cannot improve MRRW. In sign uncertainty, neural optimization instead encounters a persistent plateau; independent verification reveals severe ill-conditioning, and analysis of that failure exposes a convex reformulation that produces new certified bounds. Thus, computation contributes not only by finding better objects, but also by identifying obstructions and more appropriate representations. The numerical degree ceiling encountered in the final problem mirrors the high-degree regime that asymptotic theory predicts is necessary for further progress.

### Objective landscape and trajectory search

The last-sign-change objective  $\rho(\tau_1, \tau_2)$  on a slice through the certified record configuration ( $d = 1$ , degree 38,  $m = 9$ ,  $\rho = 0.5725887$ ).

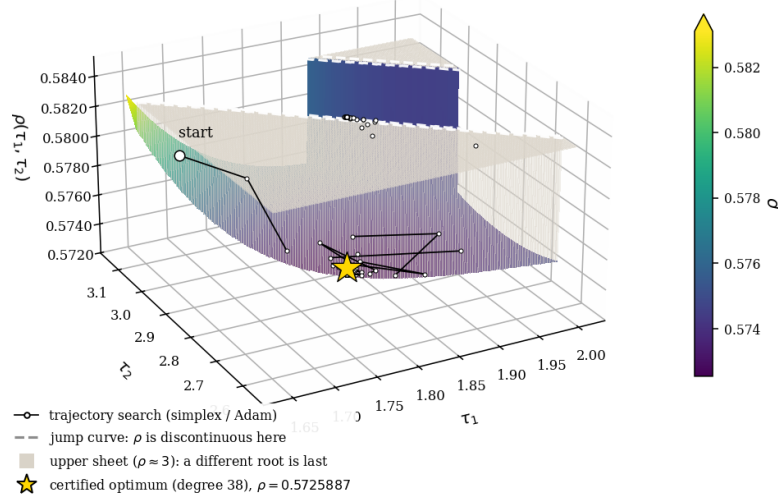


Figure 2: **Objective landscape and trajectory search.** The last-sign-change functional  $\rho(\tau_1, \tau_2)$  on a slice through the certified record configuration ( $d = 1$ , degree 38,  $m = 9$  contacts,  $\rho = 0.5725887$ ), obtained by varying the two innermost contacts with the remaining seven held at  $\tau^*$ . The surface is drawn from the raw grid without smoothing or interpolation, and is evaluated in the damped Laguerre basis  $\psi_n(u) = L_n^{(\alpha)}(2u)e^{-u}$ : at this degree the underlying polynomial reaches  $\sim 10^{69}$  on the grid, so a monomial evaluation loses every digit at the outer tangencies and splits them into spurious sign changes. The landscape has two sheets, a narrow valley (colour) and an upper sheet at  $\rho \approx 3$  (translucent), with no intermediate values — crossing the dashed curve, a different root of  $P$  becomes the last sign change and  $\rho$  leaps. Black markers show a simplex/Adam trajectory, its polyline broken wherever the objective jumps. First-order optimality forces a change of root multiplicity, so the minimiser lies on the discontinuity set of the functional that root-parametrised methods differentiate; the convex reformulation avoids this by treating  $\rho$  as a bisection variable rather than an objective. The dashed curve is drawn as a high-gradient contour and indicates where  $\rho$  jumps rather than resolving the discontinuity set exactly.

## Methods

### 1 Discovery and certification of sieve constants improve prime gap bounds

#### 1.1 Maynard–Tao variational formulation

The Maynard–Tao sieve reduces bounded gaps between primes to the construction of admissible trial functions with large variational quotient [5, 10]. For the separable constructions considered here, the relevant denominator and numerator functionals can be expressed through pairwise one-dimensional convolution integrals. We write the trial function as

$$F_{\theta,c}(t_1, \dots, t_k) = \sum_{j=1}^m c_j \prod_{i=1}^k g_{\theta,j}(t_i), \quad (3)$$

where  $g_{\theta,j}$  are one-dimensional channels and  $c \in \mathbb{R}^m$  is a linear mixing vector. For each pair,

$$h_{j\ell}(x) = g_{\theta,j}(x)g_{\theta,\ell}(x), \quad (4)$$

and the  $k$ -dimensional functionals reduce to quadratic forms

$$I_{k,\varepsilon}(F_{\theta,c}) = c^\top A(\theta)c, \quad J_{k,\varepsilon}^{(1)}(F_{\theta,c}) = c^\top B(\theta)c, \quad (5)$$

whose entries depend on one-dimensional integrals involving  $h_{j\ell}^{*(k-1)}$ . At fixed channels the objective is therefore the generalized Rayleigh quotient

$$R(\theta) = k \lambda_{\max}(B(\theta), A(\theta)) = k \max_{c \neq 0} \frac{c^\top B(\theta)c}{c^\top A(\theta)c}. \quad (6)$$

The nonlinear optimizer consequently searches only over the channel manifold; the optimal linear combination is re-solved as a generalized Ritz vector at each objective evaluation [11, 12]. The precise normalization of  $M_k$ , its  $\varepsilon$ -enlarged analogue, and the conversion of thresholds in  $M_k$  to bounds on  $H_m$  are given in Supplementary Methods, Section 6.1.

## 1.2 Neural discovery of separable trial functions

The discovery stage is used only to identify strong explicit trial functions and does not establish a rigorous inequality. To expose the increasingly sharp boundary structure at large  $k$ , each channel is parameterized as

$$g_{\theta,j}(x) = r_{\theta,j}(x)e^{-\rho_j x}, \quad \rho_j > 0, \quad (7)$$

where the residual functions are produced by a shared multilayer perceptron with channel-specific outputs. The network receives  $(x, e^{-kx})$  as input, explicitly exposing the natural  $O(k^{-1})$  boundary-layer scale. In the positive-channel solver,  $r_{\theta,j}(x) = \exp(f_{\theta,j}(x, e^{-kx}))$ , allowing the convolution recursion to be performed without cancellation in logarithmic coordinates.

The leading generalized Ritz vector is profiled out rather than learned jointly with the channel parameters. If

$$B(\theta)v_\theta = \lambda_\theta A(\theta)v_\theta, \quad v_\theta^\top A(\theta)v_\theta = 1, \quad (8)$$

we freeze  $v_\theta$  and differentiate

$$q_{v_\theta}(\theta) = k \frac{v_\theta^\top B(\theta)v_\theta}{v_\theta^\top A(\theta)v_\theta}. \quad (9)$$

Under the usual nondegeneracy condition, the Hellmann–Feynman/envelope identity [13, 14] gives the derivative of the profiled leading eigenvalue without differentiating through the eigendecomposition. Adam [15] is used for exploration and a block L-BFGS minorize–maximize procedure for polishing [16]. Channel-rank and grid-resolution continuation are used during search. Network architecture, optimization schedules, finite-difference gradient controls and continuation rules are specified in Supplementary Methods, Sections 6.4–6.10.

## 1.3 Stable evaluation of high-order convolution powers

Direct recursion on  $h^{*p}$  becomes unstable as  $p$  grows because algebraic concentration, exponential decay and global amplitude must otherwise be represented simultaneously. For each channel pair we write

$$h_{j\ell}(x) = r_{j\ell}(x)e^{-a_{j\ell}x}, \quad a_{j\ell} = \rho_j + \rho_\ell, \quad (10)$$

and factor its convolution powers as

$$\nu_p^{(j\ell)}(s) = s^{p-1} e^{-a_{j\ell}s} e^{\sigma_p^{(j\ell)}} \psi_p^{(j\ell)}(s). \quad (11)$$

The algebraic and exponential terms are handled analytically, the scalar  $\sigma_p$  carries the global amplitude, and only the residual shape  $\psi_p$  is represented numerically. Substitution into the convolution identity and the change of variables  $x = su$  yield a beta-weighted recursion,

$$\psi_{p+q}(s) = B(p, q) \int_0^1 \psi_p(su) \psi_q(s(1-u)) \frac{u^{p-1}(1-u)^{q-1}}{B(p, q)} du, \quad (12)$$

which is evaluated with Gauss–Jacobi quadrature [17]. The concentration associated with increasing  $p$  and  $q$  is thereby moved into a known quadrature weight rather than left as unresolved structure in the represented residual.

For positive channels we store  $\xi_p = \log \psi_p$  and evaluate the positive quadrature sum by log-sum-exp, preventing underflow across repeated convolutions. The required power  $h^{*(k-1)}$  is assembled through a binary addition chain, reducing convolution depth from  $O(k)$  to  $O(\log k)$ . Representation, convolution, primitive-integration and outer-integration grids are kept distinct. Full recursion formulas, grid definitions and quadrature construction are given in Supplementary Methods, Sections 6.5–6.7.

## 1.4 Structure-preserving Rayleigh–Ritz optimization

The denominator matrix  $A$  is mathematically a Gram matrix and must satisfy  $A \succeq 0$ . Entrywise accurate discretization does not automatically preserve this property. We therefore use positive two-point interpolation in the convolution recursion and a shared positive-weight outer integration mesh across all channel pairs. Whenever the inner quadrature nodes are common to all pairs, the discrete denominator has the form

$$A_{j\ell}^{(N)} = \sum_{\alpha} w_{\alpha} \Phi_{\alpha j} \Phi_{\alpha \ell}, \quad w_{\alpha} > 0, \quad (13)$$

and therefore retains the positive-semidefinite Gram structure up to floating-point round-off. In the saturated regime, where the inner rule for the channel primitives becomes pair-dependent, this common representation is preserved only up to inner-quadrature error; numerical adequacy is then assessed by the positive-semidefiniteness, two-grid and refinement diagnostics of Section 1.5, which are safeguards rather than rigorous bounds on that error. No bound reported here depends on this discrete structure: the certified values are produced by the independent exact certifier.

Because channel norms may span hundreds of orders of magnitude, the Gram pencil is diagonally equilibrated by the exact congruence transformation

$$A \mapsto D^{-1/2} A D^{-1/2}, \quad B \mapsto D^{-1/2} B D^{-1/2}, \quad D = \text{diag}(A_{11}, \dots, A_{mm}). \quad (14)$$

Numerically unresolved directions are removed by rank-revealing spectral whitening; no ridge term is added to the variational denominator. The interpolation, shared-mesh construction, equilibration and rank-revealing Ritz solve are detailed in Supplementary Methods, Sections 6.8–6.9.

## 1.5 Validation of neural candidates

Validation is embedded into model selection because a flexible optimizer can exploit structured discretization error. Candidate states are therefore required to reproduce the independently recomputed Rayleigh quotient, preserve numerical positive semidefiniteness of the denominator, retain sufficient effective rank, remain stable under a more conservative rank threshold, respect applicable rigorous upper bounds for both individual channels and the full quotient (rejecting numerical states that spuriously exceed the Cauchy–Schwarz ceiling [5]), and reproduce on a finer representation grid. Closed-form controls based on  $g(x) = 1$  and  $g(x) = x$  exercise the convolution and integration paths before optimization. Final candidates are re-evaluated on multiple grids; extrapolation is used only diagnostically and never as a certified lower bound. Exact validation criteria, controls and refinement procedures are provided in Supplementary Methods, Sections 6.11–6.12.

## 1.6 Exact certification of neural candidates

A discovered candidate is projected onto rational polynomial channels,

$$F(t_1, \dots, t_k) = \sum_{j=1}^m c_j \prod_{i=1}^k q_j \left( \frac{t_i}{1 + \varepsilon} \right), \quad (15)$$

and the mixing vector is transferred from the preconditioned discovery frame to the corresponding exact channel frame. For fixed rational channels and rational coefficients the associated Maynard quotient is rational; certification therefore reduces to exact evaluation of the two quadratic forms.

Rather than constructing and reconstructing every exact Gram-matrix entry, the certifier accumulates the pair contributions to  $c^\top A c$  and  $c^\top B c$  modulo machine-word primes. For each prime  $p$  it computes only two aggregate residues,

$$S_A(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_\ell A_{j\ell}^{\text{int}} \pmod{p}, \quad S_B(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_\ell B_{j\ell}^{\text{int}} \pmod{p}. \quad (16)$$

Only the aggregate integers  $S_A$  and  $S_B$  are reconstructed, with analytically known positive denominators  $D_A, D_B$ :

$$c^\top A c = \frac{S_A}{D_A}, \quad c^\top B c = \frac{S_B}{D_B}. \quad (17)$$

Rigorous a priori integer bounds  $|S_A| \leq \mathcal{B}_A$  and  $|S_B| \leq \mathcal{B}_B$  determine the reconstruction budget. If  $M_P = \prod_{p \in P} p$  and

$$M_P > 2 \max(\mathcal{B}_A, \mathcal{B}_B), \quad (18)$$

the centered Chinese-remainder representative is unique. The final quotient is then formed exactly from the reconstructed integers and known denominators. Thus the neural optimizer, numerical quadrature and grid extrapolation no longer enter the proof once the rational trial function has been fixed.

Before exact evaluation, the candidate is compressed by loss-controlled backward elimination: channels are removed only while the re-optimized Rayleigh quotient remains within a prescribed relative loss budget. The exact backend certifies the explicit reduced function, so pruning affects certificate strength and computational cost but not validity. Detailed denominator formulas, coefficient-extraction identities, rational representation transfer, pruning rules, integer bounds and deterministic CRT reconstruction are given in Supplementary Methods, Sections 6.13–6.17.

## 1.7 Large- $k$ rational construction

At large  $k$ , neural optimization is used as a structural discovery instrument rather than as the source of a certified numerical value. The neural candidate is compressed into clustered rational families

$$g_{\text{rat}}(t) = \sum_{a=1}^r \sum_{j=1}^{p_{\max}} u_{a,j} (c_a + nt)^{-j}, \quad n = k - 1, \quad (19)$$

where  $c_a > 0$  are pole locations and the higher powers represent confluent directions. For fixed  $c_a$ , the coefficients  $u_{a,j}$  are eliminated by variable projection, so only the nonlinear pole locations are optimized. The confluent basis has a direct geometric interpretation because

$$\frac{\partial^q}{\partial c^q} \frac{1}{c + nt} = (-1)^q q! (c + nt)^{-(q+1)}. \quad (20)$$

The quality of a distilled representation is not determined by pointwise fitting error alone. Each candidate is rebuilt in an independent deterministic evaluator, the corresponding Gram pencil is re-computed, and the generalized Rayleigh problem is solved afresh. Candidates are rejected if they fail probability-mass conservation, analytic moment checks, Fourier-inversion stability, Gram-rank diagnostics, cross-cluster conditioning, or the rigorous ceiling

$$R_k \leq \frac{k}{k-1} \log k. \quad (21)$$

Surviving pole locations are then locally refined under the deterministic Rayleigh objective.

In the large- $k$  regime, additional rational clusters can reduce the pointwise approximation error while failing these distributional checks. The stable deterministic representation collapses to the first-order family

$$g(t) = \frac{1}{c + nt}, \quad n = k - 1, \quad (22)$$

which is subsequently optimized independently of the neural model. This distillation step therefore uses neural search to identify structure, while the numerical value submitted for certification is obtained from a separate rational representation. The clustered fit, deterministic Fourier evaluator, failure diagnostics and local refinement are detailed in Supplementary Methods, Section 6.18.

## 1.8 Certified large- $k$ evaluation in ball arithmetic

For the first-order family in Eq. (22), certification can be reduced to a one-dimensional probabilistic Fourier calculation. Let  $w = g^2$ ,  $m_0 = \int_0^1 w(t) dt = [c(c+n)]^{-1}$ , and let  $S$  be the sum of  $n = k - 1$  independent draws from the probability density  $w/m_0$  on  $[0, 1]$ . With

$$G(\rho) = \int_0^\rho g(t) dt, \quad H(\rho) = \int_0^\rho w(t) dt, \quad (23)$$

the separable Maynard quotient reduces exactly to

$$R = \frac{kN}{D}, \quad N = \mathbb{E}[G(1-S)^2 \mathbf{1}_{S < 1}], \quad D = \mathbb{E}[H(1-S) \mathbf{1}_{S < 1}]. \quad (24)$$

Writing

$$\varphi(\theta) = \left( \frac{\widehat{w}(\theta)}{m_0} \right)^n, \quad \Psi_X(\theta) = \int_0^1 X(\rho) e^{i\theta\rho} d\rho, \quad (25)$$

for  $X \in \{G^2, H\}$ , Fourier inversion gives

$$I_X = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi(\theta) e^{-i\theta} \Psi_X(\theta) d\theta. \quad (26)$$

The certifier evaluates a truncated trapezoidal form of Eq. (26) using Arb ball arithmetic. The difference from the exact integral is bounded by three separately controlled contributions: periodic aliasing, omitted Fourier mass, and rigorous numerical enclosure of the transforms and auxiliary integrals. Aliasing is one-sided because the corresponding convolution density is nonnegative and compactly supported; its residual mass is bounded by certified moment inequalities. The omitted Fourier range is divided into a finite band controlled by Arb wide-ball suprema and an analytic tail obtained from integration-by-parts decay.

All error budgets remain in ball arithmetic until the final quotient is assembled. If  $N_{\text{ball}}$  and  $D_{\text{ball}}$  are the certified truncated-sum enclosures and  $E_N, E_D$  the rigorous aliasing and truncation budgets, the exported lower bound is

$$R_{\text{low}} = \text{lb} \left( \frac{k \text{ lb}(N_{\text{ball}} - E_N)}{\text{ub}(D_{\text{ball}} + E_D)} \right), \quad (27)$$

with all conversions explicitly rounded outward. The stored claim is rounded downward, so changes in valid Arb subdivision or enclosure radii cannot strengthen the reported inequality. The complete reduction, Fourier lemmas, moment bounds, tail estimates, special-function evaluation and directed rounding procedure are given in Supplementary Methods, Sections 6.20–6.23.

## 1.9 Finite-range certification and stable-law asymptotics

The first-order rational family also permits a certified bound over a continuous range of integer dimensions. Let  $R_k^{(1)}$  denote the Rayleigh quotient of  $g_c(t) = (c + (k-1)t)^{-1}$  after selection of  $c$ . To avoid optimizing  $c$  independently at every integer  $k$ , we use the standardized simplex headroom

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)} \quad (28)$$

as a scale coordinate. Its moments are available in closed form, and the large- $k$  optimizers empirically concentrate near a fixed value of  $\eta$ . A log-uniform ladder beginning at  $k = 100$  was therefore generated by solving  $\eta(c, k) = -0.4746$  and certifying the resulting fixed rank-one trial at each rung. For consecutive certified values  $k_i < k_{i+1}$ , monotonicity of  $M_k$  gives, for every integer  $k \in [k_i, k_{i+1}]$ ,

$$M_k \geq R_{\text{cert}}^{(1)}(k_i) \geq \log k - \left[ \log k_{i+1} - R_{\text{cert}}^{(1)}(k_i) \right]. \quad (29)$$

The maximum interval deficit over the complete ladder is 0.306754936, yielding the uniform certified statement

$$M_k \geq \log k - 0.307, \quad 100 \leq k \leq 1.87 \times 10^9. \quad (30)$$

The ladder construction, scale surrogate and interval-by-interval calculation are given in Supplementary Methods, Sections 6.24–6.25.

The asymptotic behavior of the same rational family can be analysed independently of the numerical discovery pipeline. Put  $n = k - 1$  and choose the fixed-shift scaling

$$c_k^{-1} = \log n + a, \quad a \in \mathbb{R}. \quad (31)$$

After rescaling the squared rational channel, the associated triangular array has Lévy density converging to  $x^{-2} dx$ . The centered boundary variable converges in distribution to

$$X_a = (a + 1) - \Lambda, \quad (32)$$

where  $\Lambda$  is the spectrally positive 1-stable random variable with characteristic function

$$\mathbb{E}e^{is\Lambda} = \exp \left\{ \int_0^\infty (e^{isx} - 1 - isx\mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2} \right\}. \quad (33)$$

Uniform density and logarithmic-integrability estimates permit passage from weak convergence to the logarithmic observables in the Rayleigh quotient. The resulting asymptotic defect is

$$R(F_{k,c_k}) = \log k - \mathcal{C}(a) + o(1), \quad \mathcal{C}(a) = a - 2\mathbb{E}[\log X_a \mid X_a > 0]. \quad (34)$$

Thus, with

$$\mathcal{C}_* = \inf_{a \in \mathbb{R}} \mathcal{C}(a), \quad (35)$$

the Maynard constant satisfies

$$M_k \geq \log k - \mathcal{C}_* - o(1). \quad (36)$$

Rigorous interval evaluation of the limiting expression gives the numerical constant reported in the Results,  $\mathcal{C}_* \simeq 0.3343$ . This is a lower-bound construction obtained from fixed-shift scalings and is not asserted here to be the asymptotic optimum over all possible sequences  $c_k$ . The stable-limit proof, logarithmic uniform-integrability argument and numerical evaluation of  $\mathcal{C}_*$  are detailed in Supplementary Methods, Section 6.26.

### 1.10 $\varepsilon$ -enlarged sieve calculations

We additionally evaluate the enlarged-domain variant of the Maynard variational problem. For each computed pair  $(k, \varepsilon)$ , the same discovery-to-certification separation is retained: a candidate is optimized numerically, transferred to the explicit representation used by the  $\varepsilon$ -certifier, and the resulting lower bound is recomputed independently. To compare the enlarged and vanilla searches we define

$$\Delta_{\text{cert}}(k, \varepsilon) = L_{k,\varepsilon} - L_{k,0}, \quad (37)$$

where  $L_{k,\varepsilon}$  and  $L_{k,0}$  are the certified lower bounds obtained by the two pipelines. Across the reliable sweep,  $\Delta_{\text{cert}}$  decreases with  $k$  and changes sign for the tested values of  $\varepsilon$ . Linear interpolation of the reliable sign-bracketing pairs gives the crossover estimates reported in the Results (Supplementary Table 7).

Because Eq. (37) compares two lower bounds rather than the unknown exact variational constants, a negative value does not prove  $M_{k,\varepsilon} < M_k$ . It establishes only that the certified vanilla construction found by our pipeline outperforms the certified enlarged-domain construction at that  $k$ . We therefore treat the finite crossover as an empirical property of the certified construction ladder and state its extension to the exact variational constants as a conjecture. In particular, the observed crossover locations motivate

$$\varepsilon k^*(\varepsilon) \longrightarrow \infty \quad (\varepsilon \rightarrow 0), \quad (38)$$

but the available data do not determine a unique finer scaling law. The sign brackets, excluded optimization failures, interpolation procedure and scope of the conjecture are given in Supplementary Methods, Section 6.27.

### 1.11 Independent certificate verification

Large- $k$  certificates are self-contained records containing the exact rational trial parameter, Fourier discretization parameters, error summaries and the claimed lower bound. The standalone verifier does not trust stored intermediate numerical bounds. It reconstructs the analytic constants, regenerates the far-field partition, recomputes wide-ball suprema, certified moments, Fourier nodes, trapezoidal sums, aliasing and truncation errors, and assembles a new  $R_{\text{low}}$  in ball arithmetic. The claim is accepted only when this independently recomputed lower bound supports the stored inequality.

The certificate proves the Rayleigh quotient of one fixed admissible trial function; it does not establish optimality of its parameter, re-prove the external Maynard–Tao sieve theorem, or certify the diameter of the admissible  $k$ -tuple used to convert a certified  $M_k$  value to a prime-gap bound. Monte–Carlo and small- $k$  comparisons are falsification tests rather than proof components. Certificate schema and trust boundary are specified in Supplementary Methods, Section 6.31.

## 2 Higher-order Delsarte discovery and certification

**Symmetry-reduced hierarchy.** For hierarchy level  $r$ , blocklength  $n$  and minimum distance  $d$  we construct the symmetry-reduced higher-order Delsarte LP over type vectors

$$I_{r,n} = \left\{ \alpha \in \mathbb{N}^{2^r} : \sum_{u \in \mathbb{F}_2^r} \alpha_u = n \right\},$$

where a type records the multiplicity of each column  $u \in \mathbb{F}_2^r$  in an  $r \times n$  binary matrix. For linear codes the program is invariant under  $\text{GL}(r, 2)$ , so variables are indexed by group orbits. Full Fourier constraints are represented by multivariate Krawtchouk coefficients  $K_\alpha(\beta)$ , and the Loyfer–Linial strengthening adds the corresponding partial transforms [7]. We implemented these transforms recursively and extended the symmetry reduction to  $r = 3$ , where sorting-based orbit representatives are no longer valid; construction, orbit enumeration and validation anchors are given in Supplementary Methods 7.1.

**Exact dual certification.** Floating-point optimization is used only to locate competitive dual solutions. Candidates are reconstructed at high precision, rounded to dyadic rationals and checked against every dual inequality in exact arithmetic. Residual rounding violations are repaired using

$$\sum_{\beta \in I_{r,n}} \binom{n}{\beta} K_\alpha(\beta) = 2^{rn} \mathbf{1}_{\{\alpha = n\epsilon_0\}},$$

which supplies an exactly feasible repair direction with a rational cost. Finite- $n$  bounds therefore rest on exact dual feasibility and weak duality alone, not on the numerical LP tolerance, and are upper bounds on the level- $r$  optimum (Supplementary Methods 7.2).

### Reduction to an asymptotic configuration search

We next considered the spectral dual construction of Coregliano et al., in which the complementary spectral function is supported on a single configuration orbit [8]. A normalized configuration is a probability distribution  $G$  on  $\mathbb{F}_2^\ell$ . Because a level- $\ell$  dual bounds  $|C|^\ell$ , the asymptotically relevant rate objective is

$$J_\ell(G) = \frac{H(G)}{\ell},$$

with  $H(\cdot)$  the Shannon entropy in bits; this normalization, and two discrepancies in the cited preprint that it exposes, are recorded in Supplementary Methods 7.4. For a non-zero shift  $v$  we use the Bhattacharyya translation affinity

$$\Phi(G, v) = \sum_u \sqrt{G(u)G(u+v)}.$$

Writing  $G = p^2$  with  $p \geq 0$  and  $\|p\|_2 = 1$  gives  $\Phi(G, v) = p^\top S_v p$ , where  $S_v$  is the permutation induced by translation by  $v$ , so discovery reduces to entropy minimization on the positive unit sphere under

quadratic affinity constraints. The analytic and symmetry reductions leave  $2^\ell - 1$  free probability coordinates, at most 15 for  $\ell \leq 4$ , with no remaining dependence on  $n$ . We had intended a neural parameterization of  $G$ , but at this point a generic network was unnecessary, so we searched the reduced space directly by multistart sequential quadratic programming: 28 instance solves over  $\ell \leq 4$  and  $\varepsilon \in \{0.30, 0.20, 0.10, 0.05\}$ , 13,600 local optimizations in total. No run produced a configuration below the first MRRW benchmark, and the recovered optima matched the quasirandom tensor-product family to within  $10^{-11}$  coordinatewise (Supplementary Methods 7.5).

## The tensorization obstruction

The repeated null result is explained by an inequality that holds for every admissible configuration. Let  $\varepsilon \in (0, 1)$ , let  $V$  be the set of shifts at which the spectral walk condition of [8] is verified, and let

$$h(\varepsilon) = H_2\left(\frac{1}{2}(1 - \sqrt{1 - \varepsilon^2})\right) = H_2\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right), \quad \delta = \frac{1 - \varepsilon}{2},$$

where  $H_2$  is the binary entropy function; the second form is the first MRRW expression. Two properties of the construction enter: the admissible shifts span  $\mathbb{F}_2^\ell$ , and at each such shift a finite even spectral-walk order  $m$  satisfies a lower bound of the form  $W_m(G, v) \geq c\varepsilon^m$  with  $c \geq 1$ , where  $W_m$  is the paired walk weight appearing in the configuration asymptotics of [8]. Comparing  $W_m$  termwise with the multinomial expansion of  $\Phi(G, v)^m$  gives  $W_m(G, v) \leq \Phi(G, v)^m$ , so the walk condition forces the affinity floor  $\Phi(G, v) \geq \varepsilon$  at a fixed  $m$ , with no exchange of the  $n \rightarrow \infty$  and  $m \rightarrow \infty$  limits. Conditioning each coordinate on the others turns that floor into an entropy floor: the conditional entropy of  $X_i$  given  $X_{-i}$  equals  $\mathbb{E}[h(2\sqrt{p_Y(1 - p_Y)})]$  while  $\Phi(G, e_i)$  equals  $\mathbb{E}[2\sqrt{p_Y(1 - p_Y)}]$  for the same conditional marginals, so convexity of  $h$  and Jensen's inequality give  $H(X_i | X_{-i}) \geq h(\Phi(G, e_i))$ . The entropy chain rule then yields

$$H(G) \geq \sum_{i=1}^{\ell} h(\Phi(G, e_i)) \geq \ell h(\varepsilon), \quad \text{hence} \quad J_\ell(G) \geq h(\varepsilon).$$

The tensor-product configuration  $G_\varepsilon = \text{Bernoulli}(p_\varepsilon)^{\otimes \ell}$  attains equality, identifying the object recovered computationally as the extremizer rather than as an artefact of the search. Within the fixed single-orbit spectral ansatz, optimizing the configuration  $G$  therefore cannot improve MRRW. Full hypotheses, proofs and scope are given in Supplementary Methods 7.6 (Theorem 5 and Corollary 6); the algebraic reductions underlying the theorem were verified against independent brute-force implementations (Supplementary Methods 7.7).

The obstruction constrains only the configuration. It does not constrain alternative sign functions  $\phi$ , spectral functions supported on several configuration orbits, other uses of the partial-Fourier dual families, or general dual feasible solutions, and the higher-order hierarchy itself remains complete. The computational search therefore localized the missing asymptotic freedom in those components rather than in  $G$ .

## 3 Sign-uncertainty discovery and certification

### Fourier-eigenfunction formulation

We use

$$\widehat{f}(\xi) = \int_{\mathbb{R}^d} f(x) e^{-2\pi i \langle x, \xi \rangle} dx$$

and consider the +1 sign-uncertainty constant  $A_+(d)$  over non-zero even integrable functions satisfying  $\widehat{f} = f$ ,  $f(0) = 0$ , and  $f(x) \geq 0$  for all sufficiently large  $|x|$  [18, 19]. To construct upper bounds we restrict the search to finite-dimensional radial Laguerre polynomial–Gaussian eigenspaces. With

$$u = \pi|x|^2, \quad \alpha = \frac{d}{2} - 1,$$

the functions

$$\psi_n(u) = L_n^{(\alpha)}(2u) e^{-u}$$

satisfy  $\widehat{\psi}_n = (-1)^n \psi_n$ . Hence the truncated  $+1$  eigenspace is

$$V_N = \text{span}\{\psi_0, \psi_2, \dots, \psi_{2N}\}.$$

Writing  $f = P(u)e^{-u}$  converts the Fourier constraint into the linear requirement that  $P$  lie in the corresponding even Laguerre span. For integer  $d$  these generalized Laguerre polynomials have rational coefficients, which enables exact reconstruction.

## From neural proposals to a convex search

The initial discovery stage searched a broader Fourier-eigenfunction family with trainable Gaussian scales and contact locations. Local and neural optimization rapidly improved the objective but became unreliable when the coefficient systems approached condition numbers of  $10^{17}$ – $10^{18}$ . Independent multiprecision reevaluation showed that apparently improving trajectories were not reproducible. We therefore treated learned widths and contacts only as proposals and used this failure to reconsider the search coordinates rather than to infer that the function class had been exhausted.

For the pure Laguerre family the rescaled function is simply the polynomial  $P(u) = e^u f(u)$ . Since  $e^{-u} > 0$ , fixing a candidate last-sign point  $u_0$  reduces admissibility to

$$P(0) = 0, \quad P(u) \geq 0 \quad (u \geq u_0).$$

The feasible set in the Laguerre coefficients is convex and monotone in  $u_0$ . Thus, the non-convex optimization over roots and contact locations can be replaced by a one-dimensional bisection in  $u_0$  with a convex semi-infinite feasibility problem at each step.

## Adaptive semi-infinite feasibility

At each bisection point we solve a margin-maximizing LP on an adaptive grid. A positive linear normalization,

$$\int_{u_0}^{\infty} P(u)e^{-u} du = 1,$$

removes the conic scale, and the leading Laguerre coefficient is constrained to have the sign required for positivity at infinity. The grid is only a relaxation. After each LP solve, roots of  $P'$  are located at high precision; any missed negative minimum is inserted as a cutting plane and the LP is resolved. Near the fixed-degree feasibility boundary, the active constraints identify candidate double contacts.

The exchange loop is used to locate the boundary and its contact structure, not to certify the quoted upper bound. The underlying space is not a Haar system, so uniqueness of the active contact set is not assumed. Instead, the convex semi-infinite formulation supplies a numerical bracket for sharpness within the chosen finite-dimensional space, while exact admissibility is established independently downstream. Failure of the numerical oracle can therefore reduce sharpness but cannot make an invalid certificate valid.

## Exact rational reconstruction and verification

Let  $\tau_1 < \dots < \tau_m$  denote the active tangencies returned by the convex search, rounded to nearby rational values. We reconstruct  $P$  exactly in the Laguerre basis by imposing

$$P(0) = 0, \quad P(\tau_i) = P'(\tau_i) = 0 \quad (1 \leq i \leq m),$$

together with a fixed positive top coefficient. In the square certificates,  $n_b = 2m + 2$  Laguerre coordinates are retained. Solving over  $\mathbb{Q}$  gives

$$P(u) = \prod_{i=1}^m (u - \tau_i)^2 R(u), \quad R \in \mathbb{Q}[u],$$

with every polynomial division checked to have zero remainder.

Positivity on the complete ray is then decided independently of the LP grid. We verify that the leading coefficient of  $R$  is positive, that  $R(\bar{u}) > 0$ , and that a Sturm chain has the same sign-variation count at  $\bar{u}$  and at  $+\infty$ . Hence  $R$  has no zero on  $[\bar{u}, \infty)$  and therefore

$$P(u) \geq 0 \quad (u \geq \bar{u}).$$

Since  $\hat{f} = f$  and  $f(0) = 0$  hold exactly by construction,  $f = P(u)e^{-u}$  is admissible and

$$A_+(d) \leq \sqrt{\bar{u}/\pi}.$$

For the  $d = 1$  degree-38 certificate,  $m = 9$  and

$$\bar{u} = \frac{514\,997\,856\,761}{5 \cdot 10^{11}},$$

which yields

$$A_+(1) \leq 0.572588699 \dots$$

The  $d = 2$  degree-22 certificate is constructed analogously and gives  $A_+(2) \leq 0.756206236 \dots$ . The final certificate contains only rational Laguerre data and integer/rational Sturm arithmetic; floating-point computation is confined to discovery of a useful contact structure.

## Scope

The finite-dimensional convex reformulation does not contradict the Cohn–Dong–Gonçalves asymptotic ceiling for sublinear-degree polynomial–Gaussian families [9]. It shows instead that, at small dimension, the classical family had not been numerically exhausted by the previous root-parameterized searches. Detailed tripwires, discovery experiments, convergence diagnostics, contact degeneracies and higher-dimensional numerical limits are given in the Supplementary Methods.

## 4 Use of large language models

A large language model (Claude, Anthropic; Opus 4.5, 4.8 and Fable5) was used throughout the exploratory development of the computational pipeline as an interactive programming and research-assistance tool. Its contributions were concentrated in software engineering (making the scripts into a package) and numerical debugging: proposing and revising implementation strategies, generating and refactoring code, and diagnosing numerical failure modes in the certification pipeline, including floating-point underflow (e.g. float-64 versus float-32 in GPU representations) and dynamic-range collapse, basis-conditioning and rank-instability effects, quadrature overflow, and (formulation) inconsistencies between the discovery and certification stages (see Supplementary Tables 2 and 3). The model also served as a discussant for (formal) mathematical framing.

The mathematical content of this work is the author’s. The variational formulation, the reduction underlying the pipeline, the certification scheme, and all theorem statements were conceived, derived, and verified by the author; the model’s role with respect to the mathematics was limited to exposition, sanity-checking, and the surfacing of relevant literature, all of which the author confirmed. Where the model proposed an algorithmic optimisation, the proposal was accepted only after the author verified it. For reference; PyTorch and Flint are not installed on Anthropic servers so no computations were done by a LLM.

All model outputs were critically evaluated by the author, who bears sole responsibility for the correctness of every claim. All reported numerical bounds are the output of deterministic software. Crucially, the language model is not invoked by the final certification pipeline: the validity of certified bound rests entirely on the exact-arithmetic certificate and its independent verification, which can be checked without reference to any part of the development process. A structured contribution-provenance record, and the complete certification and verification software are provided in the Supplementary Methods.

## 5 Acknowledgments

The authors would like to thank the Collaborative Cyber Lab of the Netherlands Defence Academy for computational resources.

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## 6 Supplementary Methods: Maynard–Tao sieve

### 6.1 Variational definitions and normalization

Let

$$\mathcal{R}_k = \left\{ \mathbf{t} \in \mathbb{R}_{\geq 0}^k : \sum_{i=1}^k t_i \leq 1 \right\}.$$

For an admissible square-integrable trial function  $F$  supported on  $\mathcal{R}_k$ , define

$$I_k(F) = \int_{\mathcal{R}_k} F(\mathbf{t})^2 d\mathbf{t}$$

and

$$J_k^{(m)}(F) = \int_{\mathcal{R}_{k-1}} \left( \int_0^{1-\sum_{i \neq m} t_i} F(\mathbf{t}) dt_m \right)^2 d\mathbf{t}_{-m}.$$

The Maynard variational constant is

$$M_k = \sup_{F \neq 0} \frac{\sum_{m=1}^k J_k^{(m)}(F)}{I_k(F)}.$$

For symmetric  $F$ , all marginal functionals are equal and

$$M_k = k \sup_{F \neq 0} \frac{J_k^{(1)}(F)}{I_k(F)}.$$

Throughout the computational pipeline, the reported Rayleigh quotient is normalized according to this convention. We also use the classical Cauchy–Schwarz [1] ceiling

$$M_k \leq \frac{k}{k-1} \log k,$$

as a validation constraint on both single-channel and full numerical quotients. The enlarged-support calculations use the corresponding Polymath  $\varepsilon$ -variant, with the same discovery–certification separation. Because the exact normalization of the enlarged polytope and the external conversion from a certified variational threshold to a numerical  $H_m$  bound involve the sieve theorem and an admissible  $k$ -tuple rather than the NeuralCert certificate itself, those ingredients are kept distinct from the variational certificate. In particular, the certificate proves the stated lower bound for the explicit trial function; the final  $H_m$  conversion additionally uses the external Maynard–Tao/Polymath theorem and a separately supplied admissible-tuple diameter.

### 6.2 From exact replication to neural discovery

The Maynard experiments began from an exact finite-dimensional control rather than from a neural model. Following Lemmas 8.1 and 8.2 of [2], we used the symmetric polynomial family

$$F(\mathbf{t}) = \sum_{\ell=1}^d a_\ell (1 - P_1(\mathbf{t}))^{b_\ell} P_2(\mathbf{t})^{c_\ell}, \quad P_j(\mathbf{t}) = \sum_{i=1}^k t_i^j,$$

for which both  $I_k$  and  $\sum_m J_k^{(m)}$  reduce to rational quadratic forms in the coefficient vector  $a$ . The finite-dimensional optimization is therefore a generalized Rayleigh–Ritz problem. This approach is consistent with Maynard’s observation that an extremizer should satisfy an eigenfunction equation for the underlying integral operator. The neural stage can therefore be interpreted as a nonlinear numerical search for a dominant variational eigenfunction, rather than as unconstrained function fitting. As a first control we reproduced Maynard’s explicit  $k = 5$  trial

$$F(\mathbf{t}) = (1 - P_1)P_2 + \frac{7}{10}(1 - P_1)^2 + \frac{1}{14}P_2 - \frac{3}{14}(1 - P_1),$$

for which

$$Q_5(F) = \frac{1417255}{708216} > 2.$$

We subsequently reproduced the degree-constrained generalized eigenvalue calculation at  $k = 105$ . Exact rational assembly was retained throughout; the floating-point generalized eigensolve was used only to locate a Ritz vector, which was then rationalized and reevaluated in the exact quadratic forms.

The exact polynomial controls were next used to calibrate Monte Carlo estimators before any black-box optimization. Uniform simplex samples were generated from normalized independent exponential variables. For the marginal functional, two conditionally independent samples  $U, U'$  were drawn on the same fibre. This gives the unbiased product estimator

$$\mathbb{E}[F(U, \mathbf{W})F(U', \mathbf{W}) \mid \mathbf{W}] = \left( \frac{1}{C} \int_0^C F(u, \mathbf{W}) du \right)^2,$$

where  $C = 1 - \sum W_i$ . Squaring a single inner sample would add its conditional variance and bias the estimate upward. Agreement of the Monte Carlo estimates with the exact polynomial values, expressed through standardized residuals, was used as the first stochastic validation gate.

Only after these controls were in place was the fixed polynomial ansatz replaced by a trainable symmetric function. Early neural models used scaled power-sum coordinates

$$\phi_j(\mathbf{t}) = k^{j-1} \sum_i t_i^j$$

and optimized a stochastic Rayleigh quotient. These experiments established that the optimizer could recover useful symmetric structure without being given the final polynomial representation, but also exposed the variance, conditioning and representation-transfer problems that motivated the deterministic separable pipeline used in the main analysis.

Finally, further control showed that enlargement of the admissible domain does not by itself improve a fixed trial function. For example, Maynard's Eq. (8.16), optimized for the simplex geometry, decreases from approximately 2.00 to 1.66 when evaluated at  $\varepsilon = 1/4$ . The gain from enlarged support therefore requires re-optimization to the modified geometry rather than simple reuse of a simplex-optimized extremizer. This provided an additional motivation for learning the trial function directly in each  $\varepsilon$ -geometry.

Recent computational work has also explored heuristic modifications of Maynard-type multidimensional sieves, including dynamically enlarged support regions and random-matrix-inspired perturbations [3], but their predicted prime-gap improvements remain heuristic. In contrast, the present framework separates exploratory optimization from certification of an explicit variational trial function and invokes only independently stated sieve-theoretic inputs for the final number-theoretic consequence.

## 6.3 Exploratory representation experiments

Before arriving at the final separable channel representation, five candidate compression mechanisms, most of which were ultimately rejected. These experiments were diagnostic rather than proof components: their role was to identify which representations preserved the variational structure well enough to justify further development. All bounds in the paper are certified independently of the outcomes described below.

### 6.3.1 Experiment 1: partitions

Exact Rayleigh–Ritz optimization in the full monomial-symmetric basis is effective at small  $k$ , but encounters a symmetric curse of dimensionality: although the partition basis is  $L^2$ -dense, the number of basis elements up to degree  $D$  grows as the restricted partition count  $p_k(D)$  and becomes rapidly prohibitive. This motivated attempts to retain only a small, variationally relevant subset of partitions.

Three selectors were tested: exact one-vector residual greedy expansion, a global floating-point Ritz oracle, and neural coefficient extraction through a power-sum-product representation. The residual score

$$\eta_\lambda = \frac{|\langle m_\lambda, (L - \lambda_S)F_S \rangle|}{\|m_\lambda\|_2}$$

improved the recorded exploratory bound by only about  $2.5 \times 10^{-3}$  after adding approximately 30 highest-scoring partitions, much less than adding a complete degree shell. The global floating-point and neural coefficient selectors were additionally destabilized by the ill-conditioned change of coordinates into the monomial-symmetric basis. The experiment therefore rejected sparse single-partition selection as an effective acceleration mechanism: the missing variational gain was collective and strongly correlated rather than localized in a few coordinates.

This failure suggested that the relevant structure was not sparse in the partition basis but instead closer to a low-complexity product or tensor representation, motivating the tensor-decomposition experiments below.

### 6.3.2 Experiment 2: symmetric tensor decompositions

The failure of sparse partition selection suggested that the optimizer might be compact in a tensor representation. We therefore fitted the symmetric CP family

$$F_{\text{CP}}(\mathbf{t}) = \sum_{q=1}^r c_q \prod_{i=1}^k \phi_q(t_i)$$

to the certified  $k = 5$ , degree-10 Ritz optimizer. The relative empirical  $L^2$  reconstruction errors were

| $r$             | 1      | 2      | 3      | 4      | 5      | 6      |
|-----------------|--------|--------|--------|--------|--------|--------|
| $\mathcal{E}_r$ | 0.1887 | 0.1536 | 0.1032 | 0.1028 | 0.1076 | 0.1089 |

and plateaued near 10% beyond rank three. The corresponding Monte Carlo Rayleigh estimates were statistically compatible with the exact target but noisy and non-monotone; values above the target were within the quoted Monte Carlo uncertainty and were never treated as improved lower bounds. Thus, increasing the symmetric CP rank beyond  $r = 3$  produced essentially no further reduction in reconstruction error over the tested range. This rejected the simplest low-rank CP compression as a complete representation, but not the broader tensor-network hypothesis.

### 6.3.3 Experiment 3: structural signatures of the CP residual

For the rank-six symmetric CP fit, the relative empirical reconstruction error was

$$\frac{\|F^\star - F_{\text{CP},6}\|_2}{\|F^\star\|_2} = 0.10903.$$

We projected the residual onto the full degree-10 monomial-symmetric basis. The largest fitted coordinate signatures occurred in the one-part directions

$$(4), (3), (2), (5), (1)$$

and the two-part directions

$$(2, 1), (3, 1), (1, 1).$$

Since one-part monomial-symmetric functions are power sums, this pointed to low-order power-sum structure in the component not captured by the CP fit.

As a diagnostic, we formed the diagonal coordinate proxy

$$\zeta_\lambda = b_\lambda^2 \|m_\lambda\|_{2,\text{emp}}^2.$$

Approximately 83% of this proxy was associated with one-body partitions and 17% with two-body partitions, with negligible proxy weight at higher body counts. This must not be interpreted as an orthogonal  $L^2$ -energy decomposition:

$$\sum_{\lambda} \zeta_{\lambda} = 6.71 \times 10^6,$$

whereas the empirical squared residual norm was only 1.43, revealing strong cancellation in the non-orthogonal partition basis.

The diagnostic nevertheless suggested that the residual was concentrated in structurally simple low-order directions rather than in a diffuse high-body tail. Since such components are in principle representable within a sufficiently flexible CP family, the observed rank saturation pointed more naturally to an optimization or conditioning limitation than to an immediate expressivity obstruction. One plausible explanation, explored in subsequent experiments, was that these directions were difficult to resolve by gradient-based fitting against Monte Carlo noise.

#### 6.3.4 Experiment 4: explicit power-sum corrections

The residual signature motivated the augmented model

$$F_{\text{CP+P}} = F_{\text{CP}} + \beta_0 + \sum_{j=1}^J \beta_j \tilde{P}_j, \quad \tilde{P}_j = P_j / \mathbb{E}[P_j].$$

This experiment did not yield a reliable acceleration. The revised implementation failed first to reproduce its own CP-only baseline, the Monte Carlo estimator for  $J$  became high variance for some augmented functions, and the explicit power-sum branch overlapped strongly with directions already represented by the CP factors. Consequently, the small observed changes with increasing  $J$  could not be separated cleanly from optimization and implementation signal.

#### 6.3.5 Experiment 5: body-restricted partition spaces

The final pre-separable experiment restricted the exact polynomial trial space simultaneously by total degree and partition length, or “body count”. For

$$\mathcal{V}(\{D_b\}) = \text{span}\{m_\lambda : \ell(\lambda) = b, |\lambda| \leq D_b\},$$

matrix entries were assembled by enumerating coalescences of the positive parts of two partitions rather than their full permutation orbits. The restriction is therefore specified by a degree cap  $D_b$  for each body count  $b$ ; at  $k = 50$  no run included  $b > 5$ , so every space reported below is a genuine body-restricted subspace of the full monomial-symmetric space of the same total degree. At  $k = 5$ , where  $\ell(\lambda) \leq k = 5$  holds automatically, the unrestricted degree-10 space contained 113 partitions and reproduced

$$M_5 \geq 2.0071443298,$$

approaching Bogaert’s Krylov value 2.0071451444. Restricting to partition length at most two reduced the space to 36 basis elements and gave a strictly weaker value, confirming the coalescence construction while showing that low-body truncation alone does not recover the full small- $k$  optimizer.

At  $k = 50$ , exact matrix assembly remained feasible, but the generalized eigensolve became the dominant limitation. Rank-revealing whitening retained only a fraction of the nominal Gram modes; for a common degree cap of 15, 18, 20 and 22 on every body count the reductions were

$$408 \rightarrow 191, \quad 769 \rightarrow 278, \quad 1125 \rightarrow 344, \quad 1601 \rightarrow 421.$$

The retained block consistently approached the imposed conditioning threshold,  $\kappa \sim 10^{12}$ , indicating that the effective ceiling was set by float64 numerical resolution rather than by the nominal size of the partition space. This effect is visible in the non-monotone certified ladder obtained from these increasing nominal spaces:

$$3.36957, \quad 3.36618, \quad 3.36800, \quad 3.36435.$$

This does not contradict Rayleigh–Ritz monotonicity: the nominal trial spaces are nested, whereas the independently whitened effective subspaces need not be. Exact reevaluation still certifies the returned trial vector, but cannot restore directions discarded before the eigensolve.

The precision dependence is particularly clear in a matched 408-dimensional calculation at degree cap 15. Retaining the complete pencil at high precision gave the certified value

$$M_{50} \geq 3.6347,$$

whereas float64 spectral whitening retained only 191 modes and yielded 3.3696. Clearly, the lost variational gain was already present in the polynomial trial space but fell below the numerical resolution of the whitened Gram pencil. The loss is strongly space-dependent rather than systematic: on the 441-dimensional one-two-body space the high-precision solve gave 3.1862092 against 3.1862074 from 134 whitened modes, so there the discarded directions carried almost no variational weight. A larger space admitting three-body partitions up to degree 20 similarly reached 3.3904 with all 678 modes retained at high precision, while larger float64-whitened spaces did not systematically improve the bound. Supplementary Table 1 summarizes representative calculations.

Supplementary Table 1: Representative  $k = 50$  partition-space calculations. The trial space is specified by a degree cap per body count  $b$  (partition length); no run included  $b > 5$ . Certified values refer to the returned explicit trial function. High-precision solves retain the full trial space, whereas float64 whitening removes Gram directions below the numerical resolution threshold.

| Trial space (degree cap by body count) | Eigensolver       | Dimension | Retained | Certified |
|----------------------------------------|-------------------|-----------|----------|-----------|
| $b \leq 5$ , all degrees $\leq 15$     | high precision    | 408       | 408      | 3.6347    |
| $b \leq 5$ , all degrees $\leq 15$     | float64 whitening | 408       | 191      | 3.3696    |
| $b \leq 2$ , degrees $\leq 40$         | float64 whitening | 441       | 134      | 3.1862    |
| $b \leq 3$ , degrees (40, 40, 20)      | high precision    | 678       | 678      | 3.3904    |
| $b \leq 5$ , all degrees $\leq 22$     | float64 whitening | 1601      | 421      | 3.3644    |

Note. Bogaert reported  $M_{50} \geq 3.9358660346$ .

The experiment therefore identified two distinct obstacles in the monomial-symmetric representation: rapid growth of the partition basis and loss of numerically resolvable Gram directions. The limiting issue at  $k = 50$  was not expressive power of the polynomial space itself, but the combination of basis growth and near-degeneracy. This motivated the move to representations whose complexity is controlled before the generalized eigensolve and whose structure is more directly compatible with exact certification.

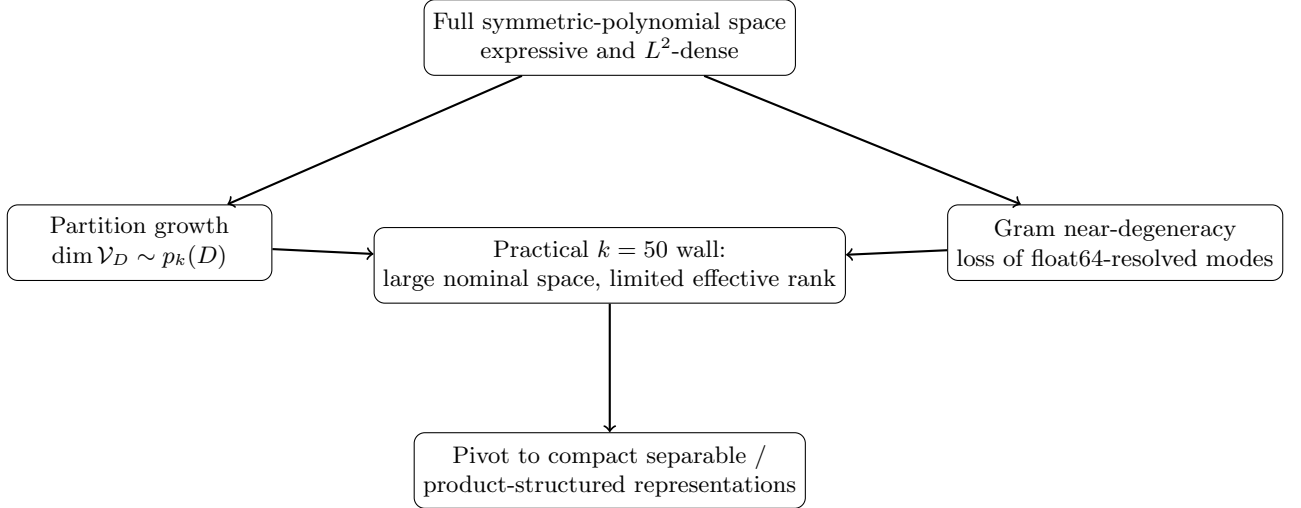


Figure 3: Scaling obstruction of the monomial-symmetric representation. Increasing polynomial degree expands an expressive symmetric trial space, but at  $k = 50$  this produces both rapid partition growth and increasingly ill-conditioned Gram pencils. The resulting loss of numerically resolved directions motivates the later separable representation.

### 6.3.6 Consequence for the final representation

The abovementioned five experiments ruled out several superficially natural routes: sparse partition support, very-low-rank symmetric CP compression, unqualified interpretation of residual coefficients, additive power-sum patches, and brute-force growth of ill-conditioned polynomial spaces. The common

Listing 1: Positive neural channel dictionary. A shared network produces the residual factors, while the exponential envelopes are represented analytically.

```

PROCEDURE CHANNELS(x, k, theta, eta, rho_max)
  rho      <- MIN(SOFTPLUS(eta), rho_max)
  features <- STACK(x, EXP(-k*x))
  log_raw <- SHARED_MLP_theta(features)
  g        <- EXP(log_raw - OUTER(x, rho))
  RETURN g, log_raw, rho
END

FUNCTION TRIAL(t, c, theta)
  RETURN SUM_j c[j] * PRODUCT_i CHANNEL_j(t[i], theta)
END

```

failure mode was not lack of a strong candidate function but loss of structure during representation or optimization. This motivated the separable-channel formulation used in the main Methods, in which a small dictionary of one-dimensional functions is optimized directly, the generalized Ritz coefficients are profiled out, and independent certification operates only after the discovered function has been reduced to an explicit mathematical representation.

Scripts for the exploratory analyses described above are not included in NeuralCert. Together with additional exploratory outputs including full residual rankings, body-count and parity summaries, and repeated fit diagnostics, they are available upon request.

## 6.4 Neural channel model

The discovery stage uses the symmetric separable representation

$$F_{\theta,c}(t_1, \dots, t_k) = \sum_{j=1}^m c_j \prod_{i=1}^k g_{\theta,j}(t_i).$$

The one-dimensional channels are written

$$g_{\theta,j}(x) = r_{\theta,j}(x)e^{-\rho_j x}, \quad \rho_j > 0,$$

with trainable rates initialized across a broad geometric range. The residual factors are produced jointly by a shared multilayer perceptron with channel-specific outputs and input

$$(x, e^{-kx}).$$

For the positive-channel model,

$$r_{\theta,j}(x) = \exp(f_{\theta,j}(x, e^{-kx})), \quad g_{\theta,j}(x) = \exp(f_{\theta,j}(x, e^{-kx}) - \rho_j x).$$

Positivity permits cancellation-free log-domain convolution. The architecture differs from a conventional DeepSets model: permutation invariance follows from the product structure of the separable expansion, while the neural network supplies a dictionary of one-dimensional factors.

## 6.5 Factored convolution powers and Gauss–Jacobi recursion

For a channel pair  $(j, \ell)$ ,

$$h_{j\ell}(x) = r_{j\ell}(x)e^{-a_{j\ell}x}, \quad r_{j\ell}(x) = r_{\theta,j}(x)r_{\theta,\ell}(x), \quad a_{j\ell} = \rho_j + \rho_\ell.$$

Every convolution power is represented as

$$\nu_p^{(j\ell)}(s) = s^{p-1} e^{-a_{j\ell}s} e^{\sigma_p^{(j\ell)}} \psi_p^{(j\ell)}(s).$$

Substituting this representation into

$$\nu_{p+q}(s) = \int_0^s \nu_p(x) \nu_q(s-x) dx$$

Listing 2: Binary-chain evaluation of the factored convolution power in logarithmic coordinates.

```

PROCEDURE LOG_CONVOLUTION_POWER(log_raw, j, l, k)
  xi[l](s) <- log_raw_j(s) + log_raw_l(s)

  FOR EACH (p,q) IN BINARY_ADDITION_CHAIN(k-1)
    (u,w) <- NORMALIZED_GAUSS_JACOBI(p,q)
    FOR EACH representation node s
      a <- LOG_POSITIVE_INTERPOLATE(xi[p], s*u)
      IF q = 1
        b <- EVALUATE_BASE_PAIR_EXACTLY(s*(1-u), j, l)
      ELSE
        b <- LOG_POSITIVE_INTERPOLATE(xi[q], s*(1-u))
      END
      xi[p+q](s) <- LOG_BETA(p,q)
        + LOGSUMEXP_r(LOG(w[r])+a[r]+b[r])
    END
  END
  RETURN xi[k-1]
END

```

and setting  $x = su$  gives

$$\psi_{p+q}(s) = B(p, q) \int_0^1 \psi_p(su) \psi_q(s(1-u)) \frac{u^{p-1}(1-u)^{q-1}}{B(p, q)} du,$$

with

$$\sigma_{p+q} = \sigma_p + \sigma_q + \log B(p, q),$$

up to an arbitrary numerical normalization of  $\psi_{p+q}$  absorbed into  $\sigma_{p+q}$ . The integral is evaluated by Gauss–Jacobi quadrature,

$$\psi_{p+q}(s) \approx B(p, q) \sum_{r=1}^{n_q} \hat{w}_r^{(p,q)} \psi_p(su_r^{(p,q)}) \psi_q(s(1-u_r^{(p,q)})),$$

where  $\sum_r \hat{w}_r^{(p,q)} = 1$ . The quadrature nodes follow the beta-weight concentration automatically. The power  $h^{*(k-1)}$  is assembled by a binary addition chain, reducing convolution depth from  $k-2$  sequential stages to  $O(\log k)$ .

## 6.6 Log-domain recursion

For positive channels define  $\xi_p(s) = \log \psi_p(s)$ . The recursion becomes

$$\xi_{p+q}(s) = \log B(p, q) + \text{logsumexp}_r \left[ \log \hat{w}_r^{(p,q)} + \xi_p(su_r^{(p,q)}) + \xi_q(s(1-u_r^{(p,q)})) \right].$$

Values that would underflow in linear arithmetic remain representable as large negative logarithms, and the positive quadrature sum is evaluated without cancellation. This is important because loss of a small component during an early convolution can propagate into entire entries of the subsequent Gram matrices.

## 6.7 Representation and integration grids

Residual functions are stored on an endpoint-inclusive Chebyshev–Lobatto grid

$$0 = s_0 < s_1 < \dots < s_{N-1} = L, \quad L = 1 + \varepsilon.$$

Every convolution target  $s_i u_r$  or  $s_i(1-u_r)$  lies in  $[0, s_i]$ , so no extrapolation is required. Four numerical point sets are kept distinct: the Chebyshev–Lobatto representation grid; Gauss–Jacobi rules for the beta-weighted convolution; Gauss–Legendre rules for the primitives

$$G_j(y) = \int_0^y g_j(x) dx, \quad H_{j\ell}(y) = \int_0^y g_j(x) g_\ell(x) dx;$$

Listing 3: Construction of the four numerical point sets used by the discovery evaluator.

```

PROCEDURE BUILD_GRIDS(k, epsilon, N, n_jacobi, n_inner, n_outer)
  L <- 1 + epsilon; U <- 1 - epsilon
  s[i] <- L*(1-COS(pi*i/(N-1)))/2, i=0,...,N-1

  FOR EACH (p,q) IN BINARY_ADDITION_CHAIN(k-1)
    jacobi[p,q] <- NORMALIZED_GAUSS_JACOBI(n_jacobi,p,q)
  END
  inner <- GAUSS_LEGENDRE(n_inner, interval=[0,1])

  x_star <- MIN(X,(k-2)/a_ref)
  edges <- GEOMETRIC_MESH(x_star/32,X)
  outer <- COMPOSITE_GAUSS_LEGENDRE(edges,n_outer)
  // The same positive-weight outer rule is used for every channel pair.
  RETURN s, jacobi, inner, outer
END

```

and a geometrically graded outer mesh for the final entries of  $A$  and  $B$ . The latter must resolve the pair-dependent factor  $x^{k-2}e^{-a_{j\ell}x}$ .

## 6.8 Preservation of Gram structure

The exact denominator matrix is a Gram matrix and therefore satisfies  $A \succeq 0$ . To preserve this structure numerically, a target  $z \in [s_i, s_{i+1}]$  is interpolated using the positive stencil

$$\psi(z) \approx \omega_i(z)\psi(s_i) + \omega_{i+1}(z)\psi(s_{i+1}), \quad \omega_i, \omega_{i+1} \geq 0, \quad \omega_i + \omega_{i+1} = 1.$$

In log-space, this interpolation is evaluated by a two-term log-sum-exp. Interpolating  $\psi$  itself, rather than  $\log \psi$ , is what makes the stencil multilinear and hence compatible with the Gram form. When the inner quadrature nodes are common to all channel pairs, the shared positive-weight outer rule yields the discrete Gram representation

$$A_{j\ell}^{(N)} = \sum_{\alpha} w_{\alpha} \Phi_{\alpha j} \Phi_{\alpha \ell}, \quad w_{\alpha} > 0,$$

and hence  $A^{(N)} \succeq 0$  up to floating-point round-off. The shared mesh therefore preserves a common discrete Gram representation rather than merely improving entrywise quadrature accuracy.

In the saturated regime, where the substitution used for the channel primitives places the inner nodes at pair-dependent positions, the common discrete Gram representation is preserved only up to inner-quadrature error. Numerical adequacy is then assessed using the positive-semidefiniteness, two-grid and refinement diagnostics described below; these diagnostics are safeguards rather than rigorous bounds on the quadrature error. Since the reported bounds come from the exact certifier and not from the discrete pencil, a failure of this structure degrades the discovery objective rather than the validity of any certificate.

## 6.9 Diagonal equilibration and rank-revealing Ritz solution

With

$$D = \text{diag}(A_{11}, \dots, A_{mm}),$$

the generalized Rayleigh quotient is invariant under

$$A \mapsto D^{-1/2} A D^{-1/2}, \quad B \mapsto D^{-1/2} B D^{-1/2}.$$

After equilibration, write

$$\tilde{A} = U \text{diag}(\alpha_1, \dots, \alpha_m) U^{\top}.$$

Directions satisfying

$$\alpha_i \leq \tau_{\text{rank}} \max_j \alpha_j$$

Listing 4: Log-domain assembly of one pair of Gram entries. With common inner quadrature nodes, the shared positive-weight outer rule gives an exact discrete Gram representation; in the saturated regime, pair-dependent inner quadrature preserves this structure only up to inner-quadrature error.

```

FUNCTION LOG_INTERPOLATE(z, xi)
  (i,i+1,omega0,omega1) <- POSITIVE_TWO_POINT_STENCIL(z)
  RETURN LOGADDEXP(LOG(omega0)+xi[i], LOG(omega1)+xi[i+1])
END

PROCEDURE GRAM_PAIR(j, l)
  a <- rho[j] + rho[l]
  xi <- LOG_CONVOLUTION_POWER(log_raw,j,l,k)
  log_Ghat, log_Hhat <- GAUSS_LEGENDRE_PRIMITIVES(j,l)
  // Saturated inner rules may depend on the channel pair (j,l).

  log_A[j,l] <- LOGSUMEXP_x((k-2)*LOG(x)-a*x
    + LOG_INTERPOLATE(x,xi)
    + LOG(L-x)+LOG_INTERPOLATE(L-x,log_Hhat)+LOG(w_x))
  log_B[j,l] <- LOGSUMEXP_x((k-2)*LOG(x)-a*x
    + LOG_INTERPOLATE(x,xi)+2*LOG(L-x)
    + LOG_INTERPOLATE(L-x,log_Ghat_j)
    + LOG_INTERPOLATE(L-x,log_Ghat_l)+LOG(w_x))
  RETURN log_A[j,l], log_B[j,l]
END

```

Listing 5: Diagonally equilibrated, rank-revealing generalized Ritz solve.

```

PROCEDURE TOP_RITZ(log_A, log_B, tau_rank)
  d[j] <- log_A[j,j]
  Ahat[j,l] <- EXP(log_A[j,l]-(d[j]+d[l])/2)
  Bhat[j,l] <- EXP(log_B[j,l]-(d[j]+d[l])/2)

  (alpha,U) <- SYMMETRIC_EIGENDECOMPOSITION(Ahat)
  keep <- {i : alpha[i] > tau_rank*MAX(alpha)}
  REQUIRE keep is nonempty
  W <- U[:,keep] * DIAG(alpha[keep]^(-1/2))

  (lambda,z) <- LEADING_EIGENPAIR(TRANPOSE(W)*Bhat*W)
  v <- W*z
  v <- v/SQRT(TRANPOSE(v)*Ahat*v)
  RETURN k*lambda, v, CARDINALITY(keep), d
END

```

are removed before whitening. On the retained space,

$$W = U_{\text{keep}} \text{diag}(\alpha_i^{-1/2}),$$

and the leading eigenpair of  $W^\top \tilde{B} W$  determines the profiled quotient and mixing vector. No ridge term is introduced; rank truncation is treated as a numerical-resolution decision and monitored by rank-stability tests.

## 6.10 Hellmann–Feynman gradients and optimization

Numerical solution of the generalized Ritz problem may require removal of directions associated with numerically negligible eigenvalues of  $A(\theta)$ . Let

$$\mathcal{S}(\theta)$$

denote the retained coefficient subspace after this rank-revealing truncation, and define the corresponding numerical Ritz value by

$$\hat{R}(\theta) = \max_{\substack{v \in \mathcal{S}(\theta) \\ v \neq 0}} k \frac{v^\top B(\theta) v}{v^\top A(\theta) v}.$$

Because  $\mathcal{S}(\theta)$  may change with  $\theta$ , the map  $\widehat{R}(\theta)$  need not be differentiable when a singular direction crosses the truncation threshold. Moreover, the standard Hellmann–Feynman or envelope identity does not automatically apply across such a parameter-dependent change of retained subspace.

We therefore freeze the retained subspace during each differentiable optimization block. At outer iteration  $s$ , let

$$\mathcal{S}_s := \mathcal{S}(\theta_s)$$

and define the fixed-subspace profile

$$R_s(\theta) = \max_{\substack{v \in \mathcal{S}_s \\ v \neq 0}} q_v(\theta), \quad q_v(\theta) = k \frac{v^\top B(\theta) v}{v^\top A(\theta) v}.$$

Let  $v_s \in \mathcal{S}_s$  be the normalized leading generalized Ritz vector at the anchor point:

$$B(\theta_s)v_s = \lambda_s A(\theta_s)v_s \quad \text{on } \mathcal{S}_s, \quad v_s^\top A(\theta_s)v_s = 1.$$

Provided that  $A(\theta)$  remains positive definite on  $\mathcal{S}_s$  in a neighbourhood of  $\theta_s$  and that the leading restricted generalized eigenvalue is simple, the fixed-subspace envelope identity gives

$$\nabla_\theta R_s(\theta_s) = \nabla_\theta q_{v_s}(\theta_s).$$

Thus the Ritz vector may be frozen when differentiating the quotient, but only while the retained subspace is held fixed.

For every fixed  $v \in \mathcal{S}_s$ ,

$$q_v(\theta) \leq R_s(\theta),$$

and at the anchor point

$$q_{v_s}(\theta_s) = R_s(\theta_s).$$

Consequently, an inner step satisfying

$$q_{v_s}(\theta_{s+1}) \geq q_{v_s}(\theta_s)$$

obeys the fixed-subspace minorize–maximize relation

$$R_s(\theta_{s+1}) \geq q_{v_s}(\theta_{s+1}) \geq q_{v_s}(\theta_s) = R_s(\theta_s).$$

This monotonicity statement concerns  $R_s$ , with  $\mathcal{S}_s$  fixed. It does not by itself imply monotonicity of the adaptively truncated value  $\widehat{R}(\theta)$  after the numerical rank is recomputed.

Neural channels are first optimized with Adam using learning-rate warm-up, cosine decay, and gradient clipping. Candidate solutions are subsequently polished by block minorize–maximize L-BFGS. Within each such block, the retained subspace, truncation mask, and frozen Ritz vector are held fixed. The rank-revealing decomposition is recomputed only between blocks.

After this recomputation, a candidate outer step is accepted only if the adaptively evaluated Ritz value satisfies

$$\widehat{R}(\theta_{s+1}) \geq \widehat{R}(\theta_s) - \tau_{\text{acc}},$$

where  $\tau_{\text{acc}}$  is a prescribed numerical acceptance tolerance. If a change in retained rank causes this test to fail, the candidate is rejected and the step size or optimization block is restarted. This acceptance test is an explicit numerical safeguard; it is not presented as a consequence of the Hellmann–Feynman identity.

Finite-difference gradient tests are performed with the retained subspace and truncation mask frozen. A finite-difference stencil that changes the retained numerical rank crosses a nonsmooth truncation boundary and is therefore not used as a test of the Hellmann–Feynman gradient. Such cases are instead flagged as rank-transition events. Away from these events, the test checks the complete differentiation path through the neural channels, factored convolution recursion, integration, and matrix assembly.

Optimization proceeds through increasing channel ranks

$$m_1 < \cdots < m_S$$

and representation resolutions

$$N_1 < \cdots < N_S.$$

Existing channels are copied when the channel rank is increased, and new channels are initialized as perturbed duplicates. Because pairwise assembly scales as  $m(m+1)/2$ , more iterations are allocated to the smaller early models.

Listing 6: Profiled optimization with a frozen Ritz vector.

```

FUNCTION FROZEN_QUOTIENT(theta, v)
  (A_theta, B_theta) <- ASSEMBLE_GRAM(theta)
  RETURN k*(TRANPOSE(v)*B_theta*v)/(TRANPOSE(v)*A_theta*v)
END

PROCEDURE OPTIMIZE_CHANNELS(theta)
  FOR EACH increasing pair (channel rank m, grid size N)
    GROW_DICTIONARY_BY_PERTURBED_DUPLICATES(theta, m)
    FOR EACH Adam iteration, with warm-up and cosine decay
      IF this is a reporting point
        VALIDATE_CURRENT_STATE(theta)  // before the update
      END
      (_, v) <- TOP_RITZ(ASSEMBLE_GRAM(theta))
      theta <- CLIPPED_ADAM_ASCENT(
        FROZEN_QUOTIENT(theta, STOP_GRADIENT(v)))
    END
  END

  REPEAT for each block minorize-maximize step
    (_, v) <- TOP_RITZ(ASSEMBLE_GRAM(theta))
    theta <- LBFGS_MAXIMIZE(FROZEN_QUOTIENT(theta, STOP_GRADIENT(v)))
    VALIDATE_CURRENT_STATE(theta)
    UPDATE incumbent only if its two-grid score improves
  END
  RETURN theta
END

```

## 6.11 Validation-aware model selection

At each reporting point the candidate must satisfy:

1. agreement between the training objective and an independently recomputed Rayleigh quotient;
2. numerical positive semidefiniteness of the denominator matrix;
3. retention of a prescribed minimum fraction of the channel span after rank truncation;
4. stability of the leading quotient under a more conservative rank threshold;
5. compliance of every diagonal ratio  $kB_{jj}/A_{jj}$  with the applicable rigorous upper bound;
6. compliance of the full quotient with the corresponding rigorous upper bound; and
7. reproduction on a finer representation grid within the prescribed snapshot tolerance.

When two grids are used during training, candidates are ranked by

$$R_{\text{score}} = \min\{R_N, R_{N_{\text{hi}}}\}.$$

A substantial negative eigenvalue of  $A$  is treated as failure of the numerical representation rather than repaired by discarding the negative branch. Rank-stability tests separately guard against whitening directions below the accuracy of the assembled matrices.

## 6.12 Closed-form controls and representation refinement

Every run begins with closed-form controls. For  $g(x) = 1$  and  $\varepsilon = 0$ ,

$$R_{\text{const}} = \frac{2k}{k+1},$$

and for  $g(x) = x$ ,

$$R_x = \frac{3k}{3k+1}.$$

Listing 7: Validation and two-grid model selection applied at each reporting point.

```

FUNCTION RIGOROUS_CEILING(k, epsilon)
  IF epsilon = 0
    RETURN k*LOG(k)/(k-1)
  ELSE
    RETURN k*LOG(2*k-1)/(k-1)
  END
END

PROCEDURE VALIDATE_CURRENT_STATE(theta)
  R_bound <- RIGOROUS_CEILING(k,epsilon)

  (R_train,A,B,v,rank) <- INDEPENDENT_RECOMPUTATION(theta)
  REQUIRE AGREES(R_train,FROZEN_QUOTIENT(theta,v))
  REQUIRE MIN_EIGENVALUE(A) >= -tau_psd*MAX_EIGENVALUE(A)
  REQUIRE rank >= required_fraction*NUMBER_OF_CHANNELS(theta)
  REQUIRE STABLE_UNDER_STRONGER_RANK_TRUNCATION(A,B)
  REQUIRE EACH k*B[j,j]/A[j,j] <= R_bound
  REQUIRE R_train <= R_bound

  R_hi <- RECOMPUTE_ON_GRID(theta,1.5*N)
  REQUIRE RELATIVE_DIFFERENCE(R_train,R_hi) <= tau_snapshot
  score <- MIN(R_train,R_hi)
  STORE theta only if score improves the validated incumbent
  RETURN score
END

```

The first exercises convolution, outer integration and scaling; the second also probes interpolation of a nonconstant residual convolution shape.

After training, candidates are evaluated on grids  $N$ ,  $1.25N$  and  $1.5N$  and fitted to

$$R_N = R_\infty - cN^{-p}.$$

The fitted limit is diagnostic only. the export is marked unconverged when the sequence is noncontracting, violates a rigorous ceiling or fails the refinement tolerance. No extrapolated discovery value is treated as a certified lower bound.

The final separable discovery solver emerged through a sequence of numerical and structural corrections. Supplementary Table 2 summarizes the principal failure modes identified during development and the corresponding fixes. These historical iterations are not separate proof components; they document how the numerical invariants enforced in the final solver were established.

Supplementary Table 2 lists the different development steps for the discovery script.

Supplementary Table 2: Reconstructed development history of the `maynard_separable_rayleigh_v9_factored` lineage (neural discovery).  
Version numbers map to successive saved iterations.

| # | Correction                                                                                                                                                                                                                                                 | Cause / symptom                                                                                                                                                                                                                    |
|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Baseline v9 (factored): $\nu_p = s^{p-1}e^{-as}e^{\sigma}\psi_p$ per pair; stable <code>inv.softplus</code> ; per-pair max-rescale; adaptive log-space GL outer windows; diagonal preconditioning ( $\text{diag } A = 1$ ); <code>parts()</code> protocol. | Foundation. <code>inv.softplus</code> fixes $\text{softplus}^{-1} = \log(\text{expm1}) \rightarrow +\infty$ for rate $\geq 710$ (all $k \geq 355$ ); per-pair rescale replaces a global max that flushed small pairs to round-off. |
| 2 | $\sigma$ double-count fix: divide exact base re-evaluations by $e^{-\sigma_1}$ .                                                                                                                                                                           | Exact base evals carried $e^{\sigma_1}$ while $\sigma_1$ was also added in <code>new.sig</code> ; the spurious factor is not of Gram form $\Rightarrow$ indefinite $A$ .                                                           |
| 3 | <code>train()</code> validation-ordering fix: validate before <code>opt.step()</code> .                                                                                                                                                                    | <code>R_train</code> read pre-step, <code>report()</code> post-step; <code>agree</code> measured the Adam step, rejecting nearly all snapshots (fallback to iter 700).                                                             |
| 4 | Log-space channel protocol: <code>log_parts</code> ; <code>--channel-sign positive/free</code> .                                                                                                                                                           | Prerequisite for the log-space recursion; needs sign-definite channels.                                                                                                                                                            |

*continued on next page*

Supplementary Table 2 – continued

| #  | Correction                                                                                                                                                                             | Cause / symptom                                                                                                                                                                                                                            |
|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5  | Log-space $\xi$ recursion: $\xi_{p+q} = \log B(p, q) + \text{logsumexp}(\log \hat{w} + \xi_p + \xi_q)$ .                                                                               | The $(\min \psi_1 / \max \psi_1)^{k-1}$ collapse drove $\psi$ past the float64 floor; round-off put random signs on diagonal pairs ( $\psi \geq 0$ ), making $A$ indefinite.                                                               |
| 6  | Shared graded outer mesh (one geometric Gauss rule for all pairs).                                                                                                                     | Per-pair windows gave each entry its own rule $\Rightarrow A$ the Gram matrix of nothing; $\lambda_{\min}/\lambda_{\max} \approx -1$ and did not improve under refinement.                                                                 |
| 7  | Log-singularity fix: $G(y) = y \hat{G}(y)$ ; interpolate $\log \hat{G}$ , reattach $\log y$ analytically.                                                                              | $G(y)$ vanishes linearly $\Rightarrow \log G \rightarrow -\infty$ (clamped); interpolating that oscillated $\Rightarrow B \approx 10^{45}$ .                                                                                               |
| 8  | Gram-preserving positive interpolation: two-point stencil, weights $\geq 0$ , via <code>logaddexp</code> ; <code>--interp positive/spectral</code> .                                   | Interpolating the log gives $\prod \psi_i^{M_i}$ (geometric), not $\sum M_i \psi_i$ (linear) $\Rightarrow$ not multilinear $\Rightarrow$ not Gram $\Rightarrow$ indefinite $A$ .                                                           |
| 9  | Hard gates (PSD, $R \leq k$ , gradient) + removed false “PSD by construction” + parameter-independent mesh.                                                                            | $R = 118, 979, 5.2 \times 10^8$ were optimised toward; positive-entrywise $\neq$ PSD; moving mesh gave a spurious HF-gradient gap.                                                                                                         |
| 10 | Resolution diagnostics ( $\xi$ span/jump), $g = x$ preflight ( $3k/(3k+1)$ ), interp-dependent $n_{\text{rep}}$ , sharp bound $\frac{k}{k-1} \ln k$ , Richardson gate, $N$ provenance. | $g \equiv 1$ is exact for any $N$ in positive mode (blind to under-resolution); $R \leq k$ too loose to catch a $6 \rightarrow 49$ climb.                                                                                                  |
| 11 | Refinement-stable snapshots + strikes + $1.5\times$ validation twin.                                                                                                                   | Optimiser maximises $R_{\text{true}} + \text{error}(\theta)$ ; refinement-instability is the exploitation signature and must gate snapshots.                                                                                               |
| 12 | Two-tier snapshots + $\min(R_N, R_{1.5N})$ ranking + fitted-order Richardson + grid continuation + result reorder.                                                                     | One tolerance did two jobs and rejected every honest state; trained-channel order $p \approx 1.6 \neq 2$ , so fixed-order Richardson mis-extrapolated.                                                                                     |
| 13 | $\varepsilon > 0$ bound fix: $R_{\text{bound}} = k$ for $\varepsilon > 0$ ; tight bound only for $\varepsilon = 0$ .                                                                   | $M_{k,\varepsilon,1/2}$ legitimately exceeds the vanilla bound ( $4.02 > 3.99$ at $k = 50$ ), so the tight bound falsely aborted. Later aligned this with proposition 6.5 from Polymath8b to                                               |
|    |                                                                                                                                                                                        | $R_{\text{bound}} = \frac{k}{k-1} \log(2k-1) \quad (\varepsilon > 0).$ <p>There is a more stringent bound in Polymath8b Remark 6.6 but this suffices as hard-coded discovery threshold / safeguard.</p>                                    |
| 14 | Per-pair diagnostic + <code>--dump-violation</code> + <code>--patience</code> ; $k$ -aware panels; excursion guard; <code>polish()</code> incumbent seeding.                           | Fixed 8-panel mesh under-resolved $x^{k-2}$ at $k \geq 250$ ( $e^{152}$ vs $2n_{\text{out}}=128$ ); <code>polish()</code> started at $-\infty$ so step 1 installed $R = 70$ as best.                                                       |
| 15 | <code>--trunc-floor</code> prevention + rank-stability detection + strike-counter fix.                                                                                                 | $R = 56$ lived in $A$ -directions at $\lambda/\lambda_{\max} \sim 10^{-6}$ , below assembly accuracy $10^{-3}$ ; whitening amplified error (again!) $\sim 10^3$ . <code>strikes</code> reset before the check $\Rightarrow$ never accrued. |
| 16 | Stability recalibration (25%, floored) + absolute $R \leq k$ backstop + excursion baseline anchor + grow-step guard + Richardson already-converged path.                               | 5% flagged honest wobble; best = $-\infty$ let a run-away pass at iter 0; a 0.375 baseline made 3.66 look 9.7 $\times$ ; a post-grow state crashed the run; a converged $k=500$ run got a false FAIL.                                      |

### 6.13 Aggregate modular evaluation

After projection, let  $q_j, q_\ell$  be rational channel polynomials and

$$h_{j\ell}(u) = q_j(u)q_\ell(u).$$

Rather than reconstructing every exact entry of the Gram matrices, the aggregate certifier accumulates the weighted pair contributions modulo machine-word primes:

$$S_A(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_\ell A_{j\ell}^{\text{int}} \pmod{p}, \quad S_B(p) = \sum_{j \leq \ell} \kappa_{j\ell} c_j c_\ell B_{j\ell}^{\text{int}} \pmod{p}.$$

Listing 8: Closed-form preflights and post-training representation refinement.

```

PROCEDURE PREFLIGHT_AND_REFINE(theta,k,epsilon,N)
  REQUIRE NUMERICAL_QUOTIENT(g=1) AGREES WITH CLOSED_FORM_CONSTANT(k,epsilon)
  IF epsilon = 0
    REQUIRE NUMERICAL_QUOTIENT(g=x) AGREES WITH 3*k/(3*k+1)
  END

  FOR Nq IN {N, CEIL(1.25*N), CEIL(1.5*N)}
    R[Nq] <- VALIDATED_QUOTIENT(theta,Nq)
  END

  IF successive differences contract with a common sign
    FIT R[Nq] = R_infinity - c*Nq^(-p)
    REQUIRE 0.8 <= p <= 4
    REQUIRE ABS(R[1.5*N] - R_infinity) / ABS(R_infinity) <= tau_refine
    REQUIRE R_infinity <= RIGOROUS_CEILING(k,epsilon)
  ELSE
    MARK candidate as numerically unresolved
  END

  RETURN R, R_infinity, p
END

```

All polynomial powers, coefficient extractions, pair weights and channel sums are evaluated in  $\mathbb{F}_p[x]$ . The machine-word primes have no denominator occurring in the modular formulas with vanishing modulo  $p$ , ensuring that every required inverse in  $\mathbb{F}_p$  is defined.

Polynomial arithmetic is performed with exact compiled integer and modular kernels. Earlier exact-certification implementations used Kronecker substitution: integer polynomial coefficients were packed into large integers, multiplied using GMP-backed arbitrary-precision arithmetic, and unpacked exactly; polynomial powers were then formed by binary exponentiation. In the later FLINT-backed implementations, polynomial multiplication, powering and modular reduction are delegated to FLINT [4], accessed through `python-flint`. FLINT selects its internal multiplication strategy according to operand size and representation. These implementation choices affect runtime and memory use but not the mathematical certificate, which depends only on the exact resulting elements of  $\mathbb{F}_p[x]$  and the subsequent CRT reconstruction.

Only the two scalar aggregate integers  $S_A, S_B$  are reconstructed. Their exact quadratic forms are

$$c^\top A c = \frac{S_A}{D_A}, \quad c^\top B c = \frac{S_B}{D_B},$$

with global denominators fixed before the modular sweep.

Let

$$N_{\text{CRT}} = k s_{\max}, \quad r_{\max} = (k-1) s_{\max}, \quad E = k-1 + r_{\max} + m_{\max},$$

and

$$L_{\text{cm}} = \text{lcm}\{k-1, \dots, k-1 + r_{\max} + m_{\max}\}.$$

The resulting global denominators are

$$D_A = (k + N_{\text{CRT}})! d_h^k d_c^2,$$

and

$$D_B = (k-2 + r_{\max})! L_{\text{cm}} \rho_d^E T_{\text{den}} d_h^{k-1} d_c^2.$$

All factors are therefore known before reconstruction.

The numerator-side coefficient sum can be expressed as a polynomial correlation. If  $T_{j\ell}(x)$  denotes the pair-dependent integrated-channel polynomial and

$$V_p(x) = \sum_{v=0}^{r_{\max} + m_{\max}} (k-1+v)^{-1} x^v \in \mathbb{F}_p[x],$$

then the required shifted inner products are obtained from multiplication with a reversed form of  $T_{j\ell}$ , implementing the transposition principle.

Polynomial powers are formed deterministically from exact arithmetic. Classical sparse-polynomial powering algorithms provide related powering recurrences [5], but in the present application the factorially weighted pair polynomials are typically dense. The principal computational gain therefore comes from aggregate modular evaluation, exact coefficient extraction and CRT reconstruction rather than from sparse polynomial storage itself.

### 6.14 Known denominators and deterministic CRT reconstruction

The modular stage reconstructs signed integer numerators rather than generic rational numbers. Rigorous bounds

$$|S_A| \leq \mathcal{B}_A, \quad |S_B| \leq \mathcal{B}_B$$

are obtained using

$$\|h^r\|_1 \leq \|h\|_1^r$$

together with explicit bounds for factorial ratios, powers of the rational support ratio, integrated channel coefficients and rational mixing weights. If

$$M_P = \prod_{p \in P} p$$

and

$$M_P > 2 \max(\mathcal{B}_A, \mathcal{B}_B),$$

the centered CRT representative in  $(-M_P/2, M_P/2]$  is unique. The implementation uses an additional safety margin, selecting primes until the product exceeds four times the larger bound. Held-out primes provide an implementation-level check but are not needed for mathematical uniqueness.

The certified quotient is

$$R_{\text{cert}} = k(1 + \varepsilon) \frac{S_B/D_B}{S_A/D_A}.$$

Once the rational channels and mixing vector are fixed, this value is independent of the neural discovery stage.

### 6.15 Development and validation history of the CRT certifier

The exact certifier was developed through a sequence of increasingly large stress tests in which numerical agreement between the floating-point proposal stage and the exact reconstruction stage was treated as a falsification criterion. Supplementary Table 3 records the principal failure modes and the corresponding corrections. These historical iterations are not independent proof components; they document how the final implementation acquired the numerical invariants and trust boundaries used throughout the reported certifications.

Supplementary Table 3: Development history of the exact multimodular (CRT) certifier for  $M_{k,\varepsilon,1/2}$ . Each version records the change and the failure mode that motivated it. With one exception (the FFT accuracy floor, v4), every defect was a silent disagreement between the floating-point pass and the exact pass at a specific weight vector  $c$ , invisible at small  $k$  and exposed only by scale, conditioning, or polynomial degree.

| Ver | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Failure mode / motivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v1  | Initial multimodular backend. $\hat{h}^{k-1}$ is never formed over $\mathbb{Z}$ ; per 62-bit prime, one <code>nmod</code> powering plus two weight-table multiplies execute both linear functionals as single coefficient extractions: $[x^{n_{\max}}](bh^k \cdot W_{\text{rev}})$ for $A$ , and for $B$ the pair-dependent $t$ -weights combined as $\sim m_{\max}$ scalar-poly ops against per-prime tables $G^{(m)}$ folding in $\Lambda = \text{lcm}(k-1, \dots, k-1+r_{\max}+m_{\max})$ , the $\rho$ -power clearing, and batched harmonic inverses. Reconstruction is aggregate-only—two integers $S_A, S_B$ with known-denominator scalings—via deterministic centered CRT against exact a priori bounds, plus held-out verification primes. Float pass decoupled from the exact engine ( $A = \int_0^1 h^{*k}$ , $B = \int_0^\rho h^{*(k-1)} Q_j Q_l (1-s)$ by renormalized binary powering). Greedy $c$ -weighted pruning on the float pencil ( <code>--prune-tol</code> ); only surviving pairs reach the CRT engine. | Establishes the architecture: avoid the one unavoidable giant integer, reconstruct only two aggregate scalars, keep the float discovery pass cheap and separate from exact arithmetic.                                                                                                                                                                                                                                                                                                                    |
| v2  | <code>_ritz</code> no longer calls the Cholesky-based generalized solver: it equilibrates by the diagonal, eigendecomposes the metric $A$ , truncates to eigenvalues above $10^{-12}$ of the top one, and solves the standard symmetric problem in whitened coordinates, embedding zeros for dead channels. Self-test adds exact agreement with <code>scipy.eigh(B,A)</code> on a PD case and a finite $\lambda$ on a singular metric.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <code>scipy.eigh(B,A)</code> requires $A$ positive definite; at $k = 500$ the float Gram is numerically semidefinite and Cholesky fails ( <code>LinAlgError: leading minor not positive definite</code> ). Restricting to $\text{range}(A)$ is the correct regularization: null-space components contribute $\approx 0$ to $c^T A c$ .                                                                                                                                                                    |
| v3  | B-side of the float DP rescaled by $u = s/\rho$ : run the $(k-1)$ -fold convolution power on $h_\rho(t) = h(\rho t)$ , so $B = \rho^{k-1} \int_0^1 h_\rho^{*(k-1)}(u) T(\rho u) du$ , with $\rho^{k-1}$ carried in the log-scale ( $\varepsilon = 0$ degenerates to the old path). Driver interlock: float-predicted $R$ checked against discovery $R_{\text{NN}}$ before any exact work.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | At $k = 500$ the certified value collapsed to $\sim 10^{-12}$ (valid but worthless). $h^{*(k-1)}$ concentrates like $s^{k-2}$ near $s = 1$ , so the whole B-region $s \leq \rho = \frac{12}{13}$ sat $\rho^k \sim e^{-40}$ below the renormalized peak—beneath float64 accuracy. $B$ was noise; the pruner dropped 48 channels “for free” and the exact engine certified a garbage frozen $c$ .                                                                                                           |
| v4  | Truncated convolution <code>_tconv</code> forced to <code>direct</code> <code>np.convolve</code> , removing the FFT path. Large- $k$ float-DP regression test against exact $g \equiv 1$ closed forms at $k = 200, 500$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | FFT convolution carries an <i>absolute</i> error $\sim 10^{-16}$ relative to the vector peak, while $h^{*j}$ spans $s^{j-1}$ internally; entries sit up to $2^{-j}$ below peak and are buried by noise by $j \sim 55$ , then amplified through each squaring (closed-form check failed by $e^{+1000}$ ). Direct summation keeps per-entry <i>relative</i> accuracy; the self-convolution integrand peaks at the interior $t = s/2$ , representable to $j \sim 511$ (float64 ceiling near $k \sim 1000$ ). |

*continued on next page*

Table 3, continued

| Ver | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Failure mode / motivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v5  | Product-exact prime budgeting ( <b>gen_primes_for_bound</b> ): accumulate the actual prime product until it exceeds $4 \cdot$ bound rather than assuming 61 bits/prime. Miller–Rabin citation corrected (least strong pseudoprime to bases $2 \dots 37$ is $3.19 \times 10^{23}$ ; Sorenson–Webster 2017, OEIS A014233). Self-test cross-checks generated primes against FLINT’s proven <b>fmpz.is_prime</b> ; new strong-cancellation test ( $c = (1, -1)$ , near-identical channels).         | Collaborator review round: tighten certification claims (exact prime count, correct primality bound, FLINT belt-and-braces), cover the small-signed-aggregate CRT branch, and <i>measure</i> which proposed speedups were real ( <b>mul_low</b> benched slower than the full FFT multiply; native modular dot kernel deferred).                                                                                                                                                                                    |
| v6  | Added <b>--float-only</b> survey mode: projection + float DP + pruning per degree, stopping before any exact/CRT work.                                                                                                                                                                                                                                                                                                                                                                          | Cheap degree scouting at large $k$ : projection $L_\infty$ and float-predicted $R$ answer “is this $k$ reachable at this basis degree” in minutes per degree, without spending the prime budget.                                                                                                                                                                                                                                                                                                                   |
| v7  | Stable Legendre-basis float evaluation. <b>project_channels_v3</b> returns both monomial integers (exact engine) and dyadic Legendre numerators (float engine); the float engine evaluates channels and the B-weight antiderivative via the shifted-Legendre recurrence, with a non-finite guard that raises with a diagnosis.                                                                                                                                                                  | At $d = 160$ the float pass produced NaN: channels were evaluated by <b>np.polyval</b> on monomial coefficients of size $\sim 8^d \sim 10^{150}$ , so Horner formed an $O(1)$ value as an alternating sum of $10^{150}$ -scale terms ( $\sim 150$ orders of cancellation), poisoning the convolutions to $\pm\infty$ and hence $\text{off}_A = \max(-\infty) = -\infty$ , $(-\infty) - (-\infty) = \text{NaN}$ . The Legendre recurrence ( $ \tilde{P}_n  \leq 1$ on $[0, 1]$ ) is well-conditioned at any degree. |
| v8  | Auto-escalating rank truncation in the whitened Ritz, anchored to $R_{\text{NN}}$ : <b>_ritz</b> gains a <b>rank_tol</b> threaded through <b>greedy_prune</b> ; the driver escalates $10^{-10} \rightarrow 10^{-8} \rightarrow 10^{-6} \rightarrow 10^{-4}$ until the predicted $R$ is physical, or uses a fixed <b>--rank-tol</b> .                                                                                                                                                            | At $d = 40$ , $k = 500$ the float $R$ came out $7.4 \times 10^{21}$ . Gram entries were cancellation-limited to $\sim 10^{-8}$ but the rank cut was $10^{-12} \lambda_{\text{max}}$ ; directions above the cut yet below the accuracy floor are noise, and whitening by $1/\sqrt{\text{noise}}$ makes $\lambda = B_{\text{noise}}/A_{\text{noise}}$ explode. The true pencil is bounded by $M_{k,\varepsilon}$ (single digits).                                                                                    |
| v9  | Worker refactor: the B-functional inner sum $S_R = \sum_m tw_m/(k-1+R+m)$ computed for all $R$ by <i>one</i> correlation multiply $IT = \text{INV} \cdot TW_{\text{rev}}$ against a single global per-prime inverse polynomial, then combined in a short loop against the base ladder. The $(m_{\text{max}}+1)$ -polynomial $G_{\text{rev}}$ tables are eliminated; per-prime tables become three ladders regardless of degree. Added <b>--resume</b> : append-only $(p, S_A, S_B)$ checkpoint. | At $d = 160$ , $m_{\text{max}} = 2(d+1) = 322$ , so the v1 $G_{\text{rev}}$ tables were $323 \times 160k \approx 413$ MB per worker and $\sim 5 \times 10^7$ Python ops per prime for table construction alone—dominating the sweep across $\sim 28k$ primes. The correlation form is degree-oblivious; resume makes the multi-hour sweep preemption-tolerant (spot instances).                                                                                                                                    |

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Table 3, continued

| Ver | Change                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Failure mode / motivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v10 | Exact mantissa-preserving $c$ rationalization plus a frozen- $c$ gate. $c_q = \lceil \text{Fraction}(\text{float}(v)) \rceil$ (full 53-bit mantissa per component) replaces uniform-absolute dyadic( $v$ , bits) rounding; the gate re-evaluates the float quotient at the rationalized $c$ and aborts if it moved by $> 1\%$ . Resume caches gain a SHA-256 payload fingerprint and refuse stale residues.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | After an 11-hour $k = 500$ sweep the certified value was $8.9 \times 10^{-5}$ . Uniform absolute $2^{-48}$ rounding annihilated small-but-essential components of the balanced Ritz $c$ , whose entries span dozens of orders ( $c_j \sim A_{jj}^{-1/2}$ , $A_{jj} \sim (\sup)^{2k}$ , so channels differing by 1.1 in sup-norm have diagonals differing by $1.1^{1000} \sim 10^{41}$ ). The float pass evaluated the true $c$ ( $\approx 4.19$ ); the exact engine certified the mutilated one. |
| v11 | Fixed the Gauss–Jacobi overflow in the shared quadrature builder: replace <code>scipy.roots_jacobi</code> with an overflow-free Golub–Welsch construction (Jacobi tridiagonal eigenproblem; weights = squared first-eigenvector components normalized to sum 1, so the diverging $\mu_0$ cancels). ( <i>Shared-infrastructure fix; gates whether large-<math>k</math> discovery npz files can be produced at all.</i> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <code>roots_jacobi</code> computes $\mu_0 = 2^{a+b+1} B(a+1, b+1)$ in linear space; with $p - 1 \sim 2499$ at $k = 2500$ the Beta/Gamma factors overflow float64 ( $\Gamma(171)$ already overflows), returning NaN weights and crashing the $g \equiv 1$ preflight. Golub–Welsch needs only recurrence coefficients (no $\Gamma$ ); validated to $\sim 10^{-14}$ vs <code>roots_jacobi</code> for $p \leq 60$ and finite/correct to $p = 6000$ .                                                 |
| v12 | Diagonal preconditioning of the float Gram: <code>matrices()</code> subtracts $(\log A_{jj} + \log A_{ii})/2$ from each log-entry (so $\hat{A}$ ’s diagonal is exactly 1, off-diagonals $O(1)$ by Cauchy–Schwarz), applies the same congruence to $B$ , and returns $\log A_{\text{diag}}$ . Regression: a 195-nat diagonal spread must keep all channels alive with float-vs-CRT agreement $10^{-5}$ .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | A single global $\exp(\log A - \text{off})$ offset underflowed entries $< \text{off} - 745$ to exactly 0; at $k = 1000$ the diagonal spans $\sim 6900$ nats, so the Ritz solve collapsed to rank 2 (observed: 30 of 32 channels dropped “for free” at 0.0 loss). Per-channel log-diagonal preconditioning keeps every channel alive regardless of $k$ .                                                                                                                                          |
| v13 | Frame conversion of the imported discovery $c$ on load. When the npz carries $\log A_{\text{diag}}$ , the loader maps discovery’s exported (preconditioned-frame) $c$ to the true channel frame via $c_{\text{true}} = c \cdot \exp(-\frac{1}{2} \log A_{\text{diag}}) \cdot \text{channel\_norms}$ (both factors were then thought necessary: the $k$ -th-power diagonal scale is independent of the $L^2$ scale). Later corrected: discovery assembles its Gram pencil on the $L^2$ -normalized channels $g_j / \ g_j\ _2$ , and the certifier rebuilds exactly these channels by projecting <code>g_fine/channel_norms</code> . The coefficient on the normalized channels is therefore $c_{\text{true}} = \exp(-\frac{1}{2} \log A_{\text{diag}}) \hat{c}$ , with no channel-norm factor. A conversion to the raw channels would require $\ g_j\ _2^{-k}$ , not $\ g_j\ _2$ . The conversion is now shared by all exact backends ( <code>certification/frames.py</code> ). Validity was never affected, since any explicit rational $c$ yields a valid lower bound, and the re-Ritz path (v14) never used the channel-norm factor. | At $k = 53$ discovery reported 3.994 but the certifier gave 3.594. The pencil’s frame-invariant top eigenvalue was 3.9948; discovery exports $c$ in the frame of its <i>preconditioned</i> Gram ( $\hat{c} = D^{1/2} c_{\text{true}}$ ), while the certifier evaluates $c$ on the <i>true</i> channels. The deflation grows with the $\log A_{\text{diag}}$ spread (308 nats at $k = 53$ , $\sim 6900$ at $k = 1000$ ).                                                                          |

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Table 3, continued

| Ver | Change                                                                                                                                                                                                                                                                                                                                                                   | Failure mode / motivation                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| v14 | Frame conversion of the re-Ritz $c$ (second instance of the same bug). The driver captures $\log A_{\text{diag}}$ from the float engine and maps <code>_ritz</code> 's output back to the true frame ( $c_{\text{true}} = \exp(-\frac{1}{2} \log A_{\text{diag}}) \hat{c}$ ) on the active channels before rationalizing; the frozen- $c$ gate is made frame-consistent. | After v13, <code>--use-nn-c</code> still gave 3.594. <code>_ritz</code> whitens the preconditioned Gram internally, so its eigenvector lives in the preconditioned frame—and the driver certified it as if true-frame. Certifying $\hat{c}$ as true-frame gives 3.5938 (the observed value); the mapped $c$ gives 3.9939, and the end-to-end run then certified 3.9944 (gap $+4 \times 10^{-4}$ to discovery). Affected <i>every</i> re-Ritz-path certification against a preconditioned-export npz. |

## 6.16 Representation-preserving transfer

The discovery solver uses diagonal preconditioning. If

$$A_{\text{disc}} = D^{-1/2} A_{\text{true}} D^{-1/2}$$

and  $\hat{c}$  is the mixing vector in discovery coordinates, then

$$c_{\text{true}} = D^{-1/2} \hat{c}.$$

Any additional channel normalization is likewise restored before certification.

During projection and floating-point evaluation the channels are represented in a shifted-Legendre basis,

$$q_j(u) = \sum_{r=0}^d a_{jr} \tilde{P}_r(u),$$

and evaluated through the three-term recurrence. For exact certification the same rational expansion is transformed symbolically into integer monomial coefficients with a common denominator. If

$$h(u) = \sum_{r=0}^s h_r u^r,$$

define its factorially weighted coefficient polynomial by

$$\hat{h}(x) = \sum_{r=0}^s r! h_r x^r.$$

For enlarged support,

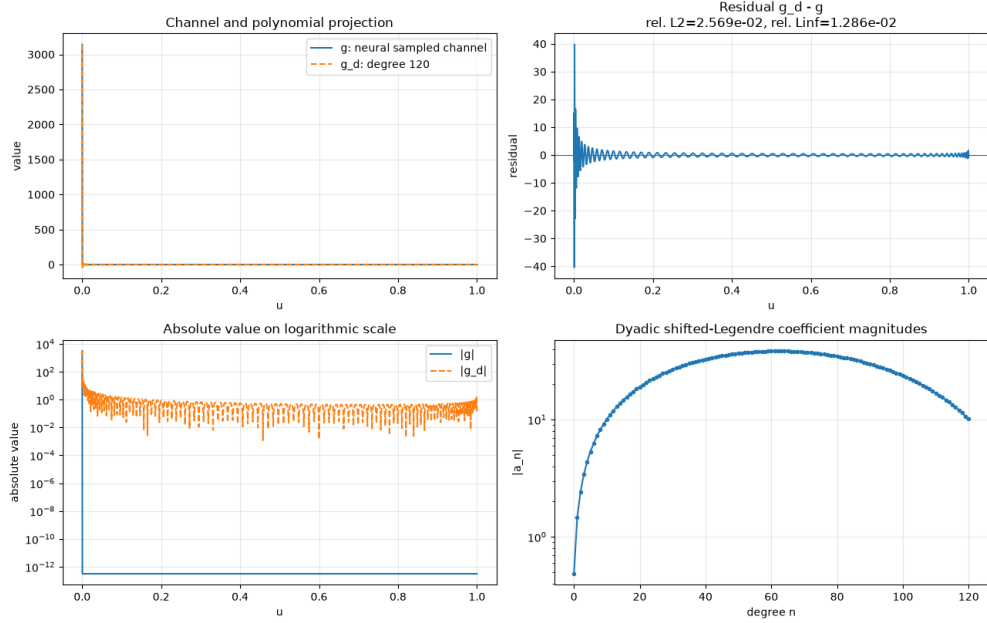
$$u = \frac{t}{1 + \varepsilon}, \quad r_\varepsilon = \frac{1 - \varepsilon}{1 + \varepsilon},$$

and the associated Jacobian and powers of  $1 + \varepsilon$  and  $r_\varepsilon$  are retained explicitly. The mixing vector is embedded into the rationals by preserving each binary floating-point component as an exact dyadic rational, avoiding loss of small but cancellation-relevant entries. Before modular evaluation, the frozen rational vector is re-evaluated in the floating-point quadratic forms and required to reproduce the Ritz candidate within tolerance. This is a transfer check, not part of the proof.

## 6.17 Loss-controlled channel reduction

Exact certification cost is quadratic in the number of surviving channels. Starting from the full floating-point Gram pencil, channels are removed by backward elimination. For each candidate removal, the generalized Rayleigh–Ritz problem is re-solved. If  $\lambda_0$  is the original value and  $\lambda_S$  the value on the active subset,

$$\frac{\lambda_0 - \lambda_S}{\lambda_0} \leq \tau_{\text{prune}}$$



Supplementary Figure 1: **Diagnostic large- $k$  failure of the early discovery pipeline.** For  $k = 3600$ , the discovery stage reported the spurious value  $R = 8.1375743146$ . Nearly all Ritz weight collapsed onto a single channel, shown here together with its degree-120 polynomial projection, residual structure, logarithmic-scale amplitude and dyadic shifted-Legendre coefficient magnitudes. Although the discovery objective was stable under grid refinement and remained below the Cauchy–Schwarz ceiling, the certification failed independent reconstruction and exposed the representation inconsistency; rejecting this candidate.

is required. At each step the channel whose removal leaves the largest quotient is discarded. This differs from coefficient thresholding because importance is measured directly in the reoptimized variational objective. The reduced rational trial function is then certified exactly; pruning changes only cost and attainable bound, not mathematical validity.

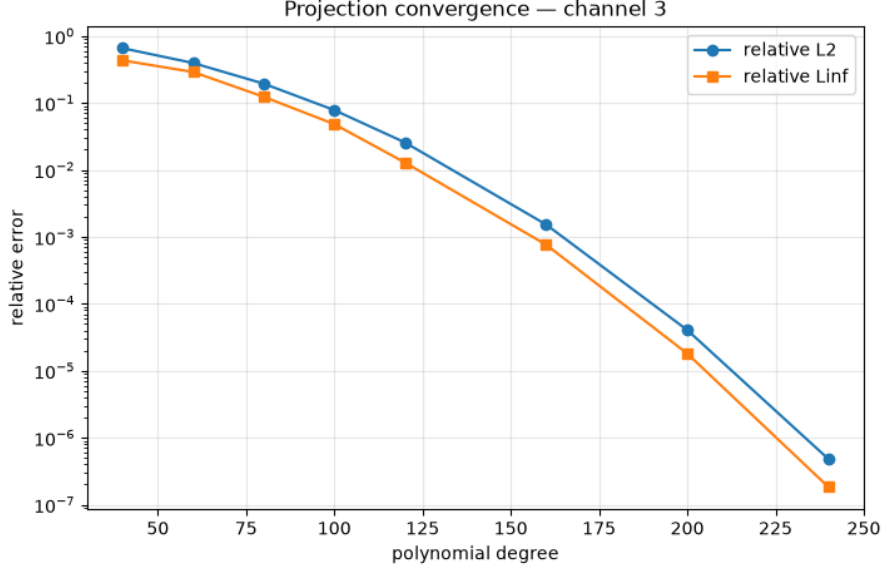
## 6.18 Diagnostic large- $k$ failure and rational distillation

The large- $k$  experiments exposed a failure mode that was not detected by the original discovery diagnostics. In an early run at  $k = 3600$ , the multi-channel optimizer collapsed onto a single sharply localized channel and reported a Rayleigh value above 8 (See Supplementary Figures 1 and 2). The value was stable under refinement of the representation grid and remained below the aforementioned ceiling ( $R_k \leq \frac{k}{k-1} \log k$ ) so the candidate initially appeared numerically credible. Also, across the validated separable runs, increasing  $k$  was accompanied by a progressive reduction in the number of channels needed to represent the best surviving candidates. This trend was empirical.

Upon certification, it appeared difficult to adequately represent the one-channel discovery solution. For a single exported channel  $q$ , the scale-invariant quantity

$$R_{1D} = k \frac{(\int q)^2}{\int q^2}$$

disagreed by orders of magnitude with the stored diagonal ratio  $kB_{jj}/A_{jj}$ . Since multiplication of  $q$  by an arbitrary scalar cannot change either ratio, this discrepancy cannot be explained by a normalization or frame convention. For the sharpest channel, the support of the exported function also gave the Cauchy–Schwarz bound  $R_{1D} \leq k\sigma$ , where  $\sigma$  is the effective support measure; the stored discovery value violated the corresponding scale by more than two orders of magnitude.



Supplementary Figure 2: **Polynomial projection convergence for the single Ritz weight holding channel of  $k = 3600$ .** Relative  $L^2$  and  $L^\infty$  projection errors decrease with polynomial degree, illustrating that the transfer error can be reduced systematically once a numerically resolved channel has been obtained.

The discrepancy was caused by a scale-law inconsistency in the discovery evaluation of (increasingly) narrow channels. The convolution recursion operated in a normalized variable, whereas the exported channel lived in the physical coordinate. In the intermediate-rate regime the error followed approximately the missing-Jacobian law

$$R_{\text{stored}} \approx \frac{\rho}{k} R_{\text{direct}},$$

with  $\rho$  the channel rate. At the highest rates a second resolution effect appeared, and the sharpest exported channel was itself truncated. The optimizer therefore concentrated its Ritz weight on the channel for which the numerical artifact was largest. The apparent rank-one structure was consequently not, by itself, evidence of a genuine extremizer.

The comparison is exact when the channel is genuinely compactly supported: if  $\text{supp}(g) \subseteq [0, \sigma]$  with  $k\sigma \leq 1$ , then  $[0, \sigma]^k \subseteq \mathcal{R}_k$ , so  $I_k(F) = (\int g^2)^k$  and  $J_k^{(1)}(F) = (\int g)^2 (\int g^2)^{k-1}$ , giving  $R = k(\int g)^2 / \int g^2$  identically. For a channel with only an *effective* support the identity acquires an uncontrolled tail term, so the scalar test is applied as an empirical safeguard against gross inflation rather than as an exact identity.

This observation is important because the artifact was close to the true large- $k$  scale. The sharp-channel artifact approached a value proportional to the first inner quadrature weight, and for the quadrature order used in the run it crossed 8 in the same  $k$ -range in which the true variational constant approaches 8. Grid stability and upper bounds were therefore insufficient to identify artifact from valid.

The final discovery solver adds an independent scalar-channel gate. For each sufficiently localized channel it compares  $kB_{jj}/A_{jj}$  with a direct one-dimensional quadrature of

$$k \frac{(\int g_j)^2}{\int g_j^2},$$

using a quadrature path independent of the convolution assembly. Wide channels are additionally checked against support-aware Cauchy–Schwarz bounds. Narrow channels get the hard  $\mu \times$  inflation rejection. Candidates failing these tests are rejected before export.

Listing 9: Checks used to diagnose unresolved or spuriously inflated neural channels before transitioning to the rational large- $k$  route. The all-width moment comparison is a conservative empirical discovery safeguard and is not part of the exact certificate.

```

FUNCTION LOG_INNER_MEAN(log_residual, rate, Y, z_cap)
  IF rate*Y < z_cap
    u <- Y*x_GL
    RETURN LOGSUMEXP_r(LOG(w_GL[r])+log_residual(u[r])-rate*u[r])
  ELSE
    z <- z_cap*x_GL
    // z=rate*u resolves the exponential boundary layer explicitly.
    RETURN LOG(z_cap/(rate*Y))
      +LOGSUMEXP_r(LOG(w_GL[r])+log_residual(z[r]/rate)-z[r])
  END
END

PROCEDURE LARGE_K_CHANNEL_CHECKS(g,A,B)
  rho <- SOFTPLUS(eta)
  (I1[j],I2[j],sigma[j]) <- FINE_GRADED_ONE_DIMENSIONAL_QUADRATURE(g_j)
  R_direct[j] <- k*I1[j]^2/I2[j]
  R_conv[j] <- k*B[j,j]/A[j,j]

  REQUIRE R_conv[j] <= (1+tau_moment)*R_direct[j] FOR EVERY j
    // conservative empirical discovery filter
  IF k*sigma[j] <= 1 AND R_conv[j] > mu*R_direct[j]
    REJECT unresolved single-channel inflation
  END
  IF k*sigma[j] > 1 AND R_conv[j] > (1+tau_support)*k*sigma[j]
    FLAG the channel as support-inconsistent
  END

  curvature <- MEAN_SQUARED_CURVATURE(LOG_RAW, coordinate=rho*x)
  RETURN curvature // Adam search penalty only
END

```

Conditional on candidates that survive these discovery checks, rational distillation provides a second and logically independent filter. Richer multi-cluster rational fits can reduce pointwise approximation error while failing probability-mass, conditioning or Fourier-stability checks. Because small one-dimensional transform errors are raised to the  $(k-1)$ st power in the large- $k$  evaluation, pointwise agreement alone does not preserve the Maynard functional.

In the stable distilled representation, the surviving family reduces to the single rational pole

$$g(t) = \frac{1}{c + (k-1)t}.$$

Its Rayleigh quotient is then optimized and evaluated independently of the neural model. Thus the rank-one collapse observed in the failed neural run and the rank-one structure of the certified rational family have opposite logical status: the former was an optimization response to a numerical artifact, whereas the latter survives reconstruction and independent certification. Neither the failed discovery value nor the neural rank collapse is used as evidence for the final bound.

## 6.19 Neural-to-rational structural distillation

At large  $k$ , the neural candidate is compressed into the clustered rational family

$$g_{\text{rat}}(t) = \sum_{a=1}^r \sum_{j=1}^{p_{\max}} u_{a,j} (c_a + nt)^{-j}, \quad n = k-1. \quad (39)$$

For fixed cluster locations  $c = (c_1, \dots, c_r)$ , the coefficients enter linearly and are determined by variable projection,

$$u^*(c) = \arg \min_u \sum_i \omega_i \left[ g_{\text{NN}}(t_i) - \sum_{a,j} u_{a,j} (c_a + nt_i)^{-j} \right]^2. \quad (40)$$

The outer optimizer therefore varies only the nonlinear pole locations. Negligible coefficients are removed and the highest retained power at each pole determines an inferred multiplicity. Higher powers are confluent directions, since

$$\frac{\partial^q}{\partial c^q}(c + nt)^{-1} = (-1)^q q! (c + nt)^{-(q+1)}.$$

Each distilled candidate is subsequently reconstructed in an independent deterministic evaluator. Fourier-domain cross terms are reduced to

$$F_p(\xi) = \int_0^1 e^{i\xi t} (c + nt)^{-p} dt, \quad (41)$$

with, for  $p \geq 2$ ,

$$F_p(\xi) = -\frac{e^{i\xi}(c + n)^{1-p} - c^{1-p} - i\xi F_{p-1}(\xi)}{n(p-1)}. \quad (42)$$

Because upward recursion can amplify round-off at large  $|\xi|$ , the Fourier computation is restricted to a stability window and requires the characteristic function to decay below a prescribed threshold before truncation. Products involving distinct clusters are reduced by partial fractions; cross-cluster cancellation is monitored and ill-conditioned configurations are rejected.

Further checks include conservation of probability mass, agreement of the first moment with its analytic value, control of the negative Fourier-inversion noise floor, rank diagnostics for the Gram matrix, and the analytic ceiling

$$R_k \leq \frac{k}{k-1} \log k.$$

A candidate that violates any check is rejected rather than assigned a numerical objective. A surviving structure is locally refined in the deterministic evaluator, so the final rational value is a newly optimized Rayleigh quotient rather than the neural objective or the pointwise fit.

## 6.20 Probabilistic reduction for the rational trial

For  $k \geq 3$ , let  $n = k - 1$ , choose an exact rational  $c > 0$ , and define

$$F(t_1, \dots, t_k) = \prod_{i=1}^k g(t_i) \mathbf{1}_{\mathcal{R}_k}(t), \quad g(t) = \frac{1}{c + nt}, \quad (43)$$

where  $\mathcal{R}_k = \{t_i \geq 0 : \sum_i t_i \leq 1\}$ . Put  $w = g^2$  and

$$m_0 = \int_0^1 w(t) dt = \frac{1}{c(c+n)}.$$

Let  $S = S_n$  denote the sum of  $n$  independent variables with density  $w/m_0$  on  $[0, 1]$ . Then

$$G(\rho) = \int_0^\rho g(t) dt = \frac{\log(1 + n\rho/c)}{n}, \quad H(\rho) = \int_0^\rho w(t) dt = \frac{1}{n} \left( \frac{1}{c} - \frac{1}{c + n\rho} \right). \quad (44)$$

Peeling one coordinate from the Maynard integrals gives

$$I_k(F) = m_0^n D, \quad J_k^{(1)}(F) = m_0^n N, \quad (45)$$

where

$$N = \mathbb{E}[G(1 - S)^2 \mathbf{1}_{S < 1}], \quad D = \mathbb{E}[H(1 - S) \mathbf{1}_{S < 1}], \quad (46)$$

and consequently  $M_k \geq kN/D$ .

Let

$$\hat{w}(\theta) = \int_0^1 e^{i\theta t} w(t) dt, \quad \varphi(\theta) = \left( \frac{\hat{w}(\theta)}{m_0} \right)^n,$$

and, for  $X \in \{G^2, H\}$ ,

$$\Psi_X(\theta) = \int_0^1 X(\rho) e^{i\theta \rho} d\rho.$$

Listing 10: Weighted variable-projection distillation of a neural channel, followed by independent Rayleigh optimization and evaluation in the resulting clustered rational family.

```

PROCEDURE DISTILL_NEURAL_CHANNEL(t, g_NN, omega, initial c, multiplicities mu, k)
  n <- k-1
  PARAMETERIZE c as positive and strictly ordered

  FUNCTION PROFILED_FIT_ERROR(c)
    X[:,a,j] <- (c[a]+n*t)^(-j), j=1,...,mu[a]
    Xw <- DIAG(SQRT(omega))*X
    yw <- SQRT(omega)*g_NN
    scale each column of Xw to unit norm
    u <- LEAST_SQUARES(Xw,yw)
    undo the column scaling in u
    RETURN SUM_i omega[i]*(g_NN[i]-(X*u)[i])^2
  END

  c <- LBFGS_MINIMIZE(PROFILED_FIT_ERROR,c)
  u <- PROFILE_LINEAR_COEFFICIENTS(c)
  PRUNE terms with negligible scaled contribution
  INFER cluster multiplicities from the retained powers
  RETURN c, u, inferred multiplicities, relative fit error
END

PROCEDURE REFINE_RATIONAL_FAMILY(k, cluster_locations c, multiplicities mu)
  n <- k-1; L <- 1+epsilon
  basis <- {(c[a]+n*t)^(-j) : j=1,...,mu[a]}
  PARAMETERIZE c as positive and strictly ordered

  FOR EACH channel pair
    IF both channels belong to one cluster
      F[1](theta) <- SINE_COSINE_INTEGRAL_FORMULA(theta,c,n,L)
      FOR p=2,...,p_max
        F[p] <- -(EXP(i*theta*L)*(c+n*L)^(1-p)-c^(1-p))
                  -i*theta*F[p-1])/(n*(p-1))
      END
    ELSE
      REDUCE the product by partial fractions
      REJECT excessive cross-cluster cancellation
    END

    theta_max <- n*(amplification_budget*(p_max-1)!)^(1/(p_max-1))
    REQUIRE the characteristic function is negligible at theta_max
    phi(theta) <- (F_pair(theta)/F_pair(0))^n FOR |theta|<=theta_max
    density <- REAL(FFT(phi))/period
    REQUIRE conservation of mass, first moment, and noise-floor control
    INTEGRATE A[j,1] and B[j,1] against density
  END

  PROFILE the linear coefficients by TOP_RITZ(A,B)
  OPTIMIZE c by Brent (one cluster) or finite-difference LBFGS (multiple)
  RETURN the best candidate re-evaluated on a finer grid
END

```

The required expectation has the Fourier representation

$$I_X = \frac{1}{2\pi} \int_{\mathbb{R}} \varphi(\theta) e^{-i\theta} \Psi_X(\theta) d\theta. \quad (47)$$

The integrability needed for inversion follows from integration-by-parts decay of both factors.

## 6.21 Aliasing bound

Fix  $P > 2$  and  $\Delta\theta = 2\pi/P$ . For the full trapezoidal sum  $T_X$ , Poisson summation and compact support imply

$$0 \leq T_X - I_X \leq X_{\max} \mathbb{P}(S > P - 1). \quad (48)$$

The one-sided sign follows from nonnegativity of the aliased density. The residual probability is bounded by certified moments. With  $\widetilde{M}_0 = 1$  and

$$\widetilde{M}_r = \sum_{j=1}^r \binom{r-1}{j-1} n \mathbb{E}[t^j] \widetilde{M}_{r-j}, \quad (49)$$

the supplied compound-Poisson domination gives

$$\mathbb{P}(S > y) \leq \min_{1 \leq r \leq r_{\max}} \frac{\widetilde{M}_r}{y^r}. \quad (50)$$

$E[S^r]$  expands over set partitions with falling-factorial coefficients  $n^{\underline{b}} \leq n^b$ , and the compound-Poisson moments have exactly  $n^{\underline{b}}$  with all terms nonnegative.

## 6.22 Fourier truncation bound

Integration by parts yields, for  $\theta \neq 0$ ,

$$\left| \frac{\widehat{w}(\theta)}{m_0} \right| \leq \frac{A}{|\theta|}, \quad A = \frac{2(c+n)}{c}, \quad (51)$$

and hence  $|\varphi(\theta)| \leq (A/|\theta|)^n$ . The omitted positive frequencies are divided into a finite band and an analytic tail. In the finite band, indices are partitioned into blocks  $[a_j, b_j]$  and Arb wide-ball evaluation supplies certified suprema  $\sigma_j$  for  $|\widehat{w}/m_0|$  on each block. Thus

$$|T_X^{\text{band}}| \leq \frac{\Delta\theta}{\pi} X_{\text{int}} \sum_j (b_j - a_j + 1) \sigma_j^n. \quad (52)$$

For

$$M_0 = \max(\lfloor \Theta_{\text{ball}}/\Delta\theta \rfloor, n_{\text{near}}), \quad \Theta_{\text{ball}} = 2A,$$

the remaining tail satisfies

$$|T_X^{\text{tail}}| \leq \frac{\Delta\theta}{\pi} X_{\text{int}} \left( \frac{A}{\Delta\theta} \right)^n \frac{M_0^{1-n}}{n-1}. \quad (53)$$

Taking  $M_0 \geq n_{\text{near}}$  ensures that the analytic tail begins where the computed sum ends.

## 6.23 Arb evaluation and directed rounding

Every scalar entering the computation is produced by Arb ball arithmetic. The transform  $\widehat{w}/m_0$  is evaluated from a closed form involving sine and cosine integrals away from the origin and by direct rigorous integration in the region where the special-function representation is ill-conditioned. The transforms  $\Psi_X$  and the moments entering the aliasing bound are likewise evaluated by rigorous integration. The pole and logarithmic branch point lie at  $t = -c/n < 0$ , outside the real integration interval; subdivision is used whenever an evaluation enclosure approaches a singularity or branch cut.

All error terms remain in ball arithmetic until the final quotient is assembled. Floating-point exports are explicitly nudged until Arb comparisons prove outward rounding. If  $N_{\text{ball}}$  and  $D_{\text{ball}}$  denote the truncated-sum enclosures and  $E_N, E_D$  the certified totals of the aliasing and truncation errors, then

$$R_{\text{low}} = \text{lb} \left( \frac{k \text{ lb}(N_{\text{ball}} - E_N)}{\text{ub}(D_{\text{ball}} + E_D)} \right). \quad (54)$$

The stored claim is additionally rounded downward so that independent recomputation with different valid Arb subdivisions or enclosure radii cannot strengthen the stated result.

## 6.24 Certified rank-one ladder and uniform finite-range bound

For  $n = k - 1$  and  $c > 0$ , let

$$g_c(t) = \frac{1}{c + nt},$$

and denote by  $R_k^{(1)}$  the value attained by this rank-one rational family. Every such value is a lower bound for  $M_k$ .

The fixed headroom value used below was selected from the independently optimized rank-one sweep rather than imposed a priori. Writing

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)},$$

the median optimized value of  $\eta$  moves steadily toward approximately  $-0.475$  as  $k$  increases:

| range of $k$         | median $\eta$ |
|----------------------|---------------|
| $k < 10^3$           | $-0.4256$     |
| $10^3 \leq k < 10^5$ | $-0.4593$     |
| $10^5 \leq k < 10^7$ | $-0.4715$     |
| $10^7 \leq k < 10^9$ | $-0.4747$     |

This convergence motivated the fixed choice

$$\eta^* = -0.4746,$$

which agrees most closely with the independently optimized scale in the large- $k$  regime containing the principal threshold results.

Fixing  $\eta$  incurs very little loss for the bulk of the optimized sweep. Interpolating the fixed-headroom ladder to the  $k$ -values of the direct rank-one sweep, the optimized certified value exceeds the corresponding fixed- $\eta$  value by a median of only

$$4.2 \times 10^{-5},$$

with maximum positive difference

$$4.2 \times 10^{-3}.$$

The comparison is not uniformly positive: the minimum difference is

$$-4.2 \times 10^{-2},$$

so at least one nominally optimized sweep row is outperformed by the fixed-headroom construction at the same  $k$ . This is consistent with the isolated high- $k$  outlier near  $\eta = -0.32$ , and is one reason the direct sweep is interpreted as a record of the numerical optimization landscape rather than as a certified sequence of exact one-parameter optima.

To obtain a uniform finite-range statement rather than isolated certified dimensions, we constructed an approximately log-uniform grid beginning at  $k = 100$ , with target spacing

$$\Delta \log k = 0.02.$$

At each grid point,  $c$  was determined by solving

$$\eta(c, k) = \eta^* = -0.4746$$

using the closed-form moments of the rank-one probability measure. The resulting explicit trial function was then certified independently using the ball-arithmetic procedure of Section 6.23. All retained rungs passed certification.

For consecutive certified grid points  $k_i < k_{i+1}$ , monotonicity of  $M_k$  implies that every integer  $k \in [k_i, k_{i+1}]$  satisfies

$$M_k \geq M_{k_i} \geq R_{\text{cert}}^{(1)}(k_i) \geq \log k - \left[ \log k_{i+1} - R_{\text{cert}}^{(1)}(k_i) \right].$$

Thus the relevant covering deficit on the interval  $[k_i, k_{i+1}]$  is

$$\delta_i = \log k_{i+1} - R_{\text{cert}}^{(1)}(k_i).$$

The largest interval deficit over the certified ladder is

$$\max_i \delta_i = 0.306754936,$$

attained on the penultimate full-size interval

$$k_i = 1,834,543,653, \quad k_{i+1} = 1,871,603,894.$$

The maximum does not occur on the final interval because the last grid step was deliberately shortened to terminate at the round endpoint

$$k = 1.88 \times 10^9.$$

Its logarithmic width is only

$$\Delta \log k = 0.004476,$$

substantially smaller than the nominal spacing 0.02. The final truncated interval therefore contributes a smaller covering deficit than the preceding full-size rung.

The distinction between pointwise and interval-covering deficits is also important. On the certified grid points themselves, the largest observed pointwise deficit is

$$\max_i \left[ \log k_i - R_{\text{cert}}^{(1)}(k_i) \right] = 0.286812601.$$

Hence the certified rungs satisfy the sharper pointwise statement

$$R_{\text{cert}}^{(1)}(k_i) \geq \log k_i - 0.287.$$

The larger constant 0.307 is required only to cover every intermediate integer  $k$  between successive certified rungs.

The interval constant is close to saturated at the upper end of the computed range. The final four full-size interval deficits are

$$0.306613, \quad 0.306660, \quad 0.306708, \quad 0.306755,$$

increasing by approximately

$$4.7 \times 10^{-5}$$

per rung. Continuing the same fixed-headroom ladder for only several further full-size steps would therefore push the covering deficit above 0.307. Accordingly, rounding the observed maximum upward gives the finite-range statement

$$\boxed{M_k \geq \log k - 0.307, \quad 100 \leq k \leq 1,880,000,000.}$$

No extension of the constant 0.307 beyond this explicitly certified range is claimed.

## 6.25 Rank-one scale selection and the headroom invariant

Under the probability measure proportional to  $g_c(t)^2 dt$ , let  $S = \sum_{i < k} t_i$  and write

$$c^{-1} = \log n + a, \quad n = k - 1.$$

Uniformly for  $a = O(\sqrt{\log n})$ , the closed-form moments of this measure imply

$$1 - \mathbb{E}[S] = c(a + 1 - \log c^{-1}) + O\left(\frac{c^2 \log n}{n}\right), \quad \text{sd}(S) = \sqrt{c} \left(1 + O\left(\frac{\log n}{n}\right)\right). \quad (55)$$

For fixed  $a$ ,  $\log c^{-1} = \log \log n + O(1/\log n)$ . Consequently,

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)} = \sqrt{c}(a + 1 - \log c^{-1}) + O\left(\frac{\log n}{n}\right). \quad (56)$$

Fixing  $\eta = -\gamma_h$ , multiplying Eq. (56) by  $c^{-1/2}$  and substituting  $a = c^{-1} - \log n$  gives the implicit scale equation

$$c^{-1} + \gamma_h c^{-1/2} - \log c^{-1} = \log n - 1 + O\left(\frac{(\log n)^{3/2}}{n}\right). \quad (57)$$

The left-hand side is strictly increasing in  $c^{-1}$  for  $c^{-1} \geq 1$ , so Eq. (57) determines the fixed-headroom scale to the stated accuracy. Writing  $x = c^{-1}$ , it gives  $x = \log n + O(\sqrt{\log n})$ , hence  $\log x = \log \log n + O((\log n)^{-1/2})$  and  $\sqrt{x} = \sqrt{\log n} - \gamma_h/2 + O(\log \log n / \sqrt{\log n})$ . Substituting these expansions yields

$$c^{-1} = \log n - \gamma_h \sqrt{\log n} + \log \log n - 1 + \frac{\gamma_h^2}{2} + O\left(\frac{\log \log n}{\sqrt{\log n}}\right). \quad (58)$$

The constant  $\gamma_h^2/2$  arises from the shift of  $c^{-1/2}$  relative to  $\sqrt{\log n}$ . The remainder decays only like  $\log \log n / \sqrt{\log n}$  and is not small over the certified range ( $\log \log n / \sqrt{\log n} \approx 0.66$  at  $\log n \approx 21.35$ ). The ladder was therefore computed from the exact closed-form moments, which agree with Eq. (57) to the stated accuracy, and not from the truncated expansion (58). Thus the empirically selected coefficient  $\gamma_h \simeq 0.4746$  is not an independent asymptotic fit: it is the same quantity as  $-\eta^*$  and follows directly from the fixed-headroom parameterization.

Equations (57) and (58) describe the fixed-headroom parameterization of the certified ladder; they are not a sharper asymptotic optimizer law. Indeed, fixed  $\eta = -\gamma_h$  corresponds to the shift

$$a(L) = c^{-1} - L = \log L - 1 - \gamma_h \sqrt{L} + \frac{\gamma_h^2}{2} + O\left(\frac{\log L}{\sqrt{L}}\right), \quad L = \log n,$$

so  $a(L) \rightarrow -\infty$  and fixed  $\eta$  eventually leaves every fixed-shift family. The leading-order part of  $a(L)$  is not monotone: its derivative  $1/L - \gamma_h/(2\sqrt{L})$  vanishes at

$$\sqrt{L} = \frac{2}{\gamma_h},$$

where it attains its maximum, and it decreases thereafter. Over the certified range, however, the induced shift obtained from Eq. (57) remains close to the small negative values where the numerical limiting defect  $\mathcal{C}(a)$  is minimized. This explains the effectiveness of the headroom surrogate over the finite range studied here: it selects rational trials lying close to the same near-optimal fixed-shift regime, even though the two parameterizations diverge asymptotically.

At the upper end of the certified ladder, the measured pointwise deficit is approximately

$$\log k - R(F_{k, c_k}) \simeq 0.2868,$$

whereas numerical evaluation of the limiting fixed-shift formula gives

$$\mathcal{C}_* \simeq 0.3343.$$

The difference is approximately

$$0.3343 - 0.2868 = 0.0475.$$

Over the finite range considered in the numerical sweep, this is close to the magnitude of the empirically fitted correction

$$\frac{1.047}{\log k} \simeq 0.049 \quad (\log k \simeq 21.35).$$

Equivalently, the observed deficits are numerically consistent with the finite-range relation

$$\log k - R(F_{k,c_k}) \approx \mathcal{C}_* - \frac{1.047}{\log k}.$$

This relation is an empirical consistency check only. In particular, the coefficient 1.047 has not been derived analytically and is not identified with the coefficient of a proved second-order expansion.

The distinction is important because the stable-law theorem concerns the fixed-shift family

$$c^{-1} = \log n + a, \quad a \in \mathbb{R} \text{ fixed},$$

whereas the fixed-headroom ladder corresponds to a shift  $a = a(\log n)$  that eventually tends to  $-\infty$ . Therefore the finite-size expansion for fixed  $a$  cannot be applied directly to the optimized headroom ladder. Even within the fixed-shift family, the proof establishes only

$$\log k - R(F_{k,c_k}) = \mathcal{C}(a) + o(1),$$

because no sufficiently sharp convergence rate has been proved for

$$\frac{L_n}{P_n} - \frac{L_a}{P_a}.$$

Consequently, neither the numerical agreement above nor the fixed-headroom parameterization establishes a rigorous  $1/\log k$  correction. The coefficient 1.047 should therefore be reported solely as a numerical finite-size observation.

## 6.26 Spectrally positive 1-stable limit

Section 8 gives the complete asymptotic analysis of the single-channel rational family. For

$$n = k - 1, \quad c_k^{-1} = \log n + a,$$

with fixed  $a \in \mathbb{R}$ , the rescaled boundary variable converges in distribution to

$$X_a = (a + 1) - \Lambda,$$

where  $\Lambda$  is the spectrally positive 1-stable random variable defined by

$$\mathbb{E}e^{is\Lambda} = \exp \left\{ \int_0^\infty (e^{isx} - 1 - isx\mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2} \right\}.$$

The proof supplements weak convergence with uniform density, small-ball, tail, and logarithmic moment estimates, which are needed because the Rayleigh quotient contains logarithmic observables at the simplex boundary.

Theorem 20 consequently gives

$$R(F_{k,c_k}) = \log k - \mathcal{C}(a) + o(1),$$

where

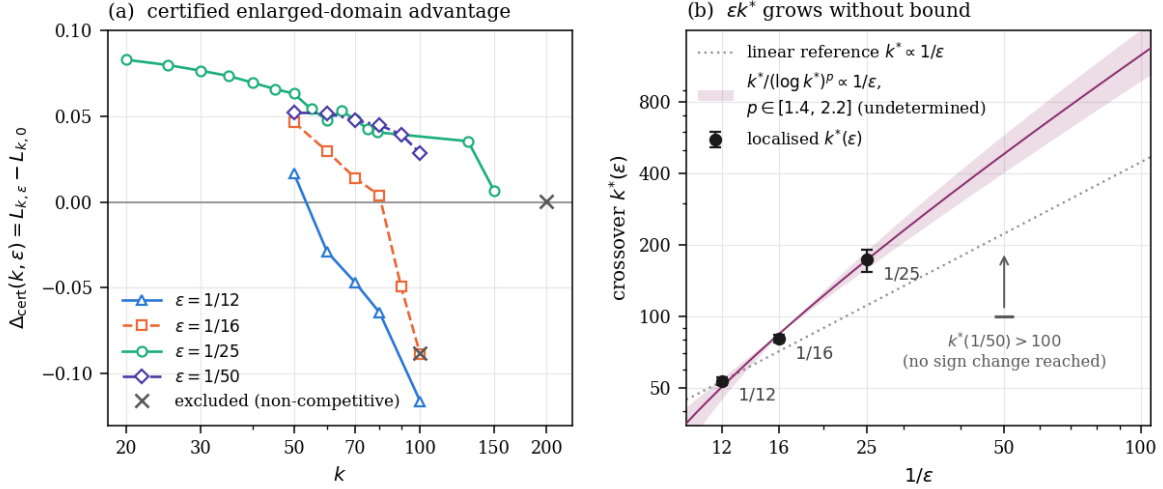
$$\mathcal{C}(a) = a - 2\mathbb{E}[\log X_a \mid X_a > 0].$$

Thus, with

$$\mathcal{C}_* := \inf_{a \in \mathbb{R}} \mathcal{C}(a),$$

Corollary 21 yields

$$M_k \geq \log k - \mathcal{C}_* - o(1).$$



Supplementary Figure 3: **Enlarged-support crossover.** (a) The certified difference  $\Delta_{\text{cert}}(k, \varepsilon) = L_{k, \varepsilon} - L_{k, 0}$ , a difference of certified lower bounds rather than of the exact constants, against  $k$  for four values of  $\varepsilon$ . It is positive at small  $k$ , decreases with  $k$ , and changes sign at a finite scale for  $\varepsilon = 1/12$  and  $1/16$ ; no sign change is reached for  $\varepsilon = 1/50$  within the computed range. Grey crosses mark certified but non-competitive runs excluded from crossover inference (Section 6.27). (b) The localised crossovers  $k^*(\varepsilon)$  against  $1/\varepsilon$  on logarithmic axes. The  $\varepsilon = 1/12$  and  $1/16$  points are linear interpolations between the certified brackets; the  $\varepsilon = 1/25$  marker is the midpoint of the extrapolative range  $[155, 192]$ , with the bar spanning that range. The points lie above the linear reference  $k^* \propto 1/\varepsilon$  (dotted) and pull away from it, giving  $\varepsilon k^* = 4.48, 5.04, 6.94$  and motivating  $\varepsilon k^*(\varepsilon) \rightarrow \infty$ . The band shows the family  $k^*/(\log k^*)^p \propto 1/\varepsilon$  over the range  $p \in [1.4, 2.2]$  admitted by the data, with the constant refitted at each  $p$ ; the exponent is not determined by these points.

Here  $\mathcal{C}_*$  is the optimal asymptotic defect only within the fixed-shift family

$$c_k^{-1} = \log(k-1) + a, \quad a \in \mathbb{R}.$$

The proof does not establish optimality over arbitrary sequences  $c_k$ , nor does it require the infimum to be attained. Numerical evaluation suggests

$$\mathcal{C}_* \approx 0.3343,$$

but this decimal value should be regarded as a numerical estimate unless the global infimum is separately certified. A rigorous enclosure of  $\mathcal{C}(a_0)$  at one fixed shift  $a_0$  already suffices for an explicit asymptotic lower bound.

Finally, although the finite- $n$  identity in Section 8 isolates a possible  $1/\log k$  correction, the required convergence rate for

$$\frac{L_n}{P_n} - \frac{L_a}{P_a}$$

has not been proved. Accordingly, no second-order asymptotic coefficient is claimed here.

**Relation to the optimized rank-one sweep.** The optimized rank-one sweep explores a different asymptotic parameter regime, including an empirically observed  $\sqrt{\log n}$  correction. Those numerical observations are not used in the fixed-shift theorem above.

## 6.27 Enlarged-support crossover analysis

Write  $L_{k, \varepsilon}$  for the certified lower bound produced for the enlarged-domain construction and  $L_{k, 0}$  for the corresponding vanilla lower bound, and define

$$\Delta_{\text{cert}}(k, \varepsilon) = L_{k, \varepsilon} - L_{k, 0}.$$

For the reliable computed configurations, this difference is positive at smaller  $k$ , decreases with  $k$ , and changes sign at a finite scale. The certified pairs used to localize the observed crossovers include

| $\varepsilon$ | $k_{\text{left}}$ | $k_{\text{right}}$ |
|---------------|-------------------|--------------------|
| 1/12          | 50                | 60                 |
| 1/16          | 80                | 90                 |

with certified differences  $+0.016961$  and  $-0.028948$  for  $\varepsilon = 1/12$ , and  $+0.003771$  and  $-0.049093$  for  $\varepsilon = 1/16$ . Linear interpolation gives

$$k^*(1/12) \approx 53.7, \quad k^*(1/16) \approx 80.7.$$

For  $\varepsilon = 1/25$ , the last reliable positive certified points are  $k = 130$  and  $150$ , with differences  $+0.035443$  and  $+0.006581$ ; the supplied analysis reports the broader extrapolative range

$$k^*(1/25) \in [155, 192]$$

rather than treating a degraded  $k = 200$  pair as a direct bracket. No sign change was reached for  $\varepsilon = 1/50$  within the reliable computed range.

All certified runs are in Supplementary Table 7. Several runs are excluded from crossover inference because they satisfy numerical certification of the particular trial function but are demonstrably noncompetitive discovery outcomes. These include a reproducible local failure at  $(k, \varepsilon) = (100, 1/25)$ , degraded paired runs at  $k = 200$ , and incomplete or nonmonotone optimization at  $k = 140$  and  $220$ . The exclusion criterion is therefore not failure of the certificate: it is failure of the discovery stage to provide a competitive construction. Monotonicity in  $k$ , the discovery-to-certification gap and retained effective rank provide inexpensive diagnostics for these cases.

The observed crossover locations give increasing values of  $\varepsilon k^*(\varepsilon)$  as  $\varepsilon$  decreases (see Supplementary Figure 3, motivating the conjecture that for each fixed  $\varepsilon > 0$  a finite crossover exists and

$$\varepsilon k^*(\varepsilon) \rightarrow \infty \quad (\varepsilon \rightarrow 0).$$

The available points are compatible with a scaling variable

$$x = \frac{\varepsilon k}{(\log k)^p},$$

but do not determine  $p$ : the supplied fits admit approximately  $p \in [1.4, 2.2]$ , and the natural  $p = 1$  and  $p = 2$  alternatives cannot be distinguished by the current data. We therefore do not assign a specific exponent.

**Scope of the crossover statement.** A crucial distinction is that  $\Delta_{\text{cert}}$  is a difference of certified lower bounds. Therefore  $\Delta_{\text{cert}}(k, \varepsilon) < 0$  does not prove  $M_{k, \varepsilon} < M_k$ ; it proves only that the certified vanilla construction found by the pipeline outperforms the certified enlarged-domain construction found at that  $k$ . Extending the observed crossover to the exact variational constants is consequently conjectural. Likewise, interpolated or extrapolated values of  $k^*(\varepsilon)$  are descriptive summaries of the computed ladder, not independently certified roots.

## 6.28 Supplementary data: optimized rank-one sweep

Supplementary Data `maynard_R_sweep_eta.csv` contains the direct rank-one optimization sweep for

$$g_c(t) = \frac{1}{c + (k-1)t},$$

with one optimized pole parameter  $c$  for each reported  $k$ . The file records the discovery value  $R$ , the independently certified lower bound, the discovery–certification gap, the Cauchy–Schwarz ceiling, the optimized scale  $c$ , the standardized simplex headroom

$$\eta(c, k) = \frac{1 - \mathbb{E}[S]}{\text{sd}(S)},$$

and numerical diagnostics used during optimization. All retained solutions have rank one in both numerator and denominator Gram representations.

The sweep contains 729 attempted configurations over  $100 \leq k \leq 2 \times 10^9$ . Of these, 727 pass certification and two are explicitly marked **FAILED**; failed rows are retained in the data for provenance but are excluded from mathematical claims. The certified discovery gap is small throughout the successful runs, with median  $9.9 \times 10^{-4}$  and maximum  $2.3 \times 10^{-3}$ . The optimized headroom is not fixed: its variation across the sweep is used diagnostically to identify the approximately constant headroom regime that motivates the geometric ladder below. This file is therefore a record of the numerical optimization landscape, not itself the basis for the uniform finite-range theorem.

## 6.29 Supplementary data: fixed-headroom geometric ladder

Supplementary Data `maynard_geometric_eta_sweep.csv` contains the certified fixed-headroom ladder used to convert isolated rank-one trial functions into a uniform bound over a continuous range of integer  $k$ . At each rung,  $c$  is chosen from

$$\eta(c, k) = -0.4746,$$

and the resulting explicit rank-one trial is certified independently. The file contains 828 certified rungs from  $k = 100$  to  $k = 1.88 \times 10^9$ ; every row has certification status **PASS**. The target headroom is reproduced to numerical precision throughout the sweep.

The ladder is approximately uniform in  $\log k$ , with asymptotic spacing  $\Delta \log k \simeq 0.02$ ; the small deviations at the lower end and final endpoint arise from integer rounding. For consecutive certified rungs  $k_i < k_{i+1}$ , monotonicity of  $M_k$  gives

$$M_k \geq R_{\text{cert}}(k_i) \quad (k_i \leq k \leq k_{i+1}),$$

and therefore

$$M_k \geq \log k - [\log k_{i+1} - R_{\text{cert}}(k_i)].$$

The largest interval deficit in the supplied data is

$$0.3067549357,$$

attained between

$$k_i = 1\,834\,543\,653, \quad k_{i+1} = 1\,871\,603\,894.$$

Rounding this deficit upward yields the reported uniform statement

$$M_k \geq \log k - 0.307$$

throughout the certified range. The file additionally records the pointwise deficit  $\log k - R_{\text{cert}}$ , the discovery–certification gap, the Cauchy–Schwarz ceiling and the scale diagnostics used to relate the finite ladder to the fixed-shift asymptotic analysis.

## 6.30 Conversion to admissible-tuple bounds

A certified inequality  $M_k > 4m$  supplies the variational input to the Maynard–Tao bounded-gap theorem. To obtain a numerical bound on  $H_m$ , one additionally requires an admissible  $k$ -tuple and its diameter. We therefore distinguish the variational certificate from the admissible-tuple construction and write  $D_k$  for the diameter of the explicit admissible  $k$ -tuple used in the conversion.

For the first threshold,  $M_{3655} > 8$ , a stronger admissible-tuple diameter is already recorded in the Engelsma–Sutherland database (<https://math.mit.edu/~primegaps/>). The entry for  $k = 3655$ , submitted by Sutherland on 27 June 2013, gives

$$D_{3655} \leq 33118.$$

Our independently constructed hybrid sieve gives the slightly weaker control

$$D_{3655} \leq 33270,$$

and is retained as a reproducibility check.

For  $k = 3655$  and  $k = 208910$ , admissible tuples were constructed with a hybrid shifted-Schinzel/greedy sieve. The algorithm first applies a Schinzel-type presieve, then greedily removes a least-populated residue class for successive primes, extracts the narrowest block of  $k$  survivors, and finally applies local diameter-reducing moves. Crucially, the discovery trajectory is not trusted: the final tuple is rechecked from scratch for every prime  $p \leq k$ .

Each constructed tuple is exported as an `admissible-tuple-certificate/1` record containing the ordered gap sequence and, for every prime  $p \leq k$ , an explicitly missing residue class. A standalone verifier shares no construction code path. It regenerates the primes up to  $k$ , reconstructs the tuple from the gap list, checks its cardinality and diameter, verifies that the witness list contains exactly the primes  $p \leq k$ , and confirms by direct enumeration that every stated residue class is absent. Since a  $k$ -element set cannot occupy all residue classes modulo any prime  $p > k$ , these checks establish admissibility.

The resulting independently verified hybrid-sieve bounds are

$$D_{3655} \leq 33270, \quad D_{208910} \leq 2718108.$$

For  $k = 3655$ , the stronger database value  $D_{3655} \leq 33118$  is used in the final bounded-gap comparison, while our independently constructed tuple serves as a reproducibility control.

For the larger thresholds, explicit window sieving is unnecessary. Let

$$\mathcal{H}_k = \{p_{\pi(k)+1}, \dots, p_{\pi(k)+k}\},$$

the first  $k$  primes strictly exceeding  $k$ . Every element of  $\mathcal{H}_k$  is prime and greater than  $k$ , so residue class  $0 \bmod p$  is empty for every prime  $p \leq k$ ; for  $p > k$ , a  $k$ -element set cannot meet all  $p$  residue classes. Hence  $\mathcal{H}_k$  is admissible by construction. Segmented prime enumeration gives

$$D_{11655069} \leq 214099720, \quad D_{644589002} \leq 14541349288.$$

The code additionally evaluates explicit prime-counting and prime-location bounds as an independent large- $k$  consistency check.

Combining these tuple diameters with the certified variational thresholds  $M_k > 4m$  gives

$$H_m \leq D_k,$$

and hence the numerical bounded-gap statements reported in the Results. The tuple-construction code, certificate files, and independent verifier are supplied as Supplementary Software/Data.

### 6.31 Standalone verification and certificate schema

The large- $k$  certificate is a self-contained JSON record containing the exact rational trial parameter  $c$ , Fourier discretization parameters, intermediate summaries and the claimed lower bound. Stored intermediate numerical enclosures are not trusted. The verifier reconstructs the analytic Fourier-tail constant, regenerates the far-field block partition, recomputes wide-ball suprema, moment bounds, Fourier nodes, trapezoidal sums, aliasing and truncation errors, and independently assembles  $R_{\text{low}}$  in ball arithmetic.

The certificate proves only the Rayleigh quotient of the fixed admissible trial function encoded by the exact rational  $c$ . It does not prove optimality of  $c$ , re-prove the external sieve criterion or general upper bounds on  $M_k$ , or certify admissible-tuple diameters. Monte-Carlo comparisons and external small- $k$  values are falsification checks rather than proof components.

## 7 Supplementary Methods: higher-order Delsarte hierarchy

Notation follows the Methods. Throughout,  $H(\cdot)$  denotes Shannon entropy in bits and  $H_2(p) = -p \log_2 p - (1-p) \log_2 (1-p)$  the binary entropy function;  $G$  is a normalized configuration on  $\mathbb{F}_2^\ell$ , i.e. a probability distribution, and  $\Phi(G, v) = \sum_u \sqrt{G(u)G(u+v)}$  is the translation affinity at a shift  $v \neq 0$ .

### 7.1 Scale-up of the hierarchy to $r \geq 3$

The finite-length hierarchy was first implemented at  $r = 2$ , where  $\text{GL}(2, 2) \cong S_3$  acts as the full symmetric group on the three non-zero vectors of  $\mathbb{F}_2^2$  and orbit representatives can be generated by sorting. This shortcut cannot be generalized to  $r = 3$ :  $|\text{GL}(3, 2)| = 168$  whereas  $|S_7| = 5040$ . Sorting therefore over-merges genuinely distinct Fano-plane orbits; at  $n = 6$ , 45 true orbits collapse to 30 sorted classes. We instead enumerate  $\text{GL}(r, 2)$  explicitly and construct genuine type orbits under the group action.

For  $S \subset [r]$ , partial Fourier transforms factor after splitting  $\mathbb{F}_2^r = \mathbb{F}_2^S \oplus \mathbb{F}_2^{S^c}$ :

$$K_\alpha^S(\beta) = \prod_{w \in \mathbb{F}_2^{S^c}} K_{\alpha(\cdot, w)}^{(|S|)}(\beta(\cdot, w)),$$

with zero value unless the  $S^c$ -marginals of  $\alpha$  and  $\beta$  agree. Each factor is a lower-dimensional full Krawtchouk transform, so the implementation recurses through the same exact row generator. Counting pairs of matrices of types  $(\alpha, \beta)$  in the two possible orders gives the reciprocity

$$\binom{n}{\beta} K_\alpha(\beta) = \binom{n}{\alpha} K_\beta(\alpha).$$

Combined with  $\text{GL}(r, 2)$  equivariance, this allows a row to be generated from one representative expansion per valid orbit. At  $r = 3$ ,  $n = 14$ ,  $d = 6$ , this reduces the number of expansions from 1235 to 112.

The resulting  $r = 3$  implementation reproduced the  $r = 2$  program when run at  $r = 2$  and produced certified strict level-3 separations, including  $V_2(9, 5) = 44/9 > V_3(9, 5) = 4$  and  $V_2(12, 5) = 24.260255 \dots > V_3(12, 5) = 16$ .

### 7.2 Exact dual certification and implementation checks

The floating-point solver is used as a proposal mechanism for an active dual set. The corresponding basic system is reconstructed at arbitrary precision, rounded to dyadic rationals, and verified against every dual inequality. The repair identity quoted in the Methods supplies an exactly feasible direction and an explicit rational repair cost, so the certified value is an exact rational upper bound on the level- $r$  optimum obtained by weak duality.

Three independent anchors were used. First, full and partial Krawtchouk rows were compared with brute-force character sums over  $\mathbb{F}_2^{r \times n}$  for small  $n$ . Second,  $r = 1$  was checked against an independent classical Delsarte implementation. Third, the published  $r = 2$  values of Loyfer and Linial were reproduced, including 24.260255... at (13, 6), 131.720587... at (16, 6), 32 at (16, 8) and (17, 8), and 256 at (17, 6) and (20, 8).

Structural invariant tests detected defects that value reproduction did not. An early active-set implementation factorized a matrix shifted by one column relative to the solved system, yielding 32.0000887... instead of 32 at (17, 8). Two precision-state errors in dyadic rounding and polishing partially masked this defect. These issues were corrected before the experiments reported here; all final bounds use the repaired exact-verification path.

### 7.3 Lift experiments and finite-length structural map

Coregliano et al. construct an explicit lift of a level-one dual to level  $\ell$  with objective  $V_1^\ell$  [6]. Because the level- $\ell$  hierarchy bounds  $|C|^\ell$ , the lift must be compared in this normalization. An initial level-one-normalized comparison reproduced  $V_1$  but violated higher-level feasibility by +9949, which served as a normalization control before the systematic sweep.

The lift has no mass on partial families below the full transform. We therefore compared the full-Fourier-only hierarchy with the complete partial-Fourier program. Across twenty tested instances the full-only optimum never exceeded  $V_1^r$ . At  $r = 2$  it was frequently exactly equal to the lift, for example

$$36 = 6^2 \quad (9, 5), \quad 576 = 24^2 \quad (11, 5), \quad 144 = 12^2 \quad (10, 5),$$

while the partial-Fourier program improved strictly. At  $r = 3$ , by contrast, the full-only hierarchy itself generally improved on the lift. The correction to a lifted level-one solution is therefore not a common object across levels.

A separate scan at approximately fixed relative distance showed why these finite instances were not used for asymptotic learning. At  $\delta = 0.40$ , the accessible finite- $n$  rates were 0.33–0.50, compared with the first-MRRW value 0.0815, and the level-one-to-level-two gain decreased with  $n$  at fixed  $d$ . Extrapolation from the accessible  $n$  range would therefore have mixed strong finite-size effects with the asymptotic target.

## 7.4 Normalization of the configuration objective

For a level- $\ell$  dual,

$$|C|^\ell \leq f(0),$$

so an asymptotic bound obtained from a normalized configuration  $G$  is

$$R(C) \leq \frac{H(G)}{\ell} + O\left(\frac{\log n}{n}\right),$$

and  $H(G)$  must not be compared directly with an ordinary code-rate bound. Re-deriving this normalization exposed two discrepancies in the cited preprint. First, the leading small- $\varepsilon$  coefficient of the first MRRW expression is  $\frac{1}{2}\varepsilon^2 \log_2(1/\varepsilon)$  rather than  $\frac{1}{4}\varepsilon^2 \log_2(1/\varepsilon)$ : with  $p_\varepsilon = \frac{\varepsilon^2}{4} + O(\varepsilon^4)$  and  $\log_2(1/p_\varepsilon) = 2 \log_2(1/\varepsilon) + 2 + O(\varepsilon^2)$ , one gets  $h(\varepsilon) = \frac{1}{2}\varepsilon^2 \log_2(1/\varepsilon) + O(\varepsilon^2)$ . Second, the closed form stated in their Lemma 6.10(i) is a lower bound on the walk count rather than an equality, since its proof passes through  $\left(\binom{m/2}{F}\right)^2 \geq \binom{m/2}{F}$ . Neither affects the qualitative conclusion of that work: the stated rate bound is off by a factor  $2\ell$  relative to  $H(G)/\ell$  — the  $\ell$  being the rate normalization above and the 2 the same halving — but the factor is applied on both sides of the comparison made there.

## 7.5 Reduced configuration search

With  $G = p^2$ , the configuration problem becomes

$$\min_{\|p\|_2=1, p \geq 0} \frac{1}{\ell} H(p^2)$$

subject to quadratic translation-affinity constraints  $p^T S_v p \geq \varepsilon$ . The largest search space considered ( $\ell = 4$ ) has 15 free probability coordinates. We therefore used direct multistart sequential quadratic programming with analytic gradients rather than a neural network. The campaign comprised 28 instance solves over  $\ell \leq 4$  and  $\varepsilon \in \{0.30, 0.20, 0.10, 0.05\}$ , using 400–600 randomized starts per instance (13,600 local optimizations in total). One start per instance used the CJJ vertex-uniform configuration and the remaining starts were randomized positive-sphere points with varying concentration.

No run produced a configuration below the first MRRW benchmark. Recovered objectives differed from MRRW by between  $-3.2 \times 10^{-14}$  and  $-2.1 \times 10^{-16}$ , consistent with floating-point noise, and the recovered configurations agreed coordinatewise with the tensor-product quasirandom family to within  $10^{-11}$ .

## 7.6 The tensorization obstruction

### 7.6.1 Setting and hypotheses

For even  $m$  and  $v \neq 0$ , the *paired walk weight* is

$$W_m(G, v) = \sum_F \frac{m!}{\prod_u F(u)!} \prod_u G(u)^{F(u)}, \quad (59)$$

the sum running over  $F : \mathbb{F}_2^\ell \rightarrow \mathbb{N}$  with  $\sum_u F(u) = m$  and  $F(u) = F(u+v)$  for all  $u$ . Quantity (59) is the leading coefficient in the configuration asymptotics of [6],

$$A_v^m \Lambda_{n,G}(X) = n^m W_m(G, v) + o(n^m), \quad (60)$$

where  $A_v$  is the adjacency operator of the column-translation graph and  $\Lambda_{n,G}$  the indicator of the configuration fibre. In (59) it is defined purely combinatorially, so no limit in  $n$  enters Lemmas 1–4 or Theorem 5.

For (60) to be available at a fixed  $G$  one needs infinitely many admissible blocklengths. Let

$$N_G = \{n \in \mathbb{N} : nG \in \text{Config}_{n,\ell}\}.$$

Along an infinite subsequence  $N_G$ , the support-size assumptions required by the configuration asymptotics hold automatically for every positive entry of  $G$ , since  $nG(u) = \Omega(n)$  there.

We isolate the two properties of the single-orbit spectral construction that the argument uses. Let  $\varepsilon \in (0, 1)$ .

**(A1) Spanning.** The set  $V \subseteq \mathbb{F}_2^\ell \setminus \{0\}$  of admissible shifts spans  $\mathbb{F}_2^\ell$ .

**(A2) Finite-walk lower bound.** For every  $v \in V$  there exist an even  $m_v \geq 2$  and a constant  $c \geq 1$  with  $W_{m_v}(G, v) \geq c\varepsilon^{m_v}$ .

In [6], (A1) is the requirement that for every  $i \neq 0$  some admissible  $v$  satisfies  $\langle i, v \rangle = 1$ . This is equivalent to spanning: failure to span is equivalent to containment in  $i^\perp$  for some non-zero  $i$ . Hypothesis (A2) holds there with  $c = 2^{2^\ell - 1} \geq 2$ ; only  $c \geq 1$  is used below.

### 7.6.2 The obstruction

**Lemma 1** (Walk-affinity comparison). *For every normalized configuration  $G$ , every  $v \neq 0$  and every even  $m$ ,*

$$W_m(G, v) \leq \Phi(G, v)^m.$$

*Proof.* Translation by  $v \neq 0$  is a fixed-point-free involution of  $\mathbb{F}_2^\ell$ , so its orbits are the  $2^{\ell-1}$  unordered pairs  $P = \{u, u+v\}$ . A multiplicity function  $F$  admissible in (59) is therefore determined by  $a_P := F(u) = F(u+v)$ , subject to  $\sum_P 2a_P = m$ , i.e.  $\sum_P a_P = m/2$ . Writing  $x_P = G(u)G(u+v) \geq 0$ ,

$$W_m(G, v) = \sum_{\sum_P a_P = m/2} \frac{m!}{\prod_P (a_P!)^2} \prod_P x_P^{a_P}. \quad (61)$$

Both  $u$  and  $u+v$  contribute  $\sqrt{x_P}$  to  $\Phi(G, v)$ , so  $\Phi(G, v) = 2 \sum_P \sqrt{x_P}$  and, by the multinomial theorem,

$$\Phi(G, v)^m = 2^m \sum_{\sum_P c_P = m} \frac{m!}{\prod_P c_P!} \prod_P x_P^{c_P/2}. \quad (62)$$

Every term of (62) is non-negative. Discard those with some  $c_P$  odd and match the remaining terms to (61) by  $c_P = 2a_P$ . The ratio of the coefficient in (61) to the corresponding coefficient in (62) is

$$\frac{m! / \prod_P (a_P!)^2}{2^m m! / \prod_P (2a_P)!} = 2^{-m} \prod_P \binom{2a_P}{a_P} = \prod_P \frac{\binom{2a_P}{a_P}}{4^{a_P}} \leq 1,$$

using  $\sum_P 2a_P = m$  and  $\binom{2a}{a} \leq 4^a$ . Summing the matched terms gives  $W_m(G, v) \leq \Phi(G, v)^m$ .  $\square$

**Lemma 2** (Affinity floor). *Under (A2),  $\Phi(G, v) \geq \varepsilon$  for every  $v \in V$ .*

*Proof.* By Lemma 1 and (A2),  $\Phi(G, v)^{m_v} \geq W_{m_v}(G, v) \geq c\varepsilon^{m_v} \geq \varepsilon^{m_v}$ . Both sides are non-negative and  $m_v$  is a fixed positive integer, so taking  $m_v$ -th roots gives the claim. No limit  $m \rightarrow \infty$  is taken, and the  $n \rightarrow \infty$  limit enters only through the definition of  $W_m$ .  $\square$

**Lemma 3** (Entropy profile). *Let  $p_\varepsilon = \frac{1}{2}(1 - \sqrt{1 - \varepsilon^2})$  and  $h(\varepsilon) = H_2(p_\varepsilon)$  for  $\varepsilon \in [0, 1]$ . Then  $h$  is strictly increasing and strictly convex on  $(0, 1)$ ,  $h(0) = 0$ ,  $h(1) = 1$ , and*

$$h(\varepsilon) = H_2\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right), \quad \delta = \frac{1 - \varepsilon}{2}. \quad (63)$$

*Proof.* Put  $s = \sqrt{1 - \varepsilon^2} \in [0, 1]$ , so  $p_\varepsilon = (1 - s)/2$  and  $1 - p_\varepsilon = (1 + s)/2$ . Then  $\delta(1 - \delta) = \frac{1 - \varepsilon}{2} \cdot \frac{1 + \varepsilon}{2} = \frac{1 - \varepsilon^2}{4} = \frac{s^2}{4}$ , whence  $\frac{1}{2} - \sqrt{\delta(1 - \delta)} = (1 - s)/2 = p_\varepsilon$ , which is (63).

Since  $H'_2(p) = \log_2((1 - p)/p)$  and  $(1 - p_\varepsilon)/p_\varepsilon = (1 + s)/(1 - s)$ , we get  $H'_2(p_\varepsilon) = 2 \operatorname{artanh}(s)/\ln 2$ , while  $dp_\varepsilon/d\varepsilon = \varepsilon/(2s)$ . Hence

$$h'(\varepsilon) = \frac{\varepsilon}{\ln 2} \frac{\operatorname{artanh}(s)}{s} > 0 \quad (0 < \varepsilon < 1).$$

Differentiating once more, using  $ds/d\varepsilon = -\varepsilon/s$  and  $\varepsilon^2 = 1 - s^2$ ,

$$h''(\varepsilon) = \frac{\operatorname{artanh}(s) - s}{\ln 2 \, s^3} > 0 \quad (0 < \varepsilon < 1),$$

because  $\operatorname{artanh}(s) = s + s^3/3 + \dots > s$  for  $s \in (0, 1)$ ; the value extends continuously to  $h''(1) = 1/(3 \ln 2)$ . Finally  $p_0 = 0$  gives  $h(0) = 0$  and  $p_1 = \frac{1}{2}$  gives  $h(1) = 1$ .  $\square$

**Lemma 4** (Conditional affinity and Jensen). *Let  $X = (X_1, \dots, X_\ell) \sim G$  and, for fixed  $i$ , write  $Y = X_{-i}$  and  $p_Y = \Pr(X_i = 1 \mid Y)$ . Then*

$$\Phi(G, e_i) = \mathbb{E}_Y \left[ 2\sqrt{p_Y(1 - p_Y)} \right], \quad \mathbf{H}(X_i \mid X_{-i}) = \mathbb{E}_Y \left[ h\left(2\sqrt{p_Y(1 - p_Y)}\right) \right], \quad (64)$$

and consequently  $\mathbf{H}(X_i \mid X_{-i}) \geq h(\Phi(G, e_i))$ .

*Proof.* Identify  $u \in \mathbb{F}_2^\ell$  with  $(y, x_i)$ . For fixed  $y$  the two points  $(y, 0)$  and  $(y, 1)$  each contribute  $\sqrt{G(y, 0)G(y, 1)} = \Pr(Y = y)\sqrt{p_y(1 - p_y)}$  to  $\Phi(G, e_i)$ , giving the first identity in (64). For the second, note that  $b(p) := 2\sqrt{p(1 - p)}$  satisfies  $b(p) = b(1 - p)$  and, for  $p \leq \frac{1}{2}$ ,  $b(p) = \varepsilon$  exactly when  $p = p_\varepsilon$ ; since  $H_2(p) = H_2(1 - p)$  as well,  $H_2(p) = h(b(p))$  for every  $p \in [0, 1]$ . Averaging over  $Y$  gives the second identity. The inequality is Jensen's applied to the convex function  $h$  of Lemma 3.  $\square$

**Theorem 5** (Tensorization obstruction). *Fix  $\ell \geq 1$  and  $\varepsilon \in (0, 1)$ , and let  $G$  be a normalized configuration on  $\mathbb{F}_2^\ell$  satisfying (A1) and (A2). Then*

$$J_\ell(G) = \frac{\mathbf{H}(G)}{\ell} \geq h(\varepsilon) = H_2\left(\frac{1}{2} - \sqrt{\delta(1 - \delta)}\right), \quad \delta = \frac{1 - \varepsilon}{2},$$

the right-hand side being the first MRRW expression. The bound is attained by  $G_\varepsilon = \text{Bernoulli}(p_\varepsilon)^{\otimes \ell}$ .

*Proof.* Entropy and affinity are  $\text{GL}(\ell, 2)$ -equivariant: for  $T \in \text{GL}(\ell, 2)$  and  $G^T(u) = G(Tu)$  one has  $\mathbf{H}(G^T) = \mathbf{H}(G)$  and  $\Phi(G^T, v) = \Phi(G, Tv)$ . By (A1) the admissible shifts span  $\mathbb{F}_2^\ell$ , so they contain a basis  $v_1, \dots, v_\ell$ ; choosing  $T$  with  $Tv_i = e_i$  and replacing  $G$  by  $G^{T^{-1}}$ , we may assume  $e_1, \dots, e_\ell \in V$  without changing either side of the asserted inequality. By Lemma 2,  $\Phi(G, e_i) \geq \varepsilon$  for each  $i$ .

Let  $X \sim G$ . The chain rule together with the fact that conditioning does not increase entropy gives

$$\mathbf{H}(G) = \sum_{i=1}^{\ell} \mathbf{H}(X_i \mid X_1, \dots, X_{i-1}) \geq \sum_{i=1}^{\ell} \mathbf{H}(X_i \mid X_{-i}).$$

By Lemma 4 and the monotonicity of  $h$  (Lemma 3),

$$\mathbf{H}(X_i \mid X_{-i}) \geq h(\Phi(G, e_i)) \geq h(\varepsilon),$$

so  $\mathbf{H}(G) \geq \ell h(\varepsilon)$ . Dividing by  $\ell$  and applying (63) gives the stated form.

For sharpness, Bhattacharyya affinity is multiplicative over tensor products and  $\Phi(\cdot, 0) = 1$ , so  $\Phi(G_\varepsilon, e_i) = 2\sqrt{p_\varepsilon(1 - p_\varepsilon)} = \varepsilon$  for every  $i$ , while additivity of entropy gives  $\mathbf{H}(G_\varepsilon) = \ell H_2(p_\varepsilon) = \ell h(\varepsilon)$ .  $\square$

**Corollary 6.** *A level- $\ell$  dual bounds  $|C|^\ell$ , so a normalized configuration  $G$  yields the rate bound  $R(C) \leq J_\ell(G) + O(\log n/n)$  (Section 7.4). By Theorem 5, every admissible  $G$  gives  $J_\ell(G) \geq h(\varepsilon)$ . Hence, within the fixed single-orbit spectral ansatz, no choice of configuration yields an asymptotic rate bound below the first MRRW bound, and  $G_\varepsilon$  already attains it.*

### 7.6.3 Scope and relation to the cited construction

*Remark 7* (Scope). Theorem 5 is a statement about the configuration  $G$  alone, for fixed  $\ell$  and fixed even  $m_v$ , under (A1)–(A2). It does not constrain alternative sign functions  $\phi$ , spectral functions supported on several configuration orbits, other uses of the partial-Fourier dual families, or general dual feasible solutions; the higher-order hierarchy itself remains complete.

*Remark 8* (Sufficiency of the walk condition). Hypothesis (A2) is the sufficient condition (25) of [6], obtained there by requiring each term of the odd-cardinality expansion of  $M_m$  to be separately non-negative. A configuration violating (A2) could in principle still yield non-negative Fourier transforms through cancellation across that expansion; Theorem 5 does not address such configurations.

*Remark 9* (Finite- $m$  admissibility). Sharpness of the entropy–affinity inequality does not imply finite- $m$  admissibility: at  $G_\varepsilon$  one has  $W_m \leq \Phi^m = \varepsilon^m$ , whereas (A2) with  $c = 2^{2\ell-1}$  requires  $W_m \geq 2^{2\ell-1}\varepsilon^m$ . The equality configuration is thus the limiting object approached by the finite- $m$  construction rather than an admissible finite- $m$  point.

*Remark 10* (Equality). Strict convexity of  $h$  forces, in the equality case,  $2\sqrt{p_Y(1-p_Y)} = \varepsilon$  almost surely for every  $i$ , i.e. every conditional marginal lies in  $\{p_\varepsilon, 1-p_\varepsilon\}$ ; together with equality in the chain-rule step this is what identifies the tensor-product family recovered numerically. We do not claim a uniqueness statement, which is not needed for Corollary 6.

*Remark 11* (Relation to the hypotheses of [6]). Two hypotheses of Theorem 5 are weaker than those under which the construction of [6] is stated, so the obstruction covers that construction and somewhat more. First, the shifts verified there are those of Hamming weight one, i.e.  $V = \{e_1, \dots, e_\ell\}$ , which already spans  $\mathbb{F}_2^\ell$ ; the  $\text{GL}(\ell, 2)$  change of basis in the proof is therefore vacuous for that instantiation, and is retained so that (A1) may be assumed in the form in which it is naturally verified. Second, their construction requires  $G(u) > 0$  for every  $u \in \mathbb{F}_2^\ell$ , since the objective is estimated through a Stirling approximation of the fibre size  $|\text{config}_{n,\ell}^{-1}(nG)|$ , whereas Theorem 5 needs only  $G \geq 0$ : the affinity floor and the entropy chain are insensitive to vanishing coordinates. In particular the obstruction applies directly to configurations supported on a proper subset of  $\mathbb{F}_2^\ell$ , such as the vertex-uniform family, without the rounding adjustment used there.

## 7.7 Computational verification of the algebraic reductions

None of the computations in this section is used as part of the analytic proof. They test that the implementation computes the objects Theorem 5 concerns, which is where a hidden factor of  $m!$ , a  $2^m$ , or an ordered-versus-unordered pair convention would appear.

The literal sum (59) over multiplicity functions was enumerated in exact rational arithmetic and matched the paired form (61) as an exact rational identity, not to a tolerance, for all tested  $\ell \in \{1, 2, 3\}$ , all  $v \neq 0$  and all even  $m \leq 8$ .

All downstream computation takes place on the configuration lattice, whereas  $A_v$  acts on  $\mathbb{F}_2^{\ell \times n}$ . To confirm that this reduction loses nothing,  $A_v$  was constructed literally on the full vertex set and applied as a matrix, rather than assuming that  $A_v^m \Lambda(X)$  depends on  $X$  only through  $\text{config}(X)$ . In every tested instance the result was constant on the entire fibre  $\text{config}_{n,\ell}^{-1}(g_0)$  and agreed exactly with an independent configuration-lattice walk implementation (Supplementary Table 4). Blocklengths are necessarily small, the largest cases being  $4^5 = 1024$  and  $8^4 = 4096$  matrices; all  $v \neq 0$  were tested and all agree, with  $v = e_1$  shown for legibility.

The finite- $n$  walk count was compared with the leading term  $n^m W_m(G, v)$  of (60). The ratios converge to 1, and do so *from above*, so the finite- $n$  correction is positive:

|                     |          |        |        |        |        |
|---------------------|----------|--------|--------|--------|--------|
| $\ell = 2, m = 2 :$ | $n = 8$  | 16     | 32     | 64     | 128    |
|                     | 1.6667   | 1.3333 | 1.1667 | 1.0833 | 1.0417 |
| $\ell = 3, m = 2 :$ | $n = 12$ | 24     | 48     | 96     | 192    |
|                     | 1.7500   | 1.3750 | 1.1875 | 1.0938 | 1.0469 |

with the same behaviour at  $m = 4$  (for  $\ell = 3$ ,  $3.1067 \rightarrow 1.0998$  over the same sequence). The sign of this correction is not used anywhere in the argument, which requires only the  $o(n^m)$  bound, but it is recorded because the direction is not obvious a priori.

| $\ell$ | $n$ | $g_0$                    | $v$   | $m$ | matrix values | config. walk | agree |
|--------|-----|--------------------------|-------|-----|---------------|--------------|-------|
| 2      | 4   | (1, 1, 1, 1)             | $e_1$ | 2   | {8}           | 8            | yes   |
| 2      | 4   | (1, 1, 1, 1)             | $e_1$ | 4   | {128}         | 128          | yes   |
| 2      | 4   | (2, 1, 1, 0)             | $e_1$ | 2   | {8}           | 8            | yes   |
| 2      | 4   | (2, 1, 1, 0)             | $e_1$ | 4   | {104}         | 104          | yes   |
| 2      | 5   | (2, 1, 1, 1)             | $e_1$ | 2   | {11}          | 11           | yes   |
| 2      | 5   | (2, 1, 1, 1)             | $e_1$ | 4   | {245}         | 245          | yes   |
| 3      | 4   | (1, 1, 1, 1, 0, 0, 0, 0) | $e_1$ | 2   | {8}           | 8            | yes   |
| 3      | 4   | (1, 1, 1, 1, 0, 0, 0, 0) | $e_1$ | 4   | {128}         | 128          | yes   |
| 3      | 4   | (2, 1, 1, 0, 0, 0, 0, 0) | $e_1$ | 2   | {8}           | 8            | yes   |
| 3      | 4   | (2, 1, 1, 0, 0, 0, 0, 0) | $e_1$ | 4   | {104}         | 104          | yes   |

Supplementary Table 4: The operator  $A_v^m \Lambda$  computed literally on  $\mathbb{F}_2^{\ell \times n}$ , against the configuration-lattice walk count. “Matrix values” is the set of values taken by  $A_v^m \Lambda(X)$  over the entire fibre  $\text{config}_{n,\ell}^{-1}(g_0)$ ; that this set is a singleton in every row is precisely the assertion that the quantity depends on  $X$  only through  $\text{config}(X)$ . All entries are exact integers.

The inequality of Lemma 1 was checked on 480 exact rational cases, with largest observed ratio 0.4957. The extremal value  $\frac{1}{2}$  is attained already at  $\ell = 1$ ,  $m = 2$ , where there is a single pair and a single admissible multiplicity, so the coefficient comparison reduces exactly to  $\binom{2}{1}/4$ .

The entropy inequalities carry complete proofs, so the remaining checks are consistency tests rather than evidence. In 60-digit arithmetic, the closed forms of Lemma 3 agreed with numerical differentiation to  $10^{-57}$ ; the entropy chain of Theorem 5 held on 1600 random distributions across  $\ell = 1, \dots, 4$  with minimum gap  $-5.6 \times 10^{-61}$ , i.e. zero to working precision; and the conditional-affinity identity (64) held on 800 further distributions with maximum deviation  $4.6 \times 10^{-61}$ .

# Supplementary Methods: sign-uncertainty discovery and diagnostics

## Rigorous tripwires

Two lower bounds are evaluated at run time and used to invalidate the numerics rather than to prove anything. For  $d = 1$ ,  $s = +1$ ,

$$A_+(1) \geq \frac{1}{2(1 + \lambda_{\text{BCK}})} = 0.4107675 \dots,$$

with  $\lambda_{\text{BCK}} = -\min_x \sin x/x$  [BCK]. More generally, Theorem 2.1 of [CDG] gives

$$u_0 \geq \lambda(d, n),$$

where  $\lambda(d, n)$  is the smallest root of  $L_{\lfloor n/2 \rfloor + 1}^{(d/2-1)}$ . The latter follows from Gauss quadrature against  $e^{-u}u^{d/2-1}du$ : positivity of the quadrature weights forces a sign change at or beyond the first node. Any run reporting feasibility below either applicable bound is therefore rejected. During development, an early grid-only relaxation returned  $\rho = 0.398942 = 1/\sqrt{2\pi}$  in  $d = 1$ , immediately falsified by the BCK tripwire.

## Discovery campaign and numerical diagnosis

The initial search deliberately explored exact Fourier-eigenfunction mixtures with trainable Gaussian scales and contact locations, using local optimization and Adam-based learned proposals. Low-rank constructions reproduced the expected Bourgain–Clozel–Kahane controls and rapidly improved the objective, but the  $d = 1$  search plateaued near  $\rho = 0.5785$ , above the published Cohn–Gonçalves value.

The failure was diagnostic. When scales crowded or the imposed contact count was incompatible with the local root structure, the coefficient solve became extremely ill-conditioned. Candidate Adam trajectories that appeared to improve the objective entered systems with condition numbers of order  $10^{17}$ – $10^{18}$ . At this scale, forward accuracy could not be guaranteed in the floating-point formats used here. Multiprecision reevaluation showed that the apparent descent was dominated by numerical error rather than a reproducible improvement. Learned widths and contacts were therefore retained only as proposals, with a hard conditioning guard before verification.

The experiments also showed that the number of double contacts should not be rigidly tied to the number of basis functions. Decoupling the two made comparisons across basis sizes meaningful and showed that additional widths at fixed contact count contributed little. Multistart runs converged to distinct plateaus near 0.5785, 0.5929 and higher values, and verified evaluations could jump when the outermost root pattern changed. This behaviour is consistent with the discontinuity mechanism reported by Cohn and Gonçalves for Newton continuation at  $d \leq 2$ . We use this only as a diagnostic, not as an optimality theorem.

## Implementation details of the convex exchange solver

Three implementation choices proved load-bearing. First, bare LP feasibility with zero objective returns arbitrary vertices that hug zero at many grid points and can dip between them. Maximizing a relative margin instead produces strict positivity between most nodes and sharply reduces the number of exchange rounds. Second, column scaling by the supremum over the full grid is dominated by the far tail and can exclude genuine feasible solutions; rows are therefore equilibrated while columns are left unscaled. Third, multiple far tail anchors are nearly parallel after normalization and degrade the simplex. The tail is imposed instead through a single structural sign constraint on the leading coefficient.

At each bisection value  $u_0$ , the LP is solved on the current grid, the candidate derivative  $P'$  is evaluated at high precision, and all real critical points on the active interval are located. Any negative critical value is inserted as a new cutting row. The process stops only after the candidate passes the independent continuous scan to the prescribed numerical tolerance.

## Convergence status of the exchange loop

The oracle is Remez-like in mechanics but not in guarantee. The truncated Laguerre space  $V_N$  is not a Haar system on  $(0, \infty)$ :  $L_{2N}^{(\alpha)}(2u)$  belongs to  $V_N$  and has  $2N$  positive roots, exceeding the Haar budget of an  $(N + 1)$ -dimensional space. Consequently, uniqueness of the contact configuration and classical equioscillation arguments are not available.

The relevant framework is convex semi-infinite programming. Every finite grid gives an outer relaxation. Under compactness and a Slater point, the exchange sequence drives the maximum constraint violation to zero and the corresponding margins converge to the semi-infinite value. Together with monotonicity and continuity in  $u_0$ , this gives a numerical bracket for the fixed-degree boundary. This concerns optimality within the chosen finite space only. Validity of every quoted upper bound is established by the exact rational certificate and does not depend on convergence of the exchange loop.

## Contact degeneracy and leave-one-out reconstruction

The active contact count is discovered at the feasibility boundary. The certified  $d = 1$  degree-22, 30 and 38 constructions use  $m = 5, 7$  and  $9$  double contacts, respectively. The non-Haar structure can make the active set non-unique. At degree 38, the exchange oracle reports ten near-active contacts although the square reconstruction uses nine. Collocating all ten produces an admissible but irrelevant function whose outer sign change moves to  $u = 274$  ( $\rho = 9.34$ ).

We therefore perform leave-one-out reconstruction: each candidate contact is removed in turn, the remaining contacts are collocated in the square Laguerre system, and only reconstructions whose exact certificate agrees with the convex estimate are retained. At degree 38, four of the ten subsets certify the same 0.5725887 value to nine digits. The same repair is required at  $d = 2$ . The multiplicity of successful subsets is consistent with a degenerate optimal facet and is not interpreted as uniqueness of the extremizer.

## Exact Sturm implementation

After rational reconstruction,

$$P(u) = \prod_i (u - \tau_i)^2 R(u),$$

and only the sign of  $R$  on the ray remains to be certified. We build a primitive pseudo-remainder sequence over  $\mathbb{Z}$ , using only positive rescalings so that Sturm sign-variation counts are preserved. This controls coefficient growth substantially better than naive rational Euclidean remainders, whose bit lengths become impractical by degree 38.

The verifier checks the exact Laguerre-to-monomial expansion, zero remainder after division by every squared contact factor, the sign of the leading coefficient, the exact value  $R(\bar{u}) > 0$ , and equality of Sturm sign-variation counts at  $\bar{u}$  and  $+\infty$ . No grid, floating-point root finder or interval sampling appears in the final admissibility decision.

## Relation to the non-Gaussian search

A numerical A/B analysis explains why the earlier Gaussian-mixture search did not improve the final polynomial construction at small  $d$ . Both spaces are evaluated through the same convex machinery, so the comparison concerns function spaces rather than optimizer quality. Projecting individual mixture directions onto the degree-22 polynomial space over the active region gives relative sup-norm residuals of approximately

$$\begin{aligned} 1.3 \times 10^{-15} \ (a = 1.5), \quad 3.0 \times 10^{-10} \ (a = 2.5), \quad 9.2 \times 10^{-7} \ (a = 4), \\ 6.3 \times 10^{-5} \ (a = 6), \quad 1.6 \times 10^{-3} \ (a = 10). \end{aligned}$$

Wider directions are genuinely distinct, but their optimized coefficients are of order  $10^{-3}$  and did not improve the objective beyond the numerical noise floor. This containment analysis is diagnostic only and is not used in the certificate.

## Higher-dimensional asymptotic diagnostic

Using the same convex machinery, the scale-free quantity  $\rho\sqrt{2\pi/d}$  along one degree–dimension path was

1.435267 ( $d=1$ ), 1.340341 ( $d=2$ ), 1.212474 ( $d=4$ ), 1.079183 ( $d=8$ ), 1.026193 ( $d=12$ ), 1.005480 ( $d=16$ ).

The sequence approaches the sublinear-degree asymptote 1 rather than crossing it. This is only a diagnostic path, not an asymptotic theorem. At  $d = 8$ , increasing degree from 14 to 38 changes the reported value non-monotonically over a range of approximately  $5.6 \times 10^{-4}$  even though the exact fixed-degree optima are nested. We therefore interpret the spread as a numerical resolution limit of the discovery oracle.

## Degree ceiling of the numerical stage

The current floating-point exchange solver is reliable only to polynomial degree approximately 38. Beyond this, the raw polynomial values span an extreme dynamic range: at degree 54,  $P$  reaches about  $9.7 \times 10^{73}$  at the far end of its own grid. Row equilibration cannot repair the resulting column-direction scaling, and the LP begins returning points that violate its own constraints.

Near the ceiling, the finite grid can also acquire a recession direction that is non-negative on all nodes while dipping between them, producing an “unbounded” rather than infeasible LP, and the cut pool can accumulate near-duplicate rows across the bisection. We cap the margin variable, refresh the cut pool when necessary, and discard runs whose bisection collides with a rigorous lower tripwire. Multiplication by the positive factor  $e^{-u}$  reduces the dynamic range but relocates the failure to underflow in the far tail. Higher-dimensional asymptotic work will therefore require higher precision or iterative/exact refinement inside the optimizer. The certificates reported here are unaffected because all lie at degree at most 38 and are validated by the exact rational stage.

Supplementary Table 5: Vanilla discovery and certification results  
( $\epsilon = 0$ ).

| $k$ | Discovery      | $Ch_{\text{disc}}$ | $Ch_{\text{cert}}$ | Degree | Certification                    | $Fr_{\text{ceiling}}$ |
|-----|----------------|--------------------|--------------------|--------|----------------------------------|-----------------------|
| 20  | 3.128345991229 | 36                 | 33                 | 60     | 3.127557950496731089199291       | 0,992                 |
| 21  | 3.170106837298 | 36                 | 30                 | 60     | 3.169687744135085908710644       | 0,992                 |
| 22  | 3.210448579010 | 36                 | 29                 | 60     | 3.209791425249774301029605       | 0,991                 |
| 23  | 3.249812450248 | 36                 | 30                 | 60     | 3.249018102873413922834334       | 0,991                 |
| 24  | 3.287477573848 | 36                 | 32                 | 60     | 3.286239994066373110168097       | 0,991                 |
| 25  | 3.322508727072 | 36                 | 28                 | 60     | 3.322151135960770560379734       | 0,991                 |
| 26  | 3.358033109070 | 36                 | 30                 | 60     | 3.356593408977485398051369       | 0,991                 |
| 27  | 3.390797412499 | 36                 | 30                 | 60     | 3.389882327187024762678030       | 0,990                 |
| 28  | 3.423550501369 | 36                 | 29                 | 60     | 3.421891175367646121306316       | 0,990                 |
| 29  | 3.453912899535 | 36                 | 33                 | 60     | 3.452879774954988104889254       | 0,990                 |
| 30  | 3.484519905740 | 42                 | 35                 | 60     | 3.482920804908356499173898       | 0,990                 |
| 31  | 3.514154741570 | 42                 | 35                 | 60     | 3.512405412198264166419745       | 0,990                 |
| 32  | 3.542948097289 | 42                 | 34                 | 60     | 3.540478095710103535569937       | 0,990                 |
| 33  | 3.569408303647 | 42                 | 35                 | 60     | 3.568168082140816922057409       | 0,990                 |
| 34  | 3.595856014881 | 42                 | 36                 | 60     | 3.594862215205947831119258       | 0,989                 |
| 35  | 3.622637140460 | 42                 | 36                 | 60     | 3.620729784609115150023189       | 0,989                 |
| 36  | 3.649049376929 | 42                 | 36                 | 60     | 3.645939449133237314781111       | 0,989                 |
| 37  | 3.672649021837 | 42                 | 36                 | 60     | 3.670749358191800897953616       | 0,989                 |
| 38  | 3.697450514132 | 42                 | 38                 | 60     | 3.694212659617672335430314       | 0,989                 |
| 39  | 3.719968833432 | 42                 | 37                 | 60     | 3.718057515927273283488455       | 0,989                 |
| 40  | 3.743412144728 | 48                 | 45                 | 60     | 3.740850129189844929384497015046 | 0,989                 |
| 41  | 3.765478132249 | 48                 | 39                 | 60     | 3.763326248211363188195226978670 | 0,989                 |
| 42  | 3.786443640815 | 48                 | 40                 | 60     | 3.784836398710852090166363045263 | 0,989                 |
| 43  | 3.808393391526 | 48                 | 44                 | 60     | 3.806112899332449843513725348999 | 0,988                 |
| 44  | 3.831089249576 | 48                 | 41                 | 60     | 3.826734286114500842550587364787 | 0,988                 |
| 45  | 3.850548910793 | 48                 | 44                 | 60     | 3.847582396814691842088947101999 | 0,988                 |
| 46  | 3.870708253098 | 48                 | 45                 | 60     | 3.867062274460791570055053587204 | 0,988                 |
| 47  | 3.890981548956 | 48                 | 42                 | 60     | 3.887050039712155465458257739811 | 0,988                 |
| 48  | 3.911606976491 | 48                 | 40                 | 60     | 3.906150910904812934847425559032 | 0,988                 |
| 49  | 3.927617139801 | 48                 | 42                 | 60     | 3.925218177946628714063038556246 | 0,988                 |
| 50  | 3.943707217858 | 40                 | 33                 | 50     | 3.942445971786197344558734862254 | 0,988                 |
| 51  | 3.962201156342 | 40                 | 34                 | 50     | 3.961742909943176882384440577615 | 0,988                 |
| 52  | 3.980085920896 | 40                 | 33                 | 50     | 3.979361743352805048779887827229 | 0,988                 |
| 53  | 3.997392090465 | 40                 | 35                 | 50     | 3.996569546603520792189153865045 | 0,988                 |
| 54  | 4.014417349448 | 40                 | 33                 | 50     | 4.013421392728337468685880863364 | 0,987                 |
| 55  | 4.031158093047 | 40                 | 33                 | 50     | 4.029930315883373661648075171587 | 0,987                 |
| 56  | 4.048740196154 | 40                 | 32                 | 50     | 4.047211183498640314770985656114 | 0,987                 |
| 57  | 4.064675943291 | 40                 | 33                 | 50     | 4.063658911265930366015897090005 | 0,987                 |
| 58  | 4.080897241650 | 40                 | 33                 | 50     | 4.079662539803768949658734868762 | 0,987                 |
| 59  | 4.096107675723 | 40                 | 33                 | 50     | 4.094929027520673522821234872044 | 0,987                 |
| 60  | 4.111224221271 | 40                 | 32                 | 50     | 4.110244679900046307236517232052 | 0,987                 |
| 61  | 4.123180413889 | 40                 | 37                 | 50     | 4.122593396892570578487592704836 | 0,986                 |
| 62  | 4.141415806845 | 40                 | 35                 | 50     | 4.140421271440397388299091151343 | 0,987                 |
| 63  | 4.151452852615 | 40                 | 37                 | 50     | 4.150583732266163623322424042114 | 0,986                 |
| 64  | 4.170843665400 | 40                 | 33                 | 50     | 4.169733035246490856721916297730 | 0,987                 |
| 65  | 4.185117958392 | 40                 | 36                 | 50     | 4.184200456291009710938969100824 | 0,987                 |
| 66  | 4.199258111085 | 40                 | 34                 | 50     | 4.198212352631011423270651543778 | 0,987                 |
| 67  | 4.212801300650 | 40                 | 32                 | 50     | 4.211734375259938467462491656228 | 0,987                 |
| 68  | 4.226913707347 | 40                 | 34                 | 50     | 4.225671537232203572914298251005 | 0,987                 |
| 69  | 4.240641054970 | 40                 | 27                 | 50     | 4.239530843403299471806859841686 | 0,987                 |
| 70  | 4.252043044524 | 40                 | 34                 | 50     | 4.250709595458423509274271903750 | 0,986                 |

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Supplementary Table 5 – continued from previous page

| $k$ | Discovery      | $Ch_{\text{disc}}$ | $Ch_{\text{cert}}$ | Degree | Certification                    | $Fr_{\text{ceiling}}$ |
|-----|----------------|--------------------|--------------------|--------|----------------------------------|-----------------------|
| 71  | 4.262077283275 | 40                 | 34                 | 50     | 4.260781471180062527612810820600 | 0,985                 |
| 72  | 4.280170671493 | 40                 | 31                 | 50     | 4.278486659483071740524507466677 | 0,987                 |
| 73  | 4.291652331079 | 40                 | 36                 | 50     | 4.290439906377909296284295483681 | 0,986                 |
| 74  | 4.302039567418 | 40                 | 32                 | 50     | 4.300766657962825879583571273685 | 0,986                 |
| 75  | 4.315037290653 | 40                 | 36                 | 50     | 4.313819804163989594408982684260 | 0,986                 |
| 76  | 4.330839005745 | 40                 | 33                 | 50     | 4.328884563460976817456708107930 | 0,986                 |
| 77  | 4.342588581101 | 40                 | 32                 | 50     | 4.340903139563253839977695048758 | 0,986                 |
| 78  | 4.353490658660 | 40                 | 30                 | 50     | 4.351450026525314703393729301722 | 0,986                 |
| 79  | 4.366332032220 | 40                 | 25                 | 50     | 4.363547069467314352935761632810 | 0,986                 |
| 80  | 4.375870655363 | 50                 | 36                 | 50     | 4.375305575875195280796809950641 | 0,986                 |
| 81  | 4.388391084261 | 50                 | 34                 | 50     | 4.387822836632632171272303086405 | 0,986                 |
| 82  | 4.394774870701 | 50                 | 40                 | 50     | 4.394326901844365176252467034760 | 0,985                 |
| 83  | 4.401879201937 | 50                 | 43                 | 50     | 4.401472671518373508912545234614 | 0,984                 |
| 84  | 4.417373705040 | 50                 | 40                 | 50     | 4.416953094949757411430914506920 | 0,985                 |
| 85  | 4.346441814015 | 50                 | 41                 | 50     | 4.346494488379658185500563460563 | 0,967                 |
| 86  | 4.431105148849 | 50                 | 42                 | 50     | 4.430626969481280584146214724073 | 0,983                 |
| 87  | 4.452908961195 | 50                 | 40                 | 50     | 4.451913396990625293127698559451 | 0,985                 |
| 88  | 4.464302979612 | 50                 | 41                 | 50     | 4.463426157257708417355375315939 | 0,986                 |
| 89  | 4.472355081593 | 50                 | 42                 | 50     | 4.471988530980810754826227770313 | 0,985                 |
| 90  | 4.487119371775 | 50                 | 37                 | 50     | 4.485910734210900551045280545626 | 0,986                 |
| 91  | 4.496698527592 | 50                 | 37                 | 50     | 4.495645093532629510509939471985 | 0,986                 |
| 92  | 4.505829693079 | 50                 | 39                 | 50     | 4.504547705976628150465431574044 | 0,985                 |
| 93  | 4.518347339954 | 50                 | 36                 | 50     | 4.516339208377947949080443322071 | 0,986                 |
| 94  | 4.528059040470 | 50                 | 29                 | 50     | 4.526635937140459614406972086093 | 0,986                 |
| 95  | 4.533806772177 | 50                 | 41                 | 50     | 4.532436225927372584839614117923 | 0,985                 |
| 96  | 4.545404066027 | 50                 | 26                 | 50     | 4.544084221152406068483876743217 | 0,985                 |
| 97  | 4.554320603674 | 50                 | 31                 | 50     | 4.553588807459129165253574203320 | 0,985                 |
| 98  | 4.565278947850 | 50                 | 28                 | 50     | 4.564004527939718801984156460497 | 0,985                 |
| 99  | 4.509354418861 | 50                 | 35                 | 50     | 4.508561572862963088224836065494 | 0,971                 |
| 100 | 4.585389371115 | 50                 | 39                 | 50     | 4.583559169120379861185947389471 | 0,985                 |
| 100 | 4.582636529114 | 50                 | 39                 | 50     | 4.581993985670381852509553180491 | 0,985                 |
| 130 | 4.822341520115 | 50                 | 13                 | 50     | 4.808713824125718836287734368717 | 0,980                 |
| 150 | 4.961245663975 | 50                 | 6                  | 50     | 4.955366718182981077061200798091 | 0,982                 |
| 200 | 5.090813348651 | 50                 | 4                  | 60     | 5.087280688405278804164639536817 | 0,955                 |
| 220 | 5.258389442417 | 50                 | 12                 | 60     | 5.247997786415732591367214256322 | 0,969                 |
| 250 | 5.384166994782 | 50                 | –                  | –      | –                                | –                     |
| 350 | 5.610666510192 | 50                 | –                  | –      | –                                | –                     |
| 500 | 6.080822857000 | 50                 | –                  | –      | –                                | –                     |

*Note.* Values  $k = 25$  and  $k > 30$  have certified values above Bogaert’s Krylov subspace method and the published numbers on the Polymath8b Wiki.  $Ch_{\text{disc}}$ ; amount of channels in the discovery,  $Ch_{\text{cert}}$ ; amount of channels in the certification (after pruning).  $Fr_{\text{ceiling}}$ ; the gap to the Cauchy Schwartz ceiling.

Supplementary Table 6: Discovery and certification results for  $\varepsilon = 1/25$ .

| $k$ | Discovery      | $Ch_{\text{disc}}$ | $Ch_{\text{cert}}$ | Degree | Certification                    |
|-----|----------------|--------------------|--------------------|--------|----------------------------------|
| 20  | 3.210936069786 | 36                 | 33                 | 60     | 3.210366921521749737183691731395 |
| 21  | 3.252680335644 | 36                 | 30                 | 60     | 3.252298834127242084124289323548 |
| 22  | 3.292779266722 | 36                 | 29                 | 60     | 3.292126369692432458798975243945 |
| 23  | 3.331173442193 | 36                 | 30                 | 60     | 3.330322094752801530355410455276 |
| 24  | 3.368036770635 | 36                 | 30                 | 60     | 3.366908489372213299919519038109 |

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Supplementary Table 6 – continued from previous page

| $k$ | Discovery      | $Ch_{\text{disc}}$ | $Ch_{\text{cert}}$ | Degree | Certification                    |
|-----|----------------|--------------------|--------------------|--------|----------------------------------|
| 25  | 3.402371365754 | 36                 | 29                 | 60     | 3.402010619443212328887383163464 |
| 26  | 3.437028098636 | 36                 | 31                 | 60     | 3.436157517278351176844746714320 |
| 27  | 3.468827024244 | 36                 | 29                 | 60     | 3.468320542253183831988945583237 |
| 28  | 3.500415876452 | 36                 | 30                 | 60     | 3.499444954997692030882286887925 |
| 29  | 3.531186516502 | 36                 | 32                 | 60     | 3.530010877716944327578014764437 |
| 30  | 3.561700179340 | 42                 | 39                 | 60     | 3.559458746545551314730876537135 |
| 31  | 3.590689744972 | 42                 | 40                 | 60     | 3.588557568639265632860385653635 |
| 32  | 3.618732963879 | 42                 | 38                 | 60     | 3.616127200711337138001580976885 |
| 33  | 3.644253262445 | 42                 | 36                 | 60     | 3.642965802069061441160922851345 |
| 34  | 3.671450599407 | 42                 | 39                 | 60     | 3.668624924763460294356636286090 |
| 35  | 3.696250084782 | 42                 | 40                 | 60     | 3.694175287948157320412197179779 |
| 36  | 3.720908518093 | 42                 | 39                 | 60     | 3.718279485810969785557788969075 |
| 37  | 3.744273426646 | 42                 | 38                 | 60     | 3.742226163630752703816176588194 |
| 38  | 3.768134969368 | 42                 | 41                 | 60     | 3.766021610226650587509688536386 |
| 39  | 3.790213730678 | 42                 | 39                 | 60     | 3.788245621497978381906985116415 |
| 40  | 3.813521764234 | 48                 | 45                 | 60     | 3.810310054297441303862285913738 |
| 41  | 3.833616627961 | 48                 | 41                 | 60     | 3.831637106843201917337701123609 |
| 42  | 3.855906628402 | 48                 | 45                 | 60     | 3.852600769154481058291704449249 |
| 43  | 3.876753550031 | 48                 | 42                 | 60     | 3.873483797869255723761005318927 |
| 44  | 3.895577913875 | 48                 | 46                 | 60     | 3.893301604331862091582323108956 |
| 45  | 3.915649733022 | 48                 | 45                 | 60     | 3.913474652772154201185956494902 |
| 46  | 3.936937139139 | 48                 | 43                 | 60     | 3.932552485373460445967272733645 |
| 47  | 3.956703033242 | 48                 | 46                 | 60     | 3.952079215054289455935133148337 |
| 48  | 3.975823596168 | 48                 | 44                 | 60     | 3.970095831528526602285865965250 |
| 49  | 3.991450730929 | 48                 | 43                 | 60     | 3.988676716542276674807876834435 |
| 50  | 4.006026386327 | 40                 | 32                 | 50     | 4.005637471340251334822223632678 |
| 51  | 4.021337913283 | 40                 | 21                 | 50     | 4.020552375066701497376356751144 |
| 52  | 4.040635119278 | 40                 | 31                 | 50     | 4.040258681163316069160504782304 |
| 53  | 4.058215551546 | 40                 | 24                 | 50     | 4.057203993845967521227902017906 |
| 54  | 4.072969225060 | 40                 | 28                 | 50     | 4.072322886063011063853846972776 |
| 55  | 4.084802202882 | 40                 | 18                 | 50     | 4.084085711139175414397956372460 |
| 56  | 4.100028783679 | 40                 | 26                 | 50     | 4.099206855766737568002165290524 |
| 57  | 4.115585844764 | 40                 | 23                 | 50     | 4.115039486353743806900256901164 |
| 58  | 4.135152191312 | 40                 | 25                 | 50     | 4.134242300759367593439951066209 |
| 59  | 4.149187895198 | 40                 | 30                 | 50     | 4.148472609734504851305763152187 |
| 60  | 4.158966932600 | 40                 | 19                 | 50     | 4.157998757293132279032736615835 |
| 61  | 4.175975507879 | 40                 | 31                 | 50     | 4.175077906373857205061979626025 |
| 62  | 4.189862734973 | 40                 | 32                 | 50     | 4.188904070211809938022792508286 |
| 63  | 4.210516336726 | 40                 | 33                 | 50     | 4.208857616045980796408818491228 |
| 64  | 4.219415161437 | 40                 | 30                 | 50     | 4.218124437987688545452024305223 |
| 65  | 4.238501800615 | 40                 | 36                 | 50     | 4.237384124837738623814213247461 |
| 66  | 4.252120545298 | 40                 | 29                 | 50     | 4.250793037094066225539732443814 |
| 67  | 4.261402748608 | 40                 | 22                 | 50     | 4.259777456025320664617972564576 |
| 68  | 4.270379618942 | 40                 | 26                 | 50     | 4.269124177371744769988813084968 |
| 69  | 4.283984925481 | 40                 | 5                  | 50     | 4.277274261946282382619508640328 |
| 70  | 4.301560117306 | 40                 | 34                 | 50     | 4.298733418769167808786843497725 |
| 71  | 4.316885083242 | 40                 | 31                 | 50     | 4.314413443106201423239218270654 |
| 72  | 4.329503334605 | 40                 | 35                 | 50     | 4.327034144580560001136968528718 |
| 73  | 4.334908974772 | 40                 | 30                 | 50     | 4.332768096731747934119777486576 |
| 74  | 4.342039995848 | 40                 | 28                 | 50     | 4.339799356192207942523849650386 |
| 75  | 4.358917246793 | 40                 | 32                 | 50     | 4.356375192119249086787668845356 |
| 76  | 4.375902255118 | 40                 | 31                 | 50     | 4.372974259197422394873126180262 |
| 77  | 4.388092689456 | 40                 | 35                 | 50     | 4.385396831879956294549952588437 |

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Supplementary Table 6 – continued from previous page

| $k$ | Discovery      | $Ch_{\text{disc}}$ | $Ch_{\text{cert}}$ | Degree | Certification                    |
|-----|----------------|--------------------|--------------------|--------|----------------------------------|
| 78  | 4.399184535712 | 40                 | 22                 | 50     | 4.395990292345199428975680930752 |
| 79  | 4.407214416241 | 40                 | 30                 | 50     | 4.404183984805812816254054716642 |
| 80  | 4.195941988767 | 50                 | 40                 | 50     | 4.197608268366179620141468586531 |
| 81  | 4.075117366264 | 50                 | 35                 | 50     | 4.093066546579362186151057435374 |
| 82  | 4.192978689328 | 50                 | 43                 | 50     | 4.195259401348774146934234241013 |
| 83  | 4.425644201176 | 50                 | 35                 | 50     | 4.424852131550489892382995112584 |
| 84  | 4.096825473475 | 50                 | 31                 | 50     | 4.099802658570555784696980423382 |
| 85  | 4.407475205627 | 50                 | 39                 | 50     | 4.406760295616284828171584604987 |
| 86  | 4.457072509121 | 50                 | 31                 | 50     | 4.458412761836136236841801895336 |
| 87  | 4.453265265989 | 50                 | 45                 | 50     | 4.453060114331590189551387134208 |
| 88  | 4.344434571908 | 50                 | 40                 | 50     | 4.344422869750479566119971669552 |
| 89  | 4.410538854701 | 50                 | 39                 | 50     | 4.417151906430070350388796841891 |
| 90  | 4.470745463723 | 50                 | 45                 | 50     | 4.471149582958362588083170625461 |
| 91  | 4.175121841242 | 50                 | 41                 | 50     | 4.176550564513205748419816639538 |
| 92  | 4.381872185197 | 50                 | 45                 | 50     | 4.389037025811565058867953246127 |
| 93  | 4.312917570269 | 50                 | 44                 | 50     | 4.314963190359444288509182754059 |
| 94  | 4.272652151109 | 50                 | 38                 | 50     | 4.273951035321736008358360794466 |
| 95  | 4.420217255929 | 50                 | 38                 | 50     | 4.419415306457317943090466947241 |
| 96  | 4.311614804044 | 50                 | 34                 | 50     | 4.311428453792439589002649755954 |
| 97  | 4.203613002767 | 50                 | 35                 | 50     | 4.206802724014883123574465837334 |
| 98  | 4.217114551052 | 50                 | 39                 | 50     | 4.225342000024770175903490961201 |
| 99  | 4.548298852384 | 50                 | 39                 | 50     | 4.546923190682762183944464394853 |
| 100 | 4.496536005802 | 50                 | 42                 | 50     | 4.495512133810554663324778495290 |
| 100 | 4.466983632765 | 50                 | 40                 | 50     | 4.466802522812985639158255185770 |
| 130 | 4.847986562151 | 50                 | 10                 | 50     | 4.844156480287745811911814959650 |
| 150 | 4.977475423203 | 50                 | 8                  | 50     | 4.961947334363613966866801798777 |
| 200 | 5.094346768152 | 50                 | 7                  | 60     | 5.087393690044834674577815885368 |
| 220 | 4.978911765299 | 50                 | 8                  | 60     | 4.976764489557045481952499319142 |
| 220 | 4.978911765299 | 50                 | 8                  | 60     | 4.976764489557045481952499319142 |
| 400 | 5.381948554900 | 50                 | -                  | -      | -                                |
| 500 | 4.512115207075 | 50                 | -                  | -      | -                                |

Note.  $Ch_{\text{disc}}$ ; amount of channels in the discovery,  $Ch_{\text{cert}}$ ; amount of channels in the certification (after pruning).

Supplementary Table 7: **Certified enlarged-support sweep underlying the  $\varepsilon$ -crossover analysis.** For each  $k$ , the table reports the certified lower bound obtained for the corresponding enlarged-support parameter  $\varepsilon$ , together with its within- $k$  ranking among the tested values. Rank 1 denotes the strongest certified value at that  $k$ .

| $k$ | $\varepsilon$ | Certified value                  | Rank |
|-----|---------------|----------------------------------|------|
| 50  | 0             | 3.942127742490453604663761985558 | 5    |
| 50  | 1/5           | 3.712509239545921204317529071368 | 8    |
| 50  | 1/6           | 3.764504991324559702999466932710 | 7    |
| 50  | 1/8           | 3.874384180153464414710468398524 | 6    |
| 50  | 1/12          | 3.959088983406359921645433483059 | 4    |
| 50  | 1/16          | 3.988718438122563563392522885252 | 3    |
| 50  | 1/25          | 3.997997740337627844075583884562 | 1    |
| 50  | 1/50          | 3.994094475900581244335373138334 | 2    |
| 60  | 0             | 4.107867690167866055301464199575 | 4    |
| 60  | 1/5           | 3.823138358991218145879633189922 | 8    |
| 60  | 1/6           | 3.901108973124642823154420824199 | 7    |
| 60  | 1/8           | 4.008638429188118037606214685517 | 6    |
| 60  | 1/12          | 4.078919312220589574136195730201 | 5    |
| 60  | 1/16          | 4.137429406260009516632280091156 | 3    |
| 60  | 1/25          | 4.154030210268053992886263760513 | 2    |
| 60  | 1/50          | 4.159671531351647672765832327114 | 1    |
| 70  | 0             | 4.250768996013457035617670337865 | 4    |
| 70  | 1/5           | 3.932874621775866265776104392099 | 8    |
| 70  | 1/6           | 3.992467605762845919987563382701 | 7    |
| 70  | 1/8           | 4.097181032238870681623243665487 | 6    |
| 70  | 1/12          | 4.203873349967048026646358848602 | 5    |
| 70  | 1/16          | 4.264751558230381563655123754834 | 3    |
| 70  | 1/25          | 4.284181954725759229768882824961 | 2    |
| 70  | 1/50          | 4.298346167280743718826612134536 | 1    |
| 80  | 0             | 4.365076286689542864035828077954 | 4    |
| 80  | 1/5           | 4.010739510791432389610499056194 | 8    |
| 80  | 1/6           | 4.046713573456469114235727637437 | 7    |
| 80  | 1/8           | 4.198433911029465756611969501053 | 6    |
| 80  | 1/12          | 4.300715218425578908913133664543 | 5    |
| 80  | 1/16          | 4.368846840474345630380842204395 | 3    |
| 80  | 1/25          | 4.397641633229676353759931557760 | 2    |
| 80  | 1/50          | 4.409932699888494327027154402066 | 1    |
| 90  | 0             | 4.475394178930606300302930028315 | 3    |
| 90  | 1/5           | 4.051193459293446704008504024550 | 7    |
| 90  | 1/6           | 4.116620710367156118853873413986 | 6    |
| 90  | 1/8           | 4.026133861058892236358880667882 | 8    |
| 90  | 1/12          | 4.277640742111484913450494485256 | 5    |
| 90  | 1/16          | 4.426300947743744344607385400548 | 4    |
| 90  | 1/25          | 4.502108322122445717967006768742 | 2    |
| 90  | 1/50          | 4.514812668499281034377537700573 | 1    |
| 100 | 0             | 4.571865161766055920977957342516 | 2    |
| 100 | 1/5           | 4.131174966202319094573252086929 | 8    |
| 100 | 1/6           | 4.228661930221300547990282980216 | 7    |
| 100 | 1/8           | 4.376164198382763272282279872439 | 6    |
| 100 | 1/12          | 4.455747109584966143321615617427 | 5    |

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Supplementary Table 7 – continued from previous page

| $k$ | $\varepsilon$ | Certified value                  | Rank |
|-----|---------------|----------------------------------|------|
| 100 | 1/16          | 4.483072604316316381523061824105 | 4    |
| 100 | 1/25          | 4.520592767325824976955743719607 | 3    |
| 100 | 1/50          | 4.600398252159768512024806128765 | 1    |

## 8 Asymptotics of the single-channel rational family

We now explain analytically why the single-channel rational trial functions used in the large- $k$  computations remain within a bounded distance of the logarithmic scale  $\log k$ .

For  $k \geq 2$ , write

$$n := k - 1,$$

and consider the symmetric trial function

$$F_{k,c}(t_1, \dots, t_k) = \mathbf{1}_{\{t_i \geq 0, \sum_{i=1}^k t_i \leq 1\}} \prod_{i=1}^k \frac{1}{c + nt_i}, \quad c > 0. \quad (65)$$

Its overall scalar normalization is immaterial.

The exact probabilistic reduction established in Section ?? gives

$$R(F_{k,c}) = k \frac{N_{k,c}}{D_{k,c}}, \quad (66)$$

where

$$w(t) = \frac{1}{(c + nt)^2}, \quad m_0 = \int_0^1 w(t) dt = \frac{1}{c(c + n)},$$

and, if  $T_1, \dots, T_n$  are independent with density

$$\frac{w(t)}{m_0} = \frac{c(c + n)}{(c + nt)^2}, \quad 0 \leq t \leq 1,$$

and

$$S_n := T_1 + \dots + T_n,$$

then

$$N_{k,c} = \mathbb{E}[G(1 - S_n)^2; S_n < 1], \quad (67)$$

$$D_{k,c} = \mathbb{E}[H(1 - S_n); S_n < 1], \quad (68)$$

with

$$G(\rho) = \frac{1}{n} \log \left( 1 + \frac{n\rho}{c} \right), \quad (69)$$

$$H(\rho) = \frac{1}{n} \left( \frac{1}{c} - \frac{1}{c + n\rho} \right). \quad (70)$$

The relevant scale for  $c$  is

$$\frac{1}{c_k} = \log(k - 1) + a, \quad a \in \mathbb{R} \text{ fixed}. \quad (71)$$

The numerical optimization of the rational family independently exhibits this  $c_k \asymp 1/\log k$  scaling. The result below identifies the limiting  $O(1)$  defect from  $\log k$  for every fixed shift  $a$ .

### 8.1 The limiting stable law

We first isolate the random variable governing the boundary of the simplex. Define

$$X_n := \frac{1 - S_n}{c}. \quad (72)$$

Thus

$$S_n < 1 \iff X_n > 0.$$

It is convenient to rescale a single summand by writing

$$V := \frac{T}{c}.$$

A direct change of variables gives the density

$$q_n(v) = \frac{c+n}{(1+nv)^2}, \quad 0 \leq v \leq \frac{1}{c}. \quad (73)$$

Hence

$$X_n = \frac{1}{c} - \sum_{j=1}^n V_j, \quad (74)$$

where  $V_1, \dots, V_n$  are independent with density (73).

We use the following normalization of the totally right-skewed 1-stable law.

**Definition 12** (The limiting 1-stable variable). *Let  $\Lambda$  be the infinitely divisible random variable with characteristic function*

$$\mathbb{E}e^{is\Lambda} = \exp \left\{ \int_0^\infty (e^{isx} - 1 - isx\mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2} \right\}. \quad (75)$$

*This is a spectrally positive stable law of index 1, in the Lévy-Khintchine normalization used below. Its law has a continuous, everywhere positive density on  $\mathbb{R}$ .*

Because  $\Lambda$  has only positive jumps, its Laplace transform is finite in the damping direction. The following elementary computation is used repeatedly: it controls the left tail of  $\Lambda$ , supplies the integrability needed in Lemma 22, and provides a one-dimensional quadrature route to the constants  $P_a$ ,  $L_a$ ,  $Q_a$  introduced below.

**Lemma 13** (Laplace transform and left tail of  $\Lambda$ ). *For every  $\theta > 0$ ,*

$$\mathbb{E}e^{-\theta\Lambda} = \exp\{\theta \log \theta - \theta + \gamma_E \theta\}, \quad (76)$$

*where  $\gamma_E$  is the Euler–Mascheroni constant. To avoid collision we use  $\gamma_E$  instead of  $\gamma_h$  from Section 6.25. Consequently, for all  $y \geq 0$ ,*

$$\mathbb{P}(\Lambda \leq -y) \leq \exp\{-e^{y-\gamma_E}\}, \quad (77)$$

*and in particular*

$$\kappa := \mathbb{E}[(-\Lambda)^+] < \infty. \quad (78)$$

*Proof.* Analytic continuation of (75) to  $s = i\theta$  gives

$$J(\theta) := \log \mathbb{E}e^{-\theta\Lambda} = \int_0^\infty (e^{-\theta x} - 1 + \theta x\mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2},$$

the integral being absolutely convergent for  $\theta > 0$ . Differentiating under the integral sign,

$$J'(\theta) = \int_0^1 \frac{1 - e^{-\theta x}}{x} dx - \int_1^\infty \frac{e^{-\theta x}}{x} dx = (\log \theta + \gamma_E + E_1(\theta)) - E_1(\theta) = \log \theta + \gamma_E,$$

where  $E_1$  is the exponential integral and we used the standard identity  $\int_0^z (1 - e^{-t})t^{-1}dt = \gamma_E + \log z + E_1(z)$ . Since  $J(0^+) = 0$ , integration gives (76).

For (77), Markov's inequality applied to  $e^{-\theta\Lambda}$  yields, for every  $\theta > 0$ ,

$$\mathbb{P}(\Lambda \leq -y) \leq e^{-\theta y} \mathbb{E}e^{-\theta\Lambda} = \exp\{\theta(\log \theta - 1 + \gamma_E - y)\}.$$

The exponent is minimized at  $\theta = e^{y-\gamma_E}$ , where it equals  $-\theta$ , giving (77). Finally  $\mathbb{E}[(-\Lambda)^+] = \int_0^\infty \mathbb{P}(\Lambda \leq -y) dy \leq \int_0^\infty \exp\{-e^{y-\gamma_E}\} dy < \infty$ .  $\square$

**Lemma 14** (Stable limit). *Suppose*

$$\frac{1}{c} = \log n + a$$

*with fixed  $a \in \mathbb{R}$ . Then*

$$X_n \xrightarrow{d} X_a := (a+1) - \Lambda. \quad (79)$$

*Proof.* The characteristic function of  $X_n$  is

$$\phi_n(s) = e^{is/c} \left( \mathbb{E} e^{-isV} \right)^n.$$

Introduce the finite Lévy measure

$$\mu_n(dx) := nq_n(x) dx = \frac{n(c+n)}{(1+nx)^2} \mathbf{1}_{\{0 \leq x \leq 1/c\}} dx,$$

of total mass  $\mu_n([0, 1/c]) = n$ . For every fixed  $x > 0$ ,

$$\frac{n(c+n)}{(1+nx)^2} \longrightarrow \frac{1}{x^2}.$$

For fixed  $s$  and sufficiently large  $n$ , choose the logarithm of  $1 + z_n(s)/n$  near 1 and use the corresponding characteristic exponent for  $\phi_n(s)$ . With this convention, we write

$$\log \phi_n(s) = \frac{is}{c} + n \log \left( 1 + \frac{z_n(s)}{n} \right), \quad z_n(s) := \int (e^{-isx} - 1) \mu_n(dx). \quad (80)$$

Fix  $s$ . Using  $|e^{-isx} - 1| \leq \min(2, |s|x)$  and splitting the integral at  $x = 1/|s|$ ,

$$|z_n(s)| \leq |s| \int_0^{1/|s|} x \mu_n(dx) + 2 \mu_n((1/|s|, 1/c]) = O_s(\log n), \quad (81)$$

by the elementary evaluation (83) below. Thus  $z_n(s)/n \rightarrow 0$ , and the principal branch of the logarithm satisfies

$$n \log \left( 1 + \frac{z_n}{n} \right) = z_n + O \left( \frac{|z_n|^2}{n} \right) = z_n + O_s \left( \frac{\log^2 n}{n} \right) = z_n + o(1).$$

(The hypothesis needed here is  $z_n = o(\sqrt{n})$ , not boundedness of  $z_n$ ; (81) supplies it.) Hence

$$\begin{aligned} \log \phi_n(s) &= \frac{is}{c} + \int_0^{1/c} (e^{-isx} - 1) \mu_n(dx) + o(1) \\ &= is \left[ \frac{1}{c} - \int_0^1 x \mu_n(dx) \right] \\ &\quad + \int_0^{1/c} (e^{-isx} - 1 + isx \mathbf{1}_{\{x \leq 1\}}) \mu_n(dx) + o(1). \end{aligned} \quad (82)$$

The compensated integral converges, by dominated convergence on compact subintervals and elementary control at 0 and  $\infty$ , to

$$\int_0^\infty (e^{-isx} - 1 + isx \mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2}.$$

It remains to identify the drift. Substituting  $u = nx$ ,

$$\int_0^1 x \mu_n(dx) = \left( 1 + \frac{c}{n} \right) \int_0^n \frac{u du}{(1+u)^2} = \left( 1 + \frac{c}{n} \right) \left[ \log(1+n) - 1 + \frac{1}{1+n} \right]. \quad (83)$$

Since  $c = 1/(\log n + a)$ ,

$$\frac{1}{c} - \int_0^1 x \mu_n(dx) \longrightarrow a + 1.$$

Consequently

$$\log \phi_n(s) \longrightarrow is(a+1) + \int_0^\infty (e^{-isx} - 1 + isx \mathbf{1}_{\{x \leq 1\}}) \frac{dx}{x^2}. \quad (84)$$

By Definition 12, the limiting characteristic function is that of  $(a+1) - \Lambda$ . Lévy's continuity theorem gives (79).  $\square$

## 8.2 Uniform density and logarithmic moment control

Weak convergence alone is insufficient for the logarithmic observables appearing in the Rayleigh quotient, because  $\log x$  is singular at  $x = 0$ . We therefore record two uniform estimates.

**Lemma 15** (Uniform Fourier integrability and bounded densities). *For fixed  $a$ , with  $c^{-1} = \log n + a$ , there exists a constant  $C_a < \infty$  such that, for all sufficiently large  $n$ ,*

$$\int_{\mathbb{R}} |\phi_n(s)| ds \leq C_a. \quad (85)$$

Consequently  $X_n$  possesses a density  $f_n$  satisfying

$$\sup_{n \geq n_0} \|f_n\|_{\infty} < \infty. \quad (86)$$

In particular, uniformly for  $0 < \delta \leq 1$ ,

$$\mathbb{P}(0 < X_n \leq \delta) \ll_a \delta. \quad (87)$$

*Proof.* Since translation does not affect the modulus of the characteristic function,

$$|\phi_n(s)| = |\psi_n(s)|^n, \quad \psi_n(s) := \mathbb{E} e^{-isV},$$

and  $|\psi_n|$  is even in  $s$ , so we may assume  $s > 0$ .

An explicit two-interval bound. For  $\beta > 0$  let

$$I_1(\beta) := \left[0, \frac{\pi}{4}\beta\right], \quad I_2(\beta) := \left[\frac{3\pi}{4}\beta, \frac{5\pi}{4}\beta\right].$$

These are disjoint, and for  $v \in I_1(\beta)$ ,  $v' \in I_2(\beta)$  we have  $v' - v \in [\frac{\pi}{2}\beta, \frac{5\pi}{4}\beta]$ . Taking  $\beta = 1/s$  therefore gives  $s(v' - v) \in [\pi/2, 5\pi/4]$ , on which  $\cos \leq 0$  and hence  $1 - \cos(s(v - v')) \geq 1$ . Writing  $V'$  for an independent copy of  $V$  and discarding all other configurations,

$$1 - |\psi_n(s)|^2 = \mathbb{E}[1 - \cos(s(V - V'))] \geq 2 \mathbb{P}(V \in I_1(1/s)) \mathbb{P}(V \in I_2(1/s)). \quad (88)$$

Both intervals lie in  $[0, 1/c]$  once  $5\pi/(4s) \leq 1/c$ , which holds for all  $s \geq 1$  and  $n$  large, since  $1/c = \log n + a \rightarrow \infty$ .

From (73), for  $0 \leq \alpha < \beta \leq 1/c$ ,

$$\mathbb{P}(V \in [\alpha, \beta]) = \frac{c+n}{n} \left[ \frac{1}{1+n\alpha} - \frac{1}{1+n\beta} \right]. \quad (89)$$

Put  $b := \pi n/(4s)$ . Then (89) gives

$$\mathbb{P}(V \in I_1(1/s)) = \frac{c+n}{n} \cdot \frac{b}{1+b}, \quad \mathbb{P}(V \in I_2(1/s)) = \frac{c+n}{n} \cdot \frac{2b}{(1+3b)(1+5b)}.$$

*Low frequencies*  $1 \leq s \leq n$ . Here  $b \geq \pi/4$ , so  $b/(1+b) \geq (\pi/4)/(1+\pi/4) > \frac{1}{3}$  and  $2b/\{(1+3b)(1+5b)\} \geq 2b/\{(3+4/\pi)b \cdot (5+4/\pi)b\} \gg 1/b \gg s/n$ . With (88),

$$1 - |\psi_n(s)|^2 \gg \frac{s}{n}, \quad 1 \leq s \leq n, \quad (90)$$

and therefore, for some  $\kappa > 0$  independent of  $n$ ,

$$|\phi_n(s)| = (|\psi_n(s)|^2)^{n/2} \leq \left(1 - \frac{\kappa s}{n}\right)^{n/2} \leq e^{-\kappa s/2} \quad (1 \leq s \leq n). \quad (91)$$

*Intermediate frequencies*  $n \leq s \leq 8(c+n)$ . Now  $b \leq \pi/4$ , so  $b/(1+b) \gg b$  and  $2b/\{(1+3b)(1+5b)\} \gg b$ , whence  $1 - |\psi_n(s)|^2 \gg b^2 \gg n^2/s^2$ . Since  $s = O(n)$  throughout this range,

$$|\phi_n(s)| \leq e^{-\kappa' n}. \quad (92)$$

High frequencies. Integration by parts in

$$\psi_n(s) = \int_0^{1/c} e^{-isv} \frac{c+n}{(1+nv)^2} dv$$

gives

$$|\psi_n(s)| \leq \frac{2(c+n)}{s}. \quad (93)$$

Thus, for  $s \geq 8(c+n)$ ,

$$|\phi_n(s)| \leq \left(\frac{1}{4}\right)^n \left(\frac{8(c+n)}{s}\right)^n \leq \left(\frac{1}{4}\right)^n, \quad (94)$$

and more precisely  $|\phi_n(s)| \leq (2(c+n)/s)^n$ , whose integral over  $s \geq 8(c+n)$  is  $O((c+n)4^{-n})$ , hence uniformly bounded.

Combining (91), (92), and (94), together with the trivial bound  $|\phi_n(s)| \leq 1$  for  $|s| \leq 1$ , proves (85).

Fourier inversion then gives

$$\|f_n\|_\infty \leq \frac{1}{2\pi} \|\phi_n\|_{L^1},$$

which proves (86); (87) follows immediately.  $\square$

**Lemma 16** (Right-tail bound and logarithmic uniform integrability). *For fixed  $a$ , the random variables*

$$\log X_n \mathbf{1}_{\{X_n > 0\}} \quad \text{and} \quad (\log X_n)^2 \mathbf{1}_{\{X_n > 0\}}$$

*are uniformly integrable. Consequently, writing*

$$P_n := \mathbb{P}(X_n > 0), \quad (95)$$

$$L_n := \mathbb{E}[\log X_n; X_n > 0], \quad (96)$$

$$Q_n := \mathbb{E}[(\log X_n)^2; X_n > 0], \quad (97)$$

*we have*

$$P_n \longrightarrow P_a := \mathbb{P}(X_a > 0) > 0, \quad (98)$$

$$L_n \longrightarrow L_a := \mathbb{E}[\log X_a; X_a > 0], \quad (99)$$

$$Q_n \longrightarrow Q_a := \mathbb{E}[(\log X_a)^2; X_a > 0]. \quad (100)$$

*Proof.* The negative logarithmic tail is controlled by Lemma 15. For  $y \geq 0$ ,

$$\mathbb{P}(0 < X_n < e^{-y}) \ll e^{-y}.$$

Hence the negative parts of  $\log X_n$  have uniformly bounded moments of every fixed order.

For the positive tail, note from (74) that

$$X_n > x \iff \sum_{j=1}^n V_j < \frac{1}{c} - x.$$

For  $\lambda > 0$ , Chernoff's inequality gives

$$\mathbb{P}(X_n > x) \leq \exp\left(\lambda\left(\frac{1}{c} - x\right)\right) (\mathbb{E}e^{-\lambda V})^n. \quad (101)$$

We claim that there is a constant  $C_a$ , independent of  $n$  and  $\lambda$ , such that

$$n(1 - \mathbb{E}e^{-\lambda V}) \geq \lambda\left(\log \frac{n}{\lambda} - C_a\right), \quad 1 \leq \lambda \leq n. \quad (102)$$

Substituting  $u = nv$  in (73) and writing  $\beta := \lambda/n \in (0, 1]$ ,

$$n(1 - \mathbb{E}e^{-\lambda V}) = (c+n) \int_0^{n/c} (1 - e^{-\beta u}) \frac{du}{(1+u)^2}. \quad (103)$$

Rather than splitting the range, evaluate the integral exactly. Integration by parts gives, for the untruncated integral,

$$\int_0^\infty (1 - e^{-\beta u}) \frac{du}{(1+u)^2} = \beta \int_0^\infty \frac{e^{-\beta u}}{1+u} du = \beta e^\beta E_1(\beta).$$

To make the estimate uniform over the entire range  $0 < \beta \leq 1$ , note that  $e^\beta E_1(\beta) - \log(1/\beta)$  is continuous on  $(0, 1]$  and tends to  $-\gamma_E$  as  $\beta \downarrow 0$ . Consequently,

$$\sup_{0 < \beta \leq 1} |e^\beta E_1(\beta) - \log(1/\beta)| < \infty.$$

The discarded tail satisfies  $\int_{n/c}^\infty (1+u)^{-2} du \leq c/n$ , contributing at most  $(c+n)c/n = O(1) \leq O(\lambda)$  since  $\lambda \geq 1$ . Multiplying by  $(c+n) = n(1+c/n)$  and using  $\beta = \lambda/n$ , with  $(c/n) \log(n/\lambda) \leq (c/n) \log n = O(n^{-1})$ , yields

$$n(1 - \mathbb{E}e^{-\lambda V}) = \lambda \left( \log \frac{n}{\lambda} - \gamma_E + O(1) \right),$$

which is (102). (The coefficient 1 in front of  $\log(n/\lambda)$  is essential: it is what cancels  $1/c = \log n + a$  in (101). A cruder split using  $1 - e^{-y} \geq y/2$  produces the coefficient  $1/2$  and the argument fails.)

Since  $\log z \leq z - 1$  for  $z > 0$ ,

$$n \log \mathbb{E}e^{-\lambda V} \leq -n(1 - \mathbb{E}e^{-\lambda V}).$$

Using  $c^{-1} = \log n + a$  in (101) therefore gives

$$\mathbb{P}(X_n > x) \leq \exp \{ \lambda (\log \lambda + A_a - x) \}, \quad 1 \leq \lambda \leq n, \quad (104)$$

with  $A_a := a + C_a$ , enlarging  $C_a$  if necessary so that  $C_a \geq 0$ .

Choosing

$$\lambda = e^{x - A_a - 1}$$

whenever this lies in  $[1, n]$  makes the exponent equal to  $-\lambda$ , giving the double-exponential bound

$$\mathbb{P}(X_n > x) \leq \exp \{ -e^{x - A_a - 1} \}, \quad A_a + 1 \leq x \leq \log n + A_a + 1. \quad (105)$$

For  $x < A_a + 1$  the trivial bound  $\mathbb{P} \leq 1$  suffices, and for  $x > \log n + A_a + 1$  we have  $x > 1/c$  (as  $C_a > -1$ ), so  $\mathbb{P}(X_n > x) = 0$  because  $X_n \leq 1/c$  almost surely.

Thus both the positive and negative logarithmic tails are uniformly integrable. Combining this with Lemma 14, and noting that the limiting stable law has a continuous positive density, so that  $\mathbb{P}(X_a = 0) = 0$  and  $P_a > 0$ , gives (98)–(100).  $\square$

### 8.3 Asymptotics of the Rayleigh quotient

We now return to the exact Rayleigh quotient.

Since  $1 - S_n = cX_n$ , equations (69) and (70) give, on  $\{X_n > 0\}$ ,

$$G(1 - S_n) = \frac{1}{n} \log(1 + nX_n), \quad (106)$$

$$\frac{H(1 - S_n)}{m_0} = \frac{(c+n)X_n}{1 + nX_n}. \quad (107)$$

Therefore

$$R(F_{k,c}) = c \left( 1 + \frac{1}{n} \right) \left( 1 + \frac{c}{n} \right) \frac{\mathbb{E}[\log^2(1 + nX_n); X_n > 0]}{\mathbb{E}[\frac{(c+n)X_n}{1 + nX_n}; X_n > 0]}. \quad (108)$$

Let

$$\ell := \log n, \quad c = \frac{1}{\ell + a}.$$

**Lemma 17** (Denominator asymptotics). *For fixed  $a$ ,*

$$\mathbb{E} \left[ \frac{(c+n)X_n}{1 + nX_n}; X_n > 0 \right] = P_n + O(n^{-1/2}). \quad (109)$$

*Remark 18.* The explicit rate, rather than a bare  $o(1)$ , is what the proof of Theorem 20 requires: the denominator is divided into a numerator of size  $\ell^2$ , so an additive error  $\eta$  here propagates to an error of size  $\ell\eta$  in  $R$ . An  $o(1)$  statement would only give  $o(\ell)$ , which is insufficient.

*Proof.* For  $0 < X_n \leq 1/c$ ,

$$0 \leq 1 - \frac{(c+n)X_n}{1+nX_n} = \frac{1-cX_n}{1+nX_n} \leq \frac{1}{1+nX_n}.$$

Choose

$$\delta_n = n^{-1/2}.$$

On  $0 < X_n \leq \delta_n$ , Lemma 15 gives probability  $O(\delta_n)$ . On  $X_n > \delta_n$ ,

$$\frac{1}{1+nX_n} \leq \frac{1}{n\delta_n} = n^{-1/2}.$$

Hence the difference between the left-hand side of (109) and  $P_n$  is  $O(n^{-1/2})$ .  $\square$

**Lemma 19** (Numerator asymptotics). *For fixed  $a$ ,*

$$\mathbb{E}[\log^2(1+nX_n); X_n > 0] = \ell^2 P_n + 2\ell L_n + Q_n + O(\ell^2 n^{-1/2}). \quad (110)$$

*Proof.* Let  $\delta_n = n^{-1/2}$ .

*Region  $X_n > \delta_n$ .* Here

$$\log(1+nX_n) = \ell + \log X_n + r_n, \quad 0 \leq r_n = \log \left( 1 + \frac{1}{nX_n} \right) \leq \frac{1}{nX_n} \leq n^{-1/2},$$

so that

$$\log^2(1+nX_n) = (\ell + \log X_n)^2 + 2(\ell + \log X_n)r_n + r_n^2.$$

By Lemma 16 the quantities  $\mathbb{E}[|\log X_n|; X_n > 0]$  are bounded uniformly in  $n$ , so

$$\mathbb{E}[|2(\ell + \log X_n)r_n| + r_n^2; X_n > \delta_n] = O(\ell n^{-1/2}).$$

*Region  $0 < X_n \leq \delta_n$ .* The uniform density bound of Lemma 15 gives

$$\begin{aligned} \mathbb{E}[(\ell + \log X_n)^2; 0 < X_n \leq \delta_n] &\leq C \int_0^{\delta_n} (\ell + \log x)^2 dx \\ &= C\delta_n \left[ (\ell + \log \delta_n)^2 - 2(\ell + \log \delta_n) + 2 \right] = O(\ell^2 n^{-1/2}), \end{aligned} \quad (111)$$

using  $\ell + \log \delta_n = \ell/2$ . Likewise, since  $nX_n \leq n\delta_n = n^{1/2}$  on this region,

$$\mathbb{E}[\log^2(1+nX_n); 0 < X_n \leq \delta_n] \leq C \int_0^{\delta_n} \log^2(1+nx) dx = O(\ell^2 n^{-1/2}).$$

Adding the two regions and recognising  $\mathbb{E}[(\ell + \log X_n)^2; X_n > 0] = \ell^2 P_n + 2\ell L_n + Q_n$  proves (110).  $\square$

## 8.4 Main asymptotic theorem

**Theorem 20** (Asymptotic defect of the rational trial family). *Fix  $a \in \mathbb{R}$ . For all sufficiently large  $k$ , put*

$$n = k - 1, \quad c_k = \frac{1}{\log n + a}.$$

*For all sufficiently large  $k$ ,  $c_k > 0$ . Let  $F_{k,c_k}$  be the single-channel rational trial function (65). Then*

$$R(F_{k,c_k}) = \log k - \mathcal{C}(a) + o(1), \quad k \rightarrow \infty, \quad (112)$$

where

$$\boxed{\mathcal{C}(a) = a - 2\mathbb{E}[\log X_a \mid X_a > 0]} \quad (113)$$

and

$$X_a = (a + 1) - \Lambda$$

with  $\Lambda$  defined by (75).

Equivalently,

$$\lim_{k \rightarrow \infty} (\log k - R(F_{k,c_k})) = \mathcal{C}(a). \quad (114)$$

*Proof.* Write Num and Den for the numerator and denominator in (108). By Lemmas 17 and 19,

$$\text{Num} = \ell^2 P_n + 2\ell L_n + Q_n + O(\ell^2 n^{-1/2}) = O(\ell^2), \quad \text{Den} = P_n + O(n^{-1/2}),$$

and  $P_n \rightarrow P_a > 0$  by Lemma 16, so Den is bounded away from 0 for large  $n$ . Hence

$$\frac{\text{Num}}{\text{Den}} = \frac{\ell^2 P_n + 2\ell L_n + Q_n}{P_n} + O(\ell^2 n^{-1/2}).$$

Multiplying by  $c(1 + \frac{1}{n})(1 + \frac{c}{n}) = c(1 + O(n^{-1}))$  and using  $c = (\ell + a)^{-1}$ ,

$$R(F_{k,c}) = \frac{\ell^2 + 2\ell L_n/P_n + Q_n/P_n}{\ell + a} + O(\ell n^{-1/2}). \quad (115)$$

A direct rearrangement yields the finite- $n$  expansion

$$\begin{aligned} \ell - R(F_{k,c}) &= a - 2\frac{L_n}{P_n} \\ &\quad - \frac{Q_n/P_n - 2aL_n/P_n + a^2}{\ell + a} + O(\ell n^{-1/2}). \end{aligned} \quad (116)$$

By Lemma 16,

$$\frac{L_n}{P_n} \longrightarrow \frac{L_a}{P_a} = \mathbb{E}[\log X_a \mid X_a > 0],$$

while the fraction on the second line of (116) tends to zero. Hence

$$\ell - R(F_{k,c}) \longrightarrow a - 2\mathbb{E}[\log X_a \mid X_a > 0].$$

Finally,

$$\log k - \log(k - 1) = O(n^{-1}),$$

so  $\ell$  may be replaced by  $\log k$ . This proves (112) and (114).  $\square$

## 8.5 Consequence for $M_k$

Define

$$\mathcal{C}_* := \inf_{a \in \mathbb{R}} \mathcal{C}(a). \quad (117)$$

**Corollary 21** (Asymptotic lower bound for  $M_k$ ). *The Maynard variational constant satisfies*

$$\liminf_{k \rightarrow \infty} (M_k - \log k) \geq -\mathcal{C}_*. \quad (118)$$

Equivalently,

$$M_k \geq \log k - \mathcal{C}_* - o(1). \quad (119)$$

*Proof.* For every  $\varepsilon > 0$ , choose a fixed  $a_\varepsilon \in \mathbb{R}$  such that

$$\mathcal{C}(a_\varepsilon) \leq \mathcal{C}_* + \varepsilon.$$

Since  $M_k$  is the supremum over admissible trial functions, Theorem 20 gives

$$M_k \geq R(F_{k,c_k}) = \log k - \mathcal{C}(a_\varepsilon) + o(1)$$

for the choice

$$c_k^{-1} = \log(k-1) + a_\varepsilon.$$

Thus

$$\liminf_{k \rightarrow \infty} (M_k - \log k) \geq -\mathcal{C}_* - \varepsilon.$$

Letting  $\varepsilon \downarrow 0$  proves (118).  $\square$

The infimum in (117) may be restricted to a half-line bounded above, since  $\mathcal{C}$  grows without bound in the positive direction.

**Lemma 22** (Growth of  $\mathcal{C}$  for large positive shift). *With  $\kappa = \mathbb{E}[(-\Lambda)^+] < \infty$  as in (78), there is  $a_0$  such that for all  $a \geq a_0$ ,*

$$\mathcal{C}(a) \geq a - 2\log(2(a+1+\kappa)) \xrightarrow{a \rightarrow \infty} \infty. \quad (120)$$

In particular  $\inf_{a \in \mathbb{R}} \mathcal{C}(a) = \inf_{a \leq a_1} \mathcal{C}(a)$  for some finite  $a_1$ .

*Proof.* Since  $X_a^+ = ((a+1) - \Lambda)^+ \leq (a+1) + (-\Lambda)^+$  for  $a \geq -1$ ,

$$\mathbb{E}[X_a; X_a > 0] \leq (a+1) + \kappa.$$

Also  $P_a = \mathbb{P}(\Lambda < a+1) \rightarrow 1$ , so  $P_a \geq \frac{1}{2}$  for  $a \geq a_0$ . By Jensen's inequality applied to the concave function  $\log$  under the conditional law,

$$\mathbb{E}[\log X_a \mid X_a > 0] \leq \log \mathbb{E}[X_a \mid X_a > 0] = \log \frac{\mathbb{E}[X_a; X_a > 0]}{P_a} \leq \log(2(a+1+\kappa)),$$

which gives (120).  $\square$

*Remark 23* (Behaviour as  $a \rightarrow -\infty$ , and attainment of the infimum). Numerical results suggest that  $\mathcal{C}(a) \rightarrow \infty$  as  $a \rightarrow -\infty$  as well, and that the infimum in (117) is attained. These assertions are not proved here. The proposed mechanism is that conditioning on  $\{X_a > 0\} = \{\Lambda < a+1\}$  for very negative  $a$  forces  $X_a$  to concentrate at 0: writing  $y = -(a+1)$ , the ratio  $\mathbb{P}(\Lambda < -y-t)/\mathbb{P}(\Lambda < -y)$  is expected to behave like  $\exp\{-ce^y t\}$  for small  $t > 0$ , so that conditionally  $X_a$  is approximately exponential with rate of order  $e^{-(a+1)}$ , giving  $\mathbb{E}[\log X_a \mid X_a > 0] = (a+1) + O(1)$  and hence  $\mathcal{C}(a) = -a + O(1)$ . A rigorous proof requires a two-sided left-tail estimate for  $\Lambda$ ; Lemma 13 supplies only the upper bound (77), and the Jensen argument of Lemma 22 is too lossy in this regime because  $P_a$  is doubly-exponentially small. We therefore state (117) as an infimum; none of the results below depend on attainment.

*Remark 24* (Scope of the optimization). The quantity  $\mathcal{C}_*$  is the optimal asymptotic constant *among the fixed-shift scalings*

$$c_k^{-1} = \log(k-1) + a, \quad a \in \mathbb{R}.$$

The theorem does not by itself prove that the same constant is optimal over every possible sequence  $c_k > 0$  in the full single-channel rational family. Such a statement would additionally require showing that every asymptotically competitive sequence satisfies

$$c_k^{-1} - \log k = O(1),$$

and controlling non-convergent or divergent shift sequences. We do not need this stronger statement for the lower bound (119).

## 8.6 Numerical interpretation

Theorem 20 is an analytic statement and does not depend on the numerical discovery or certification pipeline. Numerical quadrature of the limiting stable law can, however, be used to evaluate  $\mathcal{C}(a)$  and to locate a favorable fixed shift. Two independent routes are available: direct quadrature against the density of  $\Lambda$ , and inversion of the explicit Laplace transform (76). Our numerical evaluation indicates a minimum near a small negative value of  $a$  and an asymptotic defect of approximately

$$\mathcal{C}_* \approx 0.3343. \quad (121)$$

The value in (121) is a numerical estimate of the infimum, not a certified enclosure established by this asymptotic analysis. Certifying a finite- $k$  Rayleigh quotient and certifying the limiting constant are separate tasks.

For an explicit asymptotic lower bound it suffices to choose a single fixed shift  $a_0$  and establish, by rigorous interval quadrature with controlled truncation errors, an upper bound  $\mathcal{C}(a_0) \leq U$ . Theorem 20 then gives

$$\liminf_{k \rightarrow \infty} (M_k - \log k) \geq -U,$$

without any proof of global minimality. By contrast, a rigorous enclosure of  $\mathcal{C}_*$  itself also requires a lower bound valid for every  $a \in \mathbb{R}$ . Any numerical search restricted to a finite interval must therefore be supplemented by rigorous control of both excluded tails, including  $a \rightarrow -\infty$ . The upper bound on the left tail of  $\Lambda$  proved above does not by itself supply that control.

The finite- $k$  optimization exhibits the same qualitative scaling:

$$c_k \sim \frac{1}{\log k}, \quad R(F_{k,c_k}) = \log k - O(1).$$

The fact that the discrepancy  $\log k - R(F_{k,c_k})$  varies only slowly over more than five orders of magnitude in  $k$  is therefore explained by the stable-law limit rather than being an empirical coincidence.

## 8.7 Finite-size corrections

Expansion (116) also isolates the finite-size correction. All error terms accumulated in its derivation are  $O(\ell n^{-1/2})$ , hence  $o(\ell^{-m})$  for every fixed  $m$ ; the replacement of  $\log n$  by  $\log k$  contributes  $O(n^{-1})$ . Consequently

$$\log k - R(F_{k,c}) = a - 2\frac{L_n}{P_n} - \frac{Q_n/P_n - 2aL_n/P_n + a^2}{\ell + a} + O(\ell n^{-1/2}), \quad (122)$$

Define

$$d_n(a) := \frac{L_n}{P_n} - \frac{L_a}{P_a}, \quad B(a) := \frac{Q_a}{P_a} - 2a\frac{L_a}{P_a} + a^2.$$

Since  $P_n \rightarrow P_a > 0$ ,  $L_n \rightarrow L_a$ , and  $Q_n \rightarrow Q_a$ , the expansion above implies

$$\log k - R(F_{k,c}) = \mathcal{C}(a) - 2d_n(a) - \frac{B(a)}{\ell} + o(\ell^{-1}). \quad (123)$$

Here  $d_n(a) \rightarrow 0$ , but no quantitative rate for this convergence has been proved. Thus  $d_n(a)$  may dominate the displayed  $1/\ell$  term, and (123) is not yet a two-term asymptotic expansion with a determined coefficient.

A sufficient additional estimate is

$$d_n(a) = o(\ell^{-1}),$$

in which case

$$\log k - R(F_{k,c}) = \mathcal{C}(a) - \frac{B(a)}{\log k} + o((\log k)^{-1}).$$

More generally, if  $\ell d_n(a) \rightarrow b(a)$ , the coefficient of  $1/\log k$  is  $-[B(a) + 2b(a)]$ . A bound of only  $d_n(a) = O(\ell^{-1})$  does not determine that coefficient. Accordingly, a quantitative stable-limit argument must establish an appropriate rate for these conditional logarithmic moments, or their first-order asymptotics; a distributional convergence rate alone requires an additional argument to handle the logarithmic singularity. Any fitted coefficient of  $1/\log k$  remains a numerical finite-size observation until such an estimate is supplied.

## 8.8 Interpretation

The asymptotic mechanism can be summarized as follows. Squaring the single rational channel induces the one-dimensional density

$$q_n(v) = \frac{c + n}{(1 + nv)^2},$$

whose centered  $n$ -fold sum converges to a spectrally positive 1-stable law. Spectral positivity refers to the jumps; the limiting law has support on all of  $\mathbb{R}$ . Choosing

$$c^{-1} = \log n + a$$

centers the boundary variable

$$X_n = \frac{1}{c} - \sum_{j=1}^n V_j$$

at a non-degenerate limit. Conditional on the event  $X_n > 0$ , which is exactly the simplex event  $S_n < 1$ , the numerator of the Rayleigh quotient contains

$$\log^2(1 + nX_n) = (\log n + \log X_n + o(1))^2,$$

whereas the denominator converges to the same survival probability. The leading  $\log^2 n$  term is therefore divided by  $c^{-1} \sim \log n$ , producing a Rayleigh quotient of size  $\log n$ . The conditional logarithmic moment of the limiting stable variable supplies the non-trivial  $O(1)$  defect.

Thus the rational family is not merely a finite-dimensional numerical ansatz that happens to perform well at large  $k$ . Its observed scaling is the finite- $k$  manifestation of a stable-law limit, and the family gives the explicit asymptotic lower bound

$$M_k \geq \log k - C_* - o(1).$$

## Supplementary References

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